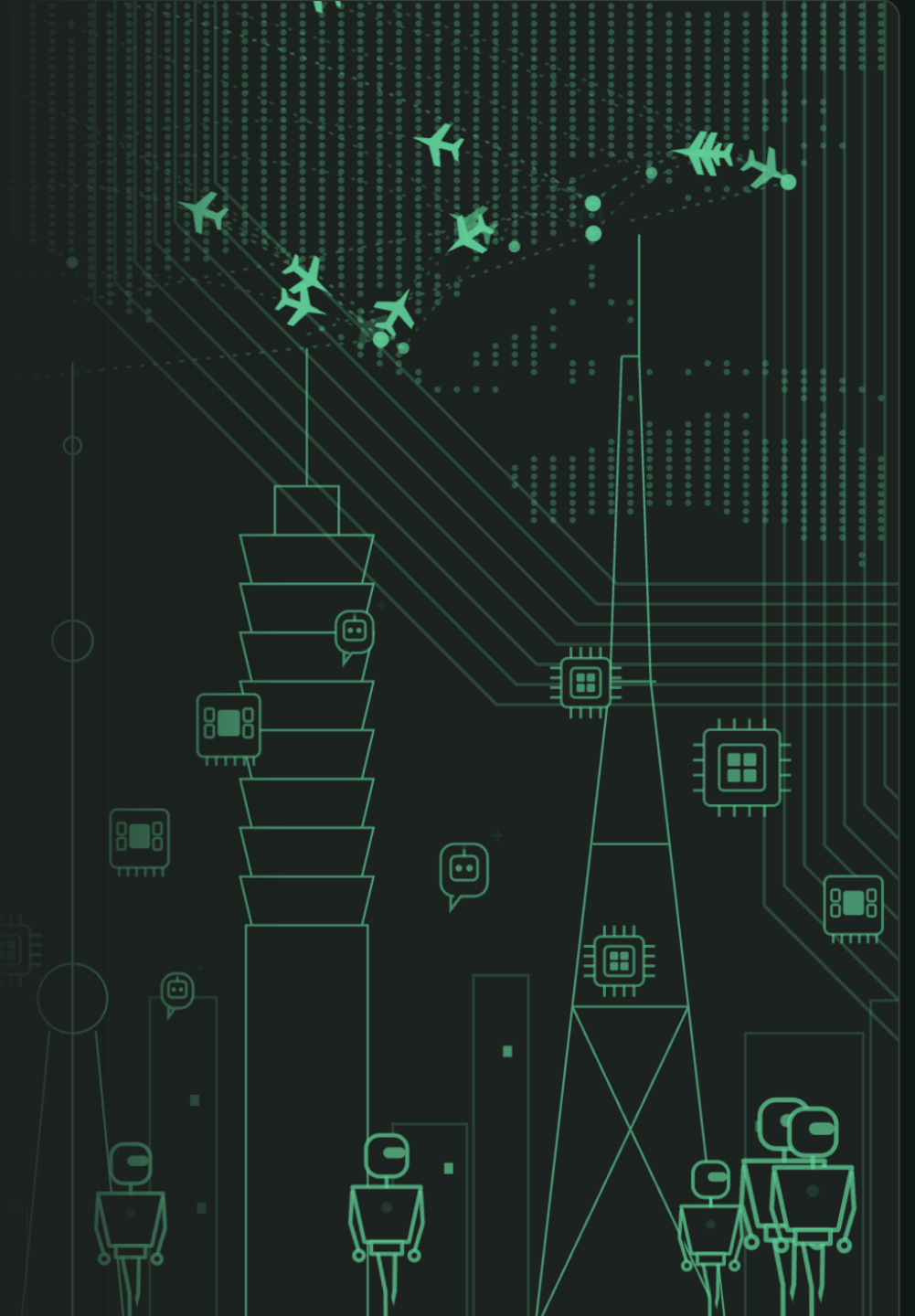
ARC RESEARCH · SPECIAL TOPICS · 11 NOVEMBER 2026

# How we got to the AI trade

A story told year by year, from the launch of ChatGPT in late 2022 to the selloff of November 2026

NVIDIA · AMD · intel · TSMC · SK hynix · SAMSUNG · micron



Data as of 11 Nov 2026 · Educational purposes only, not investment advice

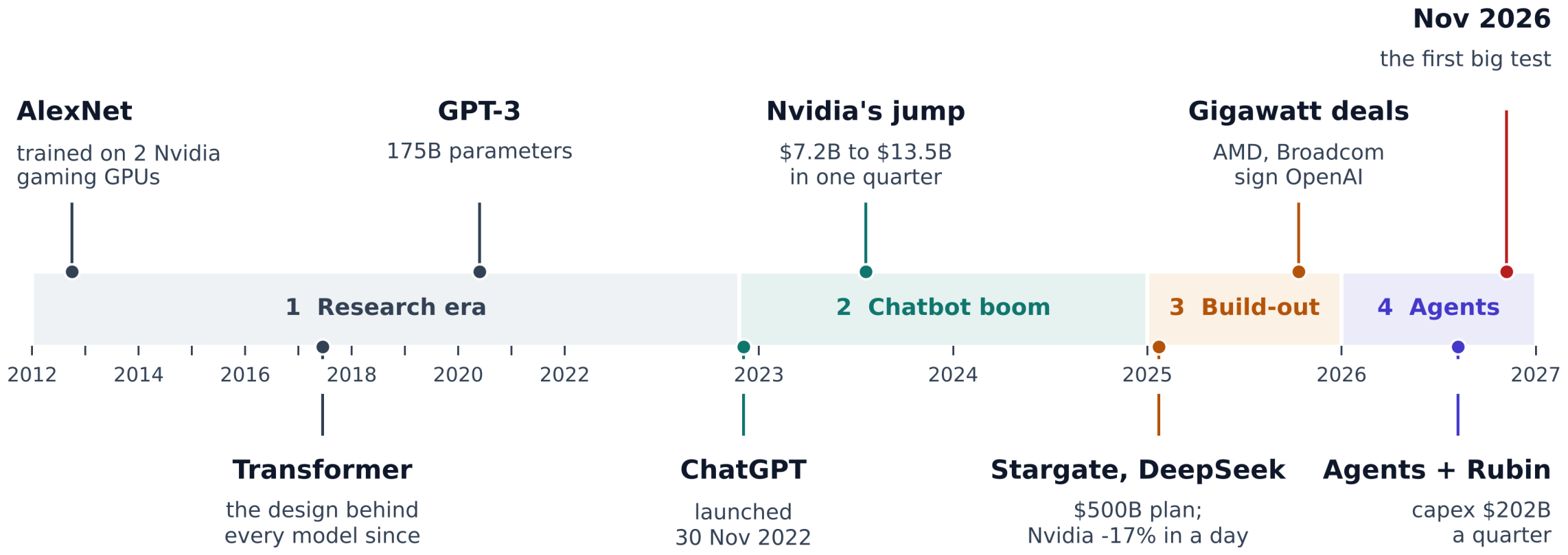
THE STORY

# Eight chapters, year by year

| | | | |
|---|---|---|---|
| 1 | **2022: ChatGPT arrives** | A bear market, then a chatbot nobody priced in | **Slides 6–8** |
| 2 | **2023: Nvidia’s year** | The market discovers who sells the picks and shovels | **Slides 9–14** |
| 3 | **2024: the trade broadens** | Power, custom chips and TSMC join Nvidia | **Slides 15–26** |
| 4 | **2025: agents and the memory squeeze** | Agents and long context land on memory | **Slides 27–38** |
| 5 | **2025: DeepSeek and the China split** | Who leads where, and how the two chains separated | **Slides 39–49** |
| 6 | **2025: gigawatts and circular money** | Deals sized in power, and money that loops back | **Slides 50–58** |
| 7 | **2026: bigger chips, bigger bets** | The rally widens, models reach trillions, capex hits records | **Slides 59–74** |
| 8 | **2026: the first big test** | Midterms, the selloff, Huang’s exit, the IPOs | **Slides 75–91** |

Before chapter 1: timeline (3), the chain (4), key terms (5) · Sources (92–95)

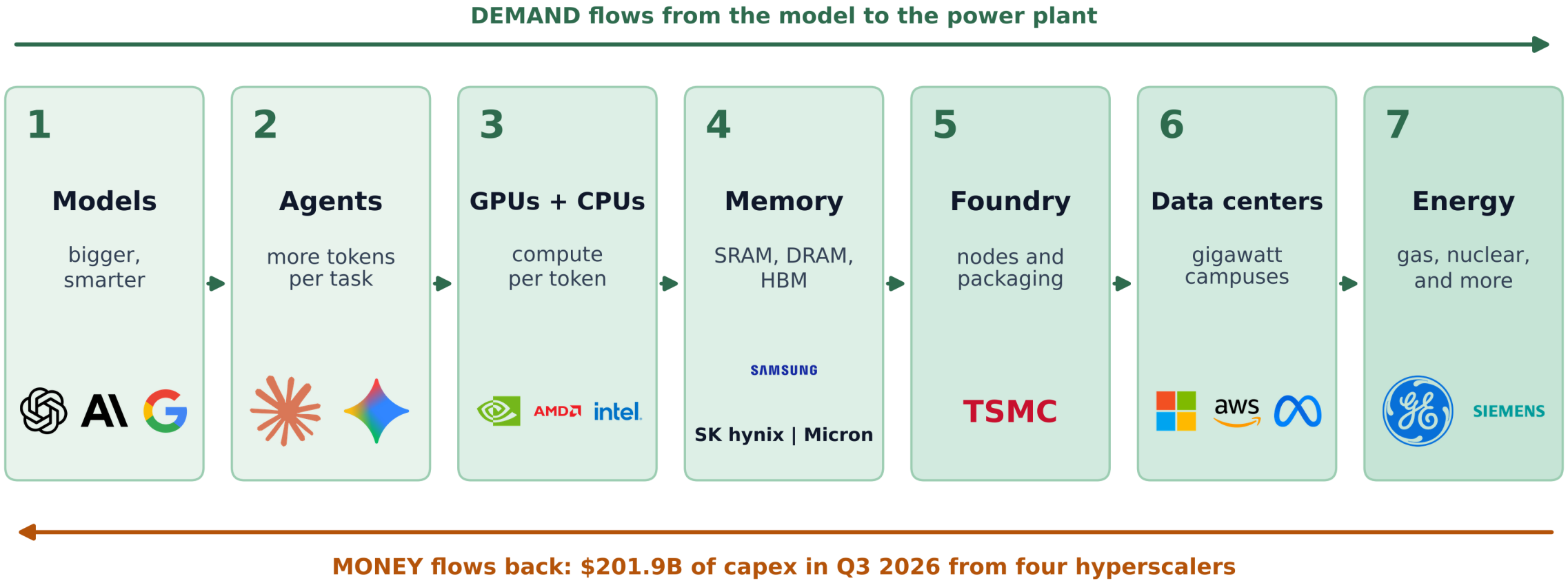
# Fourteen years from an experiment to a build-out




**Each phase moved the money one step further down the chain: models -> GPUs -> memory, foundry, power**

ARC Research timeline. Krizhevsky et al. (NeurIPS 2012); Vaswani et al. (2017); Brown et al. (2020); OpenAI (30 Nov 2022); NVIDIA results (Q1-Q2 FY2024); OpenAI Stargate (21 Jan 2025); Bloomberg (27 Jan 2025); AMD and OpenAI (6 Oct 2025); capex: SEC filings, ours.

# What the AI trade is, on one slide




ARC Research schematic; company marks are examples, not a complete list. Capex: Alphabet, Amazon, Meta, Microsoft purchases of property and equipment, calendar Q3 2026, SEC filings (our sum).

BEFORE WE START

# Ten words you will hear in this story

| Term | In plain words |
|---|---|
| GPU | Chip doing thousands of sums at once |
| CPU | General chip; runs tools and systems |
| Parameter | One number a model has learned |
| Token | A piece of a word; the unit of work |
| Agent | A model that plans and uses tools |

| Term | In plain words |
|---|---|
| KV cache | A model’s memory of one conversation |
| HBM | Memory stacked right beside the GPU |
| Capex | Spending on buildings and equipment |
| Foundry | Factory making chips others design |
| Gigawatt | A billion watts: one big power plant |

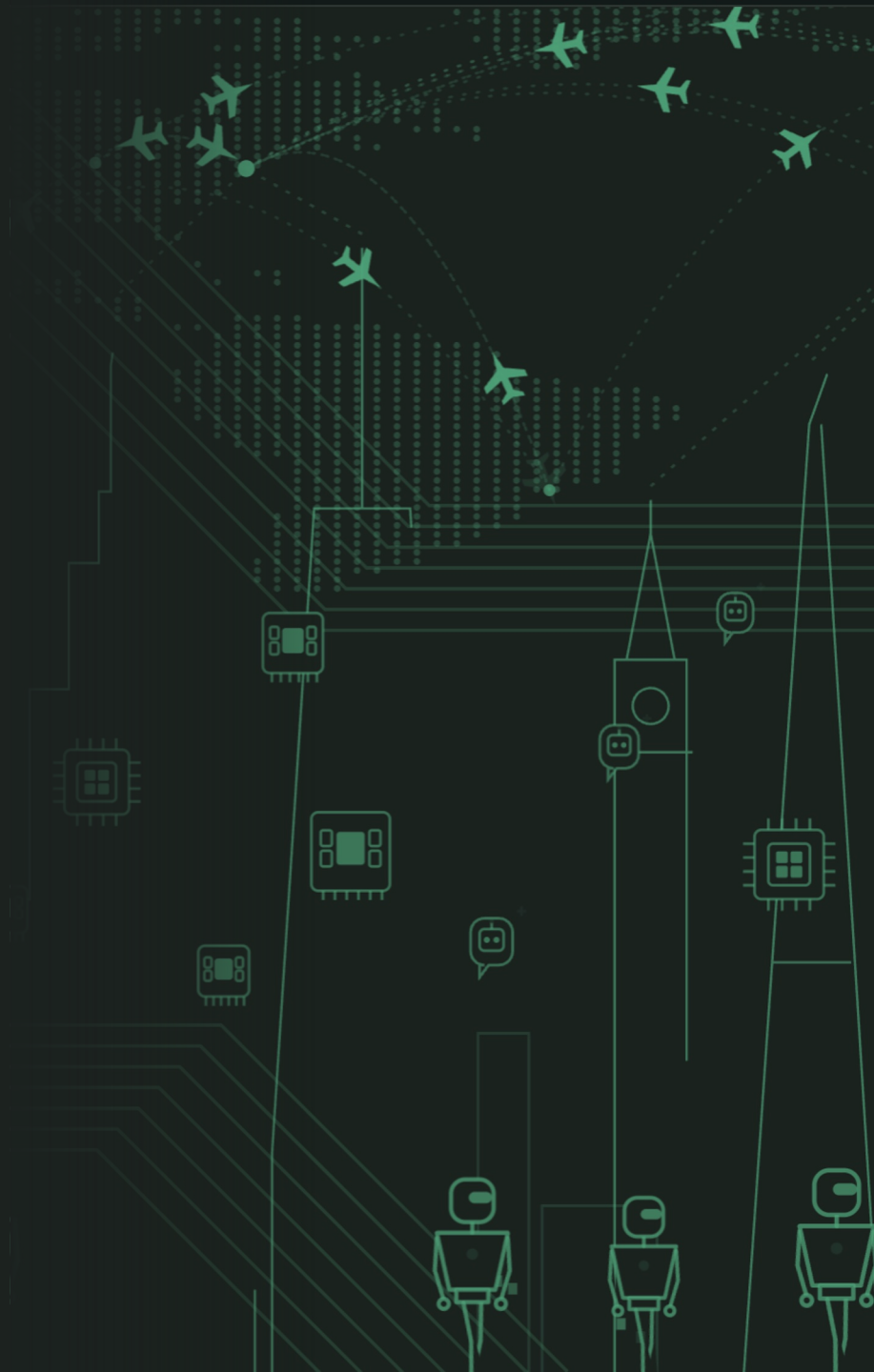
CHAPTER 1 OF 8

# 2022: ChatGPT arrives

A bear market, then a chatbot nobody priced in

COMING UP


| | |
|---|---|
| 2022 stock returns | 7 |
| ChatGPT and OpenAI | 8 |

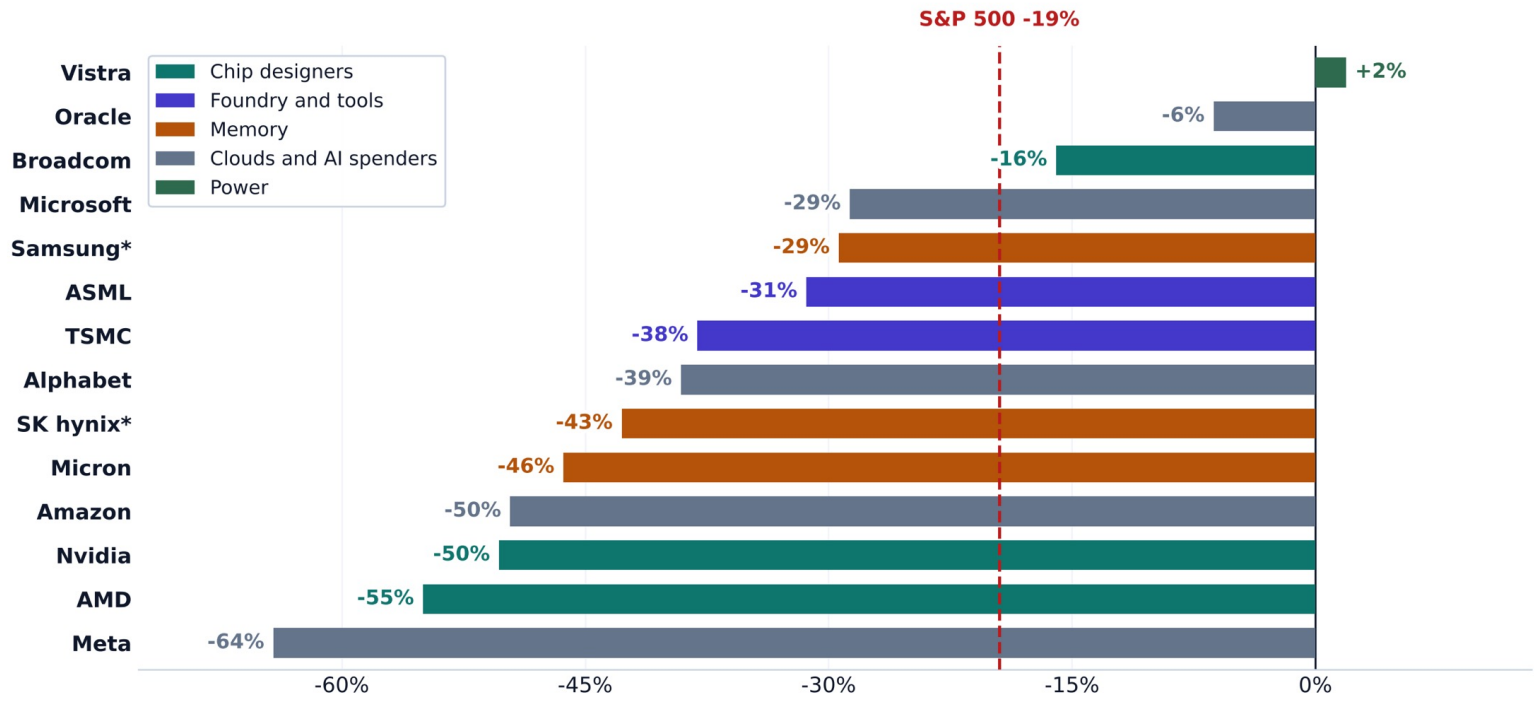



# 2022: the AI trade had not started yet

ARC RESEARCH | SPECIAL TOPICS | THE AI TRADE, YEAR BY YEAR

## 2022: everything fell, then ChatGPT arrived in November

Rate hikes crushed tech; Nvidia fell 50%. ChatGPT launched on 30 November, but markets had not noticed yet.




Source: price returns computed by us from split-adjusted closes (digrin.com; stockanalysis.com for 12 Nov 2026), cross-checked with macrotrends; not dividend-adjusted.
*In Korean won. S&P 500 via SPY. CoreWeave 2025 from its $40 IPO price (28 Mar). Names missing in a year had not listed or lack a verified start price.

### SO WHAT

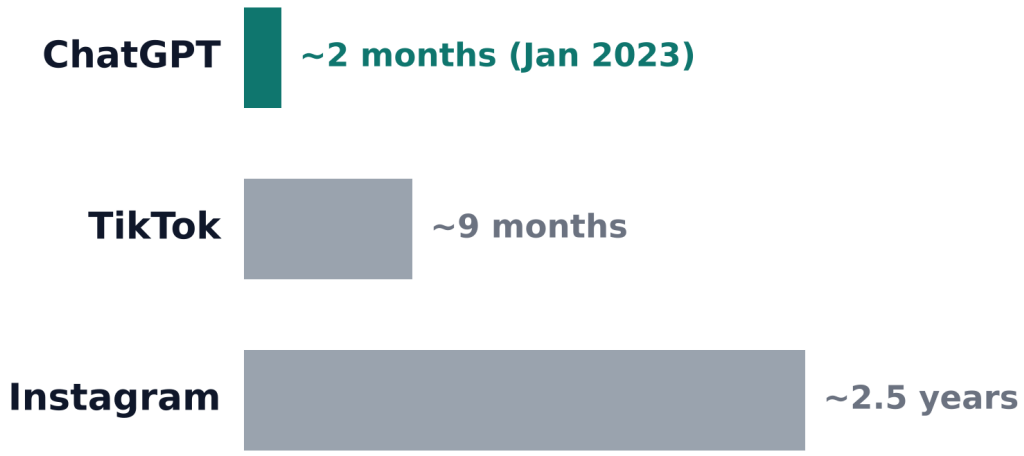
Nvidia -50% in 2022

ChatGPT: 30 November

No AI premium yet

# ChatGPT: the fastest-growing app ever, from a small lab

## Months to reach 100 million users




ChatGPT: 30 Nov 2022, free “research preview”
GPT-3.5 model, tuned with human feedback (RLHF)
Trained on Microsoft Azure supercomputers

## OpenAI, the lab behind it

| | |
|---|---|
| Dec 2015 | Founded as a non-profit research lab |
| Mar 2019 | Capped-profit company created to raise money |
| Jul 2019 | Microsoft invests $1B; Azure becomes its cloud |
| May 2020 | Azure supercomputer: 10,000 GPUs |
| 30 Nov 2022 | ChatGPT launches, free |
| Dec 2022 | Pitches investors: $200M revenue in 2023, $1B in 2024 |
| Jan 2023 | Microsoft: “multibillion” investment; ~$29B valuation |
| Feb 2023 | ChatGPT Plus at $20 a month |
| Mar 2023 | GPT-4 released |

Sources: openai.com (ChatGPT, OpenAI LP, Microsoft 2019, ChatGPT Plus, GPT-4 posts); Microsoft (May 2020, 23 Jan 2023); UBS study citing Similarweb and Sensor Tower, via Reuters, 2 Feb 2023 (users); Reuters, 15 Dec 2022 (revenue projections, hollow marker = projection); Fortune citing WSJ, 10 Jan 2023 (valuation, reported).

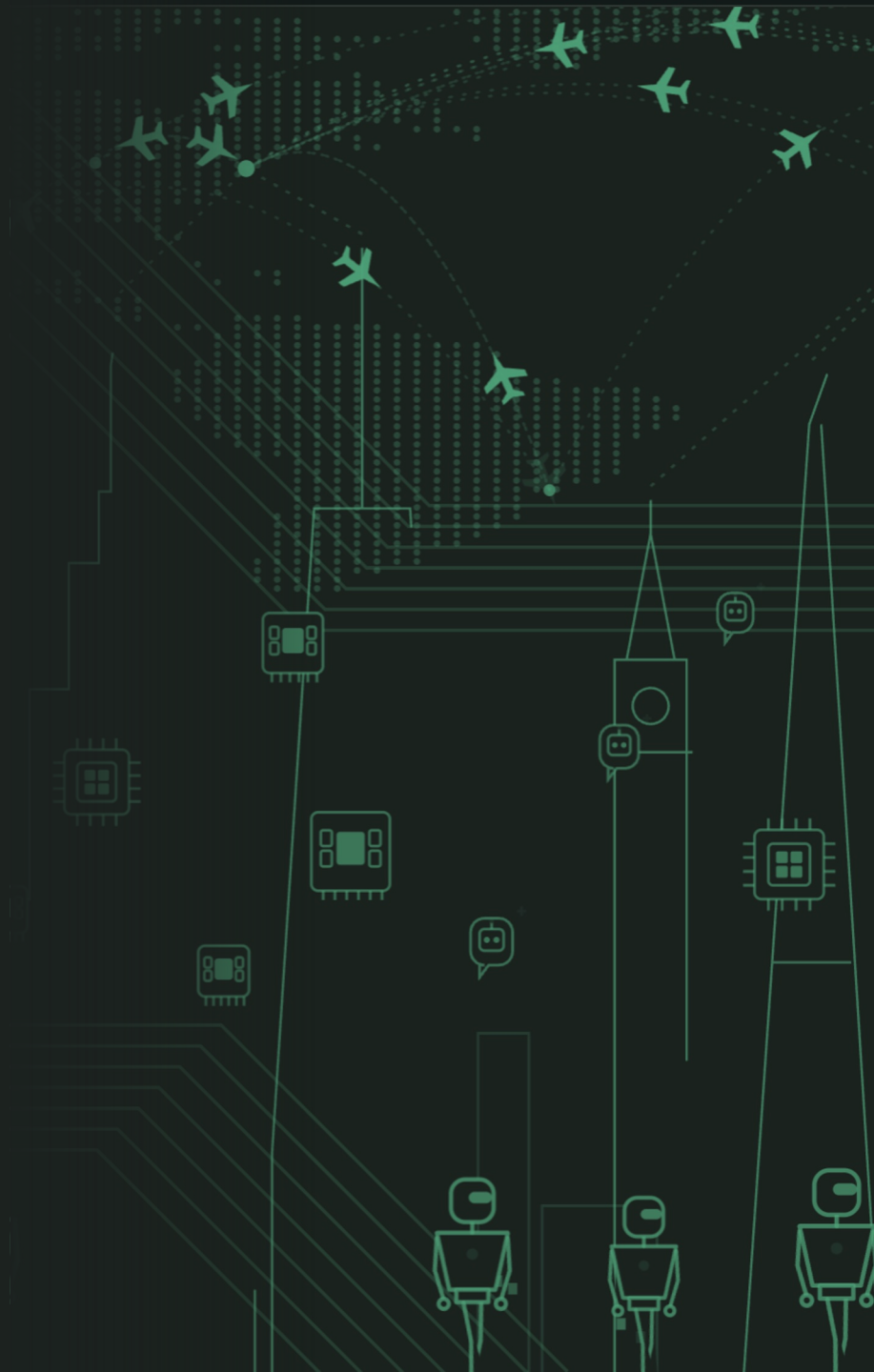
CHAPTER 2 OF 8

# 2023: Nvidia’s year

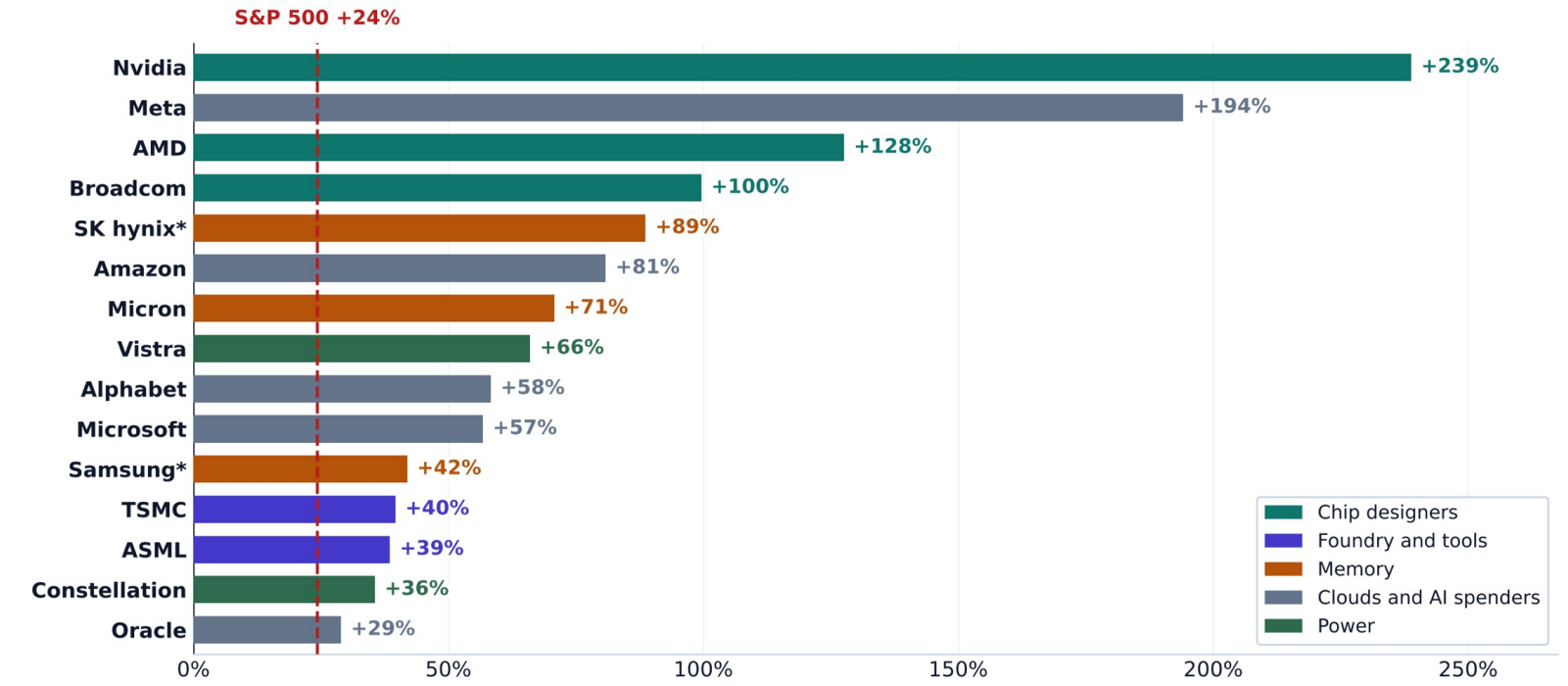
The market discovers who sells the picks and shovels

COMING UP


| | |
|---|---|
| 2023 stock returns | 10 |
| Models get bigger | 11 |
| Who pays: capex so far | 12 |
| Why Nvidia won | 13 |
| Who depends on TSMC | 14 |

# 2023: the market discovers Nvidia




## SO WHAT

Nvidia +239%

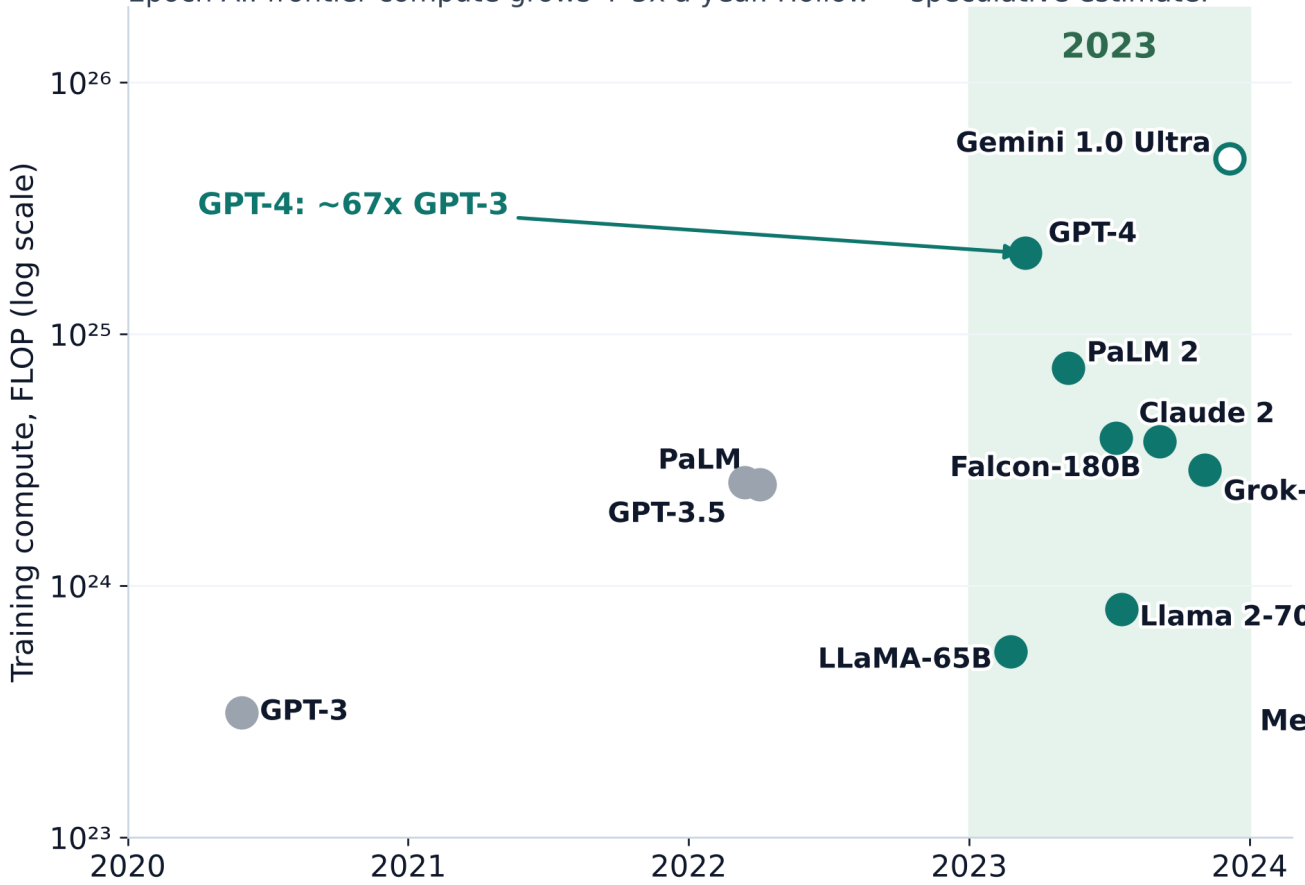
Every AI name beat the market

The spenders rallied too

# Step 1: models got 67 times bigger, and every lab wanted one

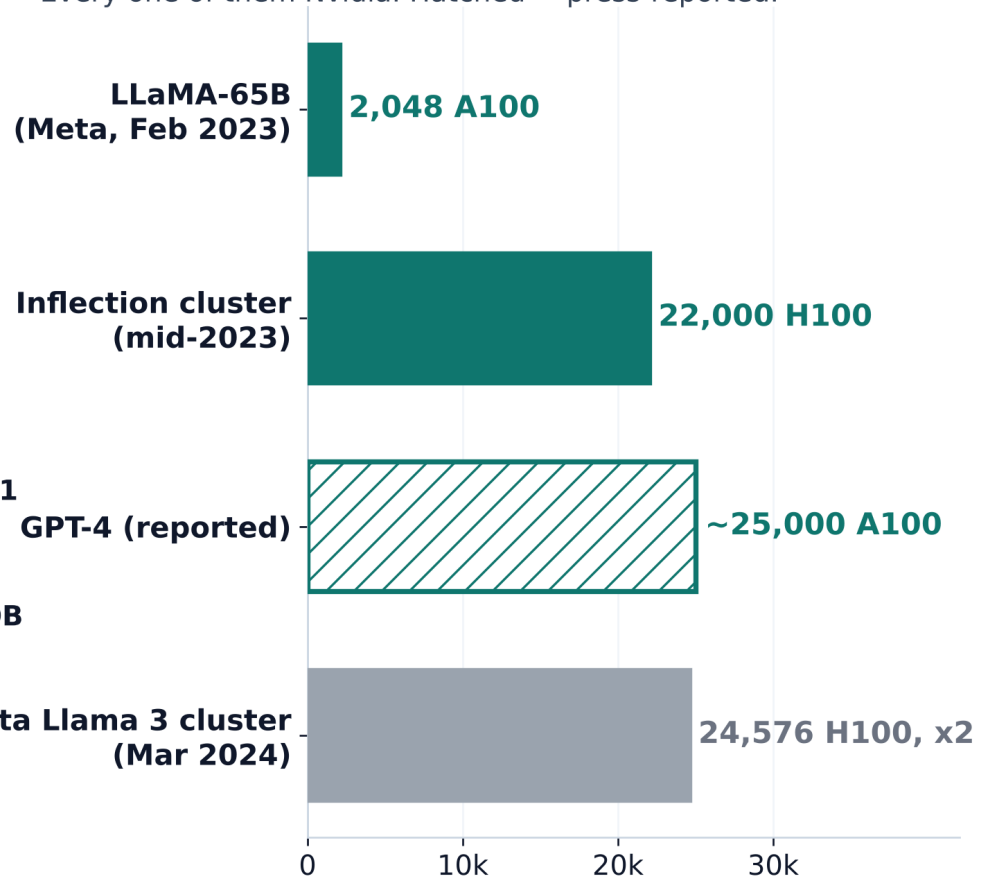
## Training compute of notable models

Epoch AI: frontier compute grows 4–5x a year. Hollow = speculative estimate.



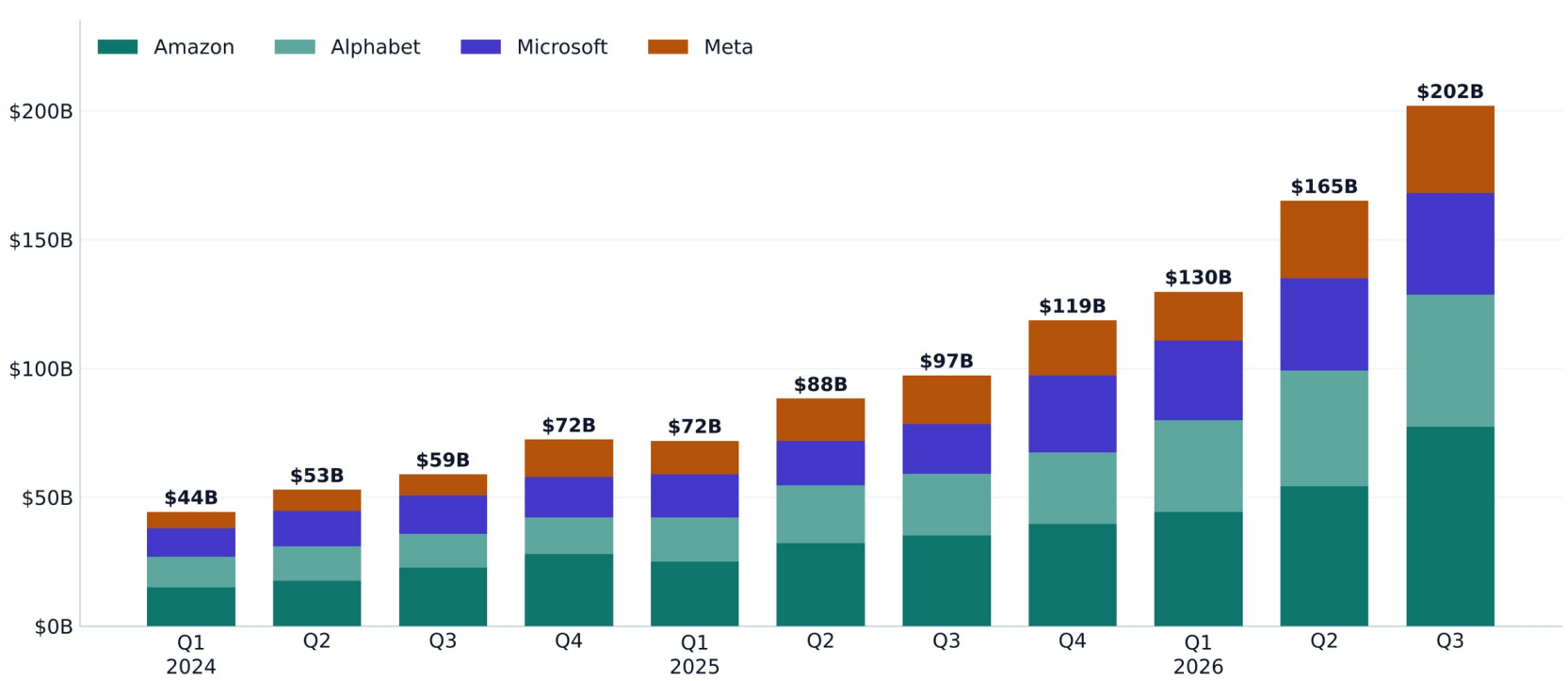

## Nvidia GPUs in one training cluster

Every one of them Nvidia. Hatched = press-reported.




Sources: Epoch AI notable AI models data (FLOP, release dates) and 'Training compute of frontier AI models grows by 4-5x per year'; ratio is ours (2.1e25 / 3.14e23). Meta LLaMA paper (arXiv 2302.13971); Inflection AI; Meta Engineering, 12 Mar 2024; GPT-4 GPU count reported by SemiAnalysis, not confirmed by OpenAI.

# Step 2: the big four started writing the checks




## SO WHAT

$202B in one quarter

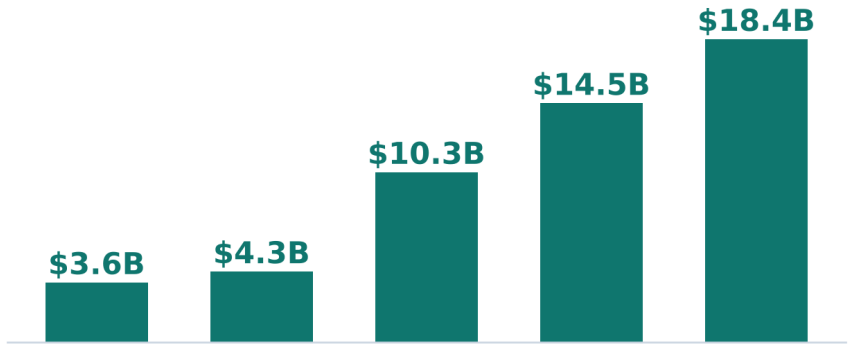
2.1x a year ago, 4.6x Q1 2024

2027 guides point higher

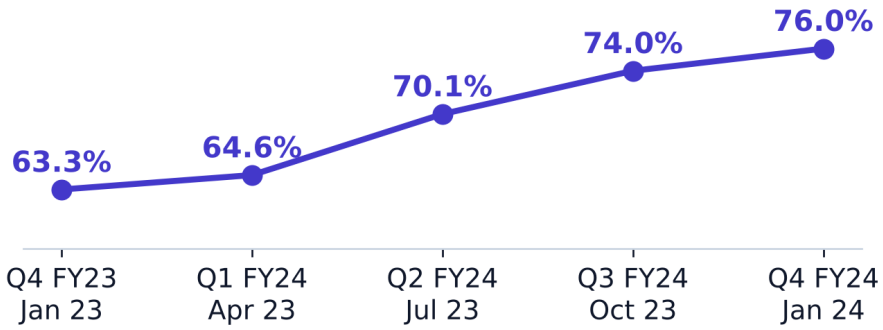
# Step 3: Nvidia was the only one ready, so revenue went 5x

## Nvidia data center revenue, per quarter

5x in one year, as the H100 shipped




## Gross margin: buyers paid up



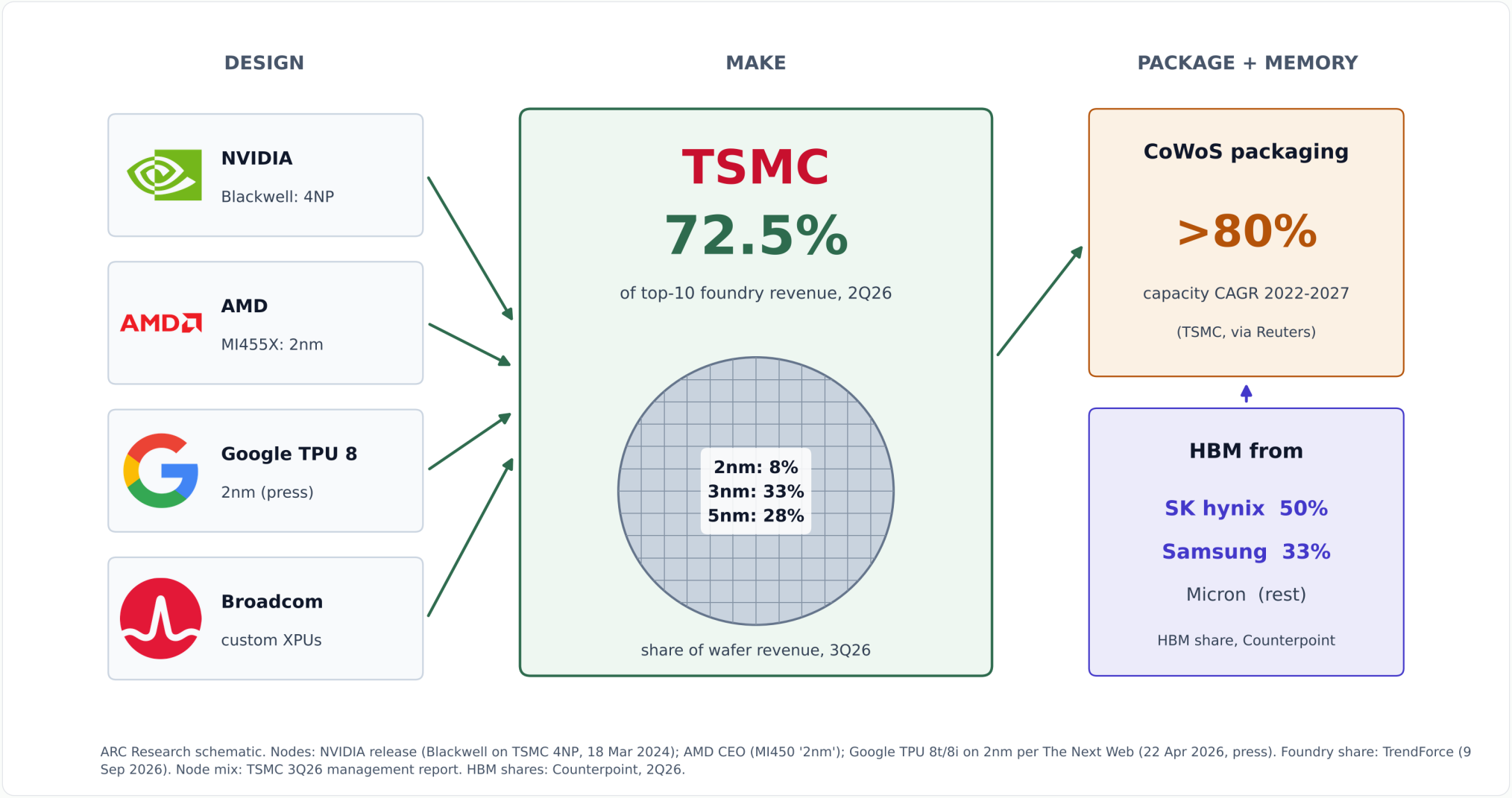

## Why Nvidia met it: it sold the whole system

| | | |
|---|---|---|
| **Software** | **CUDA: the code labs already used** | Rivals had chips but not the software stack |
| **Networking** | **NVLink + InfiniBand** | Bought Mellanox for $7B in 2020 |
| **Systems** | **DGX / HGX: 8 H100s per server** | Ordered by AWS, Google, Azure and Oracle |
| **Chip** | **H100 in full production, Sep 2022** | Shipping just as ChatGPT launched |
| **Supply** | **TSMC 4N wafers + CoWoS packaging** | Packaging was the bottleneck; TSMC planned ~2x |

**Result: ~98% of data center GPUs shipped in 2023 (TechInsights)**

Sources: Nvidia quarterly results releases, fiscal 2023–2024 (data center revenue, GAAP gross margin); Nvidia, 20 Sep 2022 (H100 in full production) and 27 Apr 2020 (Mellanox, $7B); TSMC Q2 2023 call via Counterpoint (CoWoS); TechInsights via HPCwire, 10 Jun 2024 (98% of 3.76M data center GPUs).

# Every designer’s chips pass through one foundry




ARC Research schematic. Nodes: NVIDIA release (Blackwell on TSMC 4NP, 18 Mar 2024); AMD CEO (MI450 '2nm'); Google TPU 8t/8i on 2nm per The Next Web (22 Apr 2026, press). Foundry share: TrendForce (9 Sep 2026). Node mix: TSMC 3Q26 management report. HBM shares: Counterpoint, 2Q26.

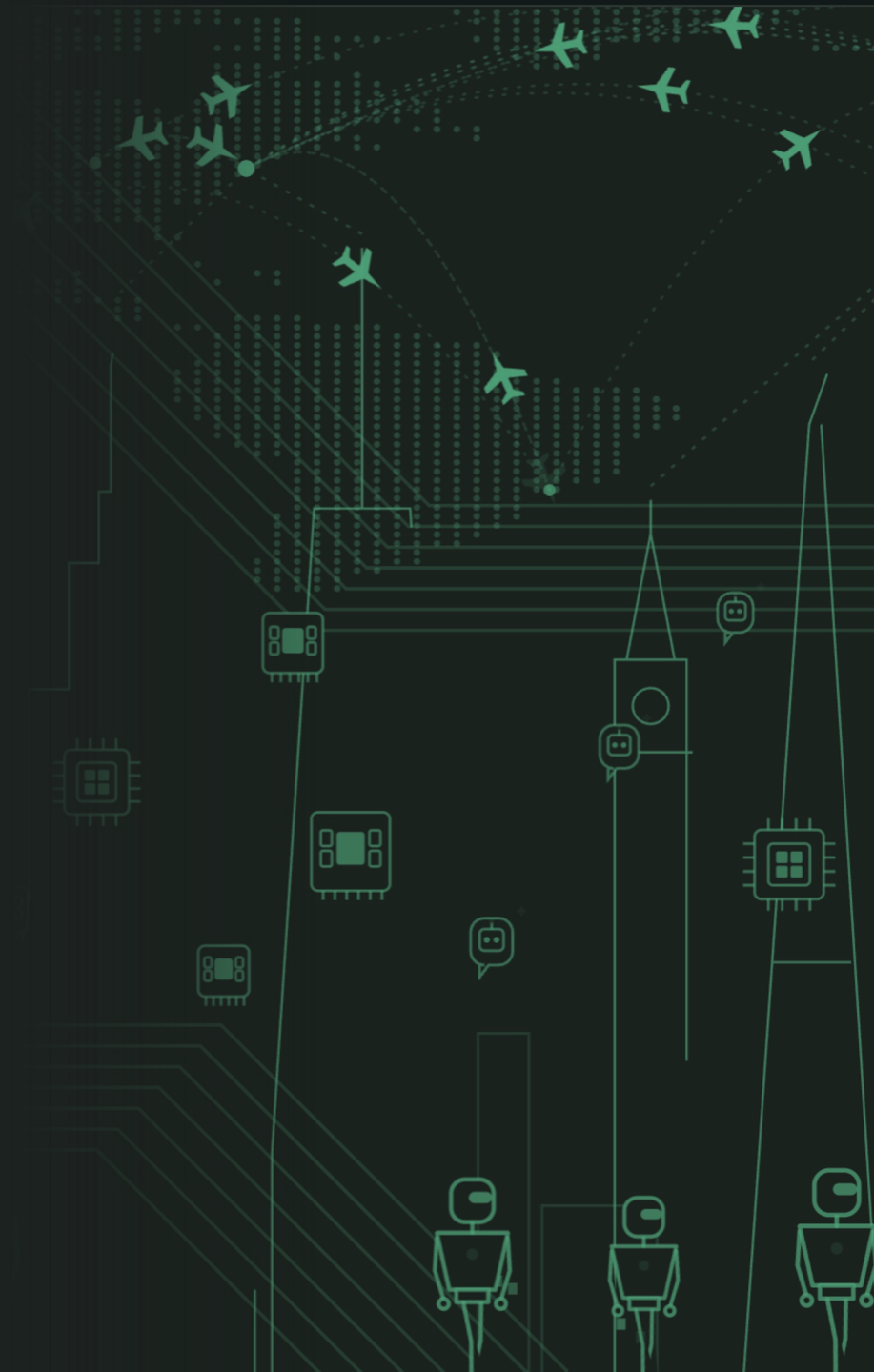
CHAPTER 3 OF 8

# 2024: the trade broadens

Power, custom chips and TSMC join Nvidia

COMING UP


| | | | |
|---|---|---|---|
| 2024 stock returns | 16 | Phones to data centers | 22 |
| The rise of AMD | 17 | Monthly revenue | 23 |
| Hyperscalers' own chips | 18 | Electricity to 2030 | 24 |
| Custom silicon | 19 | The grid is short | 25 |
| The chip lineup | 20 | The power mix | 26 |
| Seven chip programs | 21 | | |

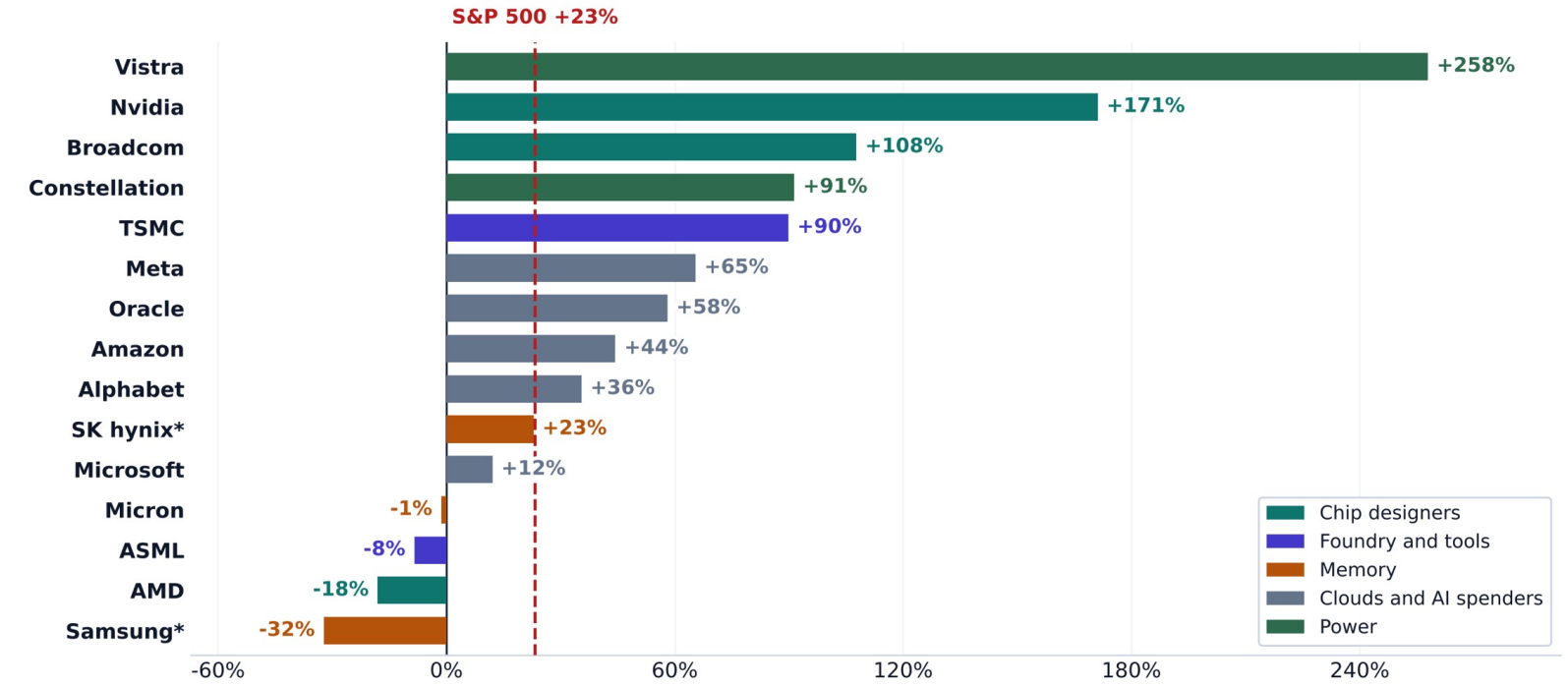



# 2024: the trade spreads beyond Nvidia

ARC RESEARCH | SPECIAL TOPICS | THE AI TRADE, YEAR BY YEAR

## 2024: the trade broadened, to power, custom chips and TSMC

Vistra beat even Nvidia; Broadcom doubled on custom chips; TSMC rose 90%. Memory and AMD lagged.




Source: price returns computed by us from split-adjusted closes (digrin.com; stockanalysis.com for 12 Nov 2026), cross-checked with macrotrends; not dividend-adjusted.
*In Korean won. S&P 500 via SPY. CoreWeave 2025 from its $40 IPO price (28 Mar). Names missing in a year had not listed or lack a verified start price.

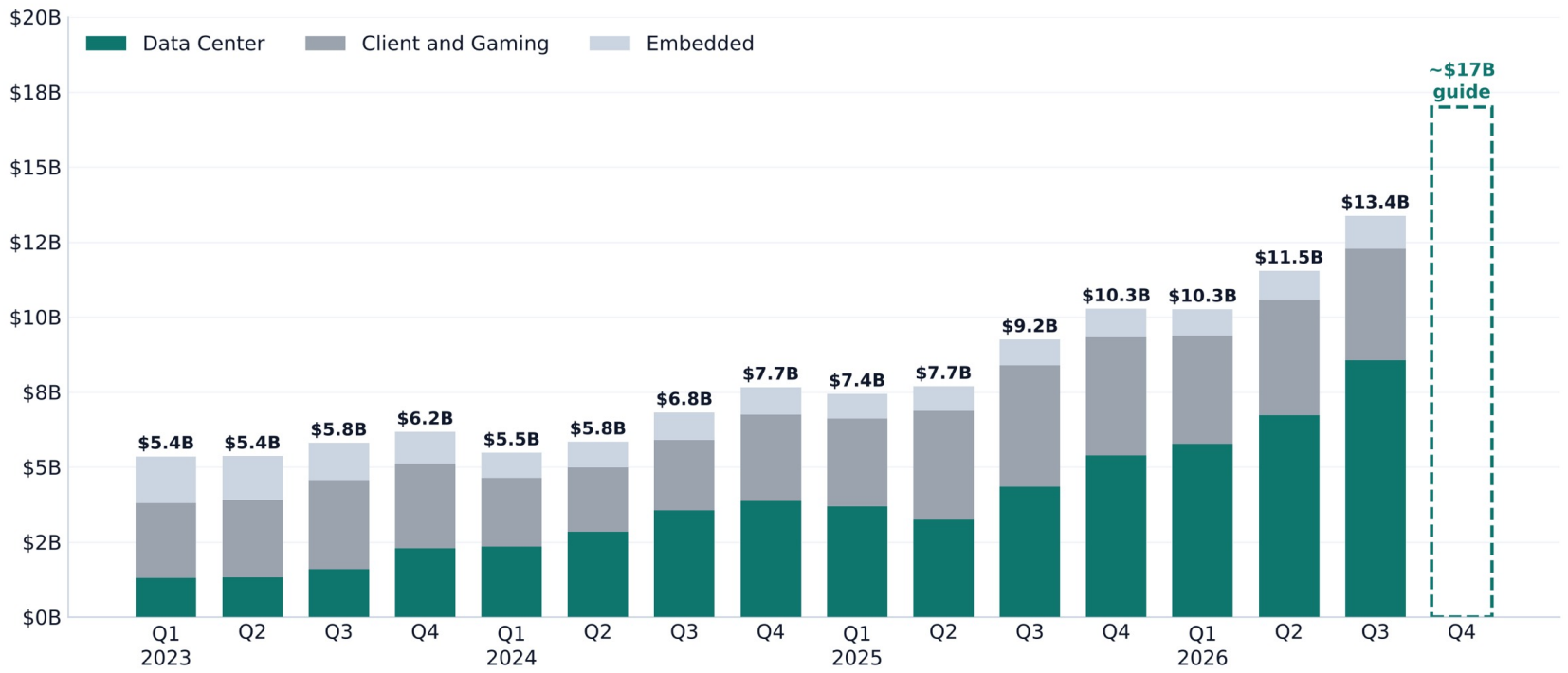
### SO WHAT

Power: Vistra +258%

Custom chips: Broadcom +108%

AMD and memory lagged

# AMD is the clearest sign of broadening




Source: AMD quarterly earnings releases (2023 from year-ago columns of 2024 releases). Client and Gaming = sum of the two segments (ours).
Dashed outline = AMD guidance for Q4 2026 of about $17B plus or minus 3% (3 Nov 2026), not a reported figure. Growth = 8,546 / 4,341 (ours).

## SO WHAT

Data center: $8.5B, 2x in a year

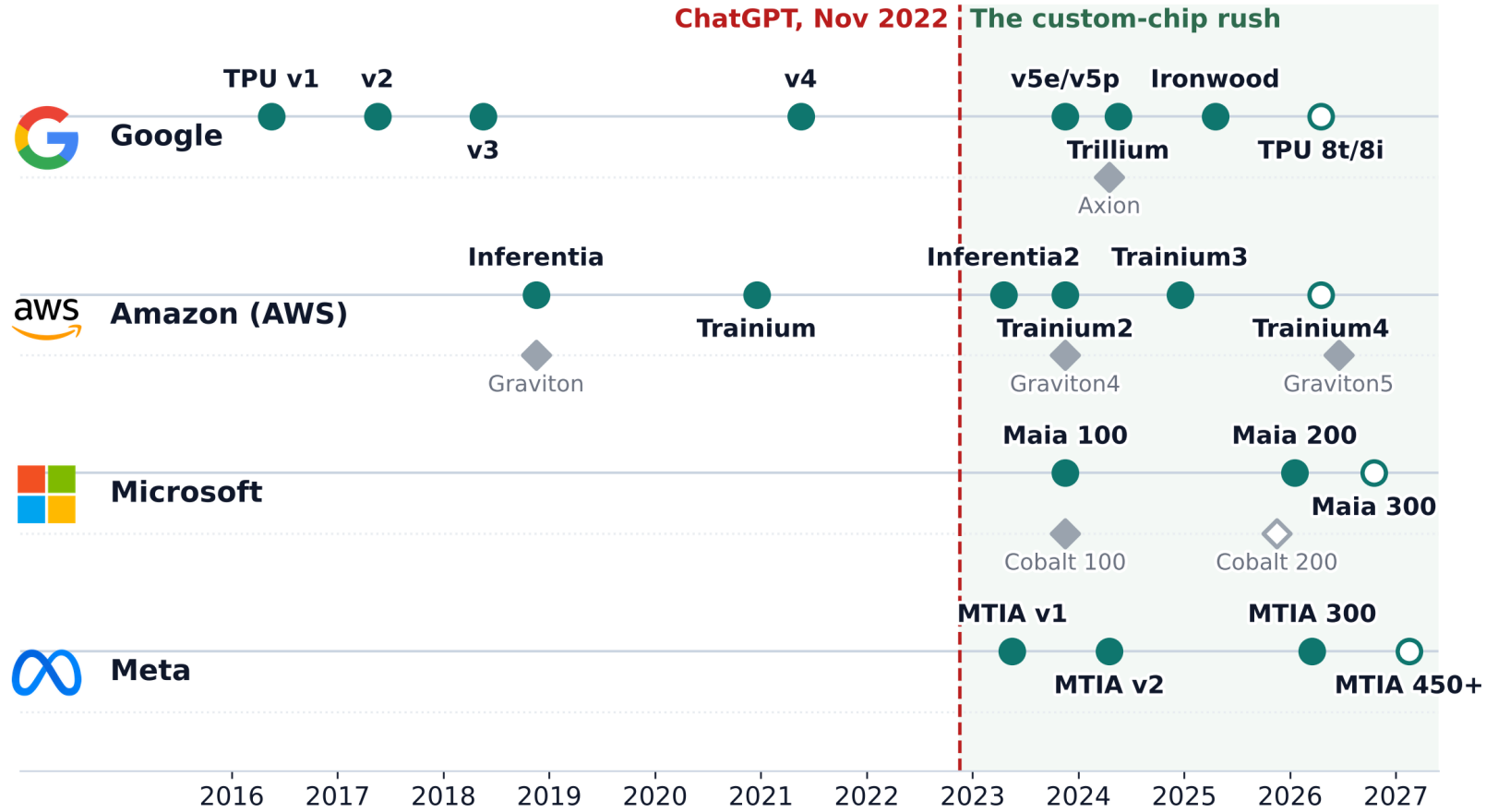
Q4 guide: about $17B

Sells both GPUs and CPUs

# The buyers become chip designers

## Every hyperscaler now designs its own chips

Circles = AI accelerators; diamonds = server CPUs. Solid = available; hollow = announced, not yet generally available.




## How big it is

**1M**

TPUs for Anthropic (up to), over 1 GW in 2026 (Google)

**~500k**

Trainium2 chips in AWS Project Rainier for Anthropic

**>$25B**

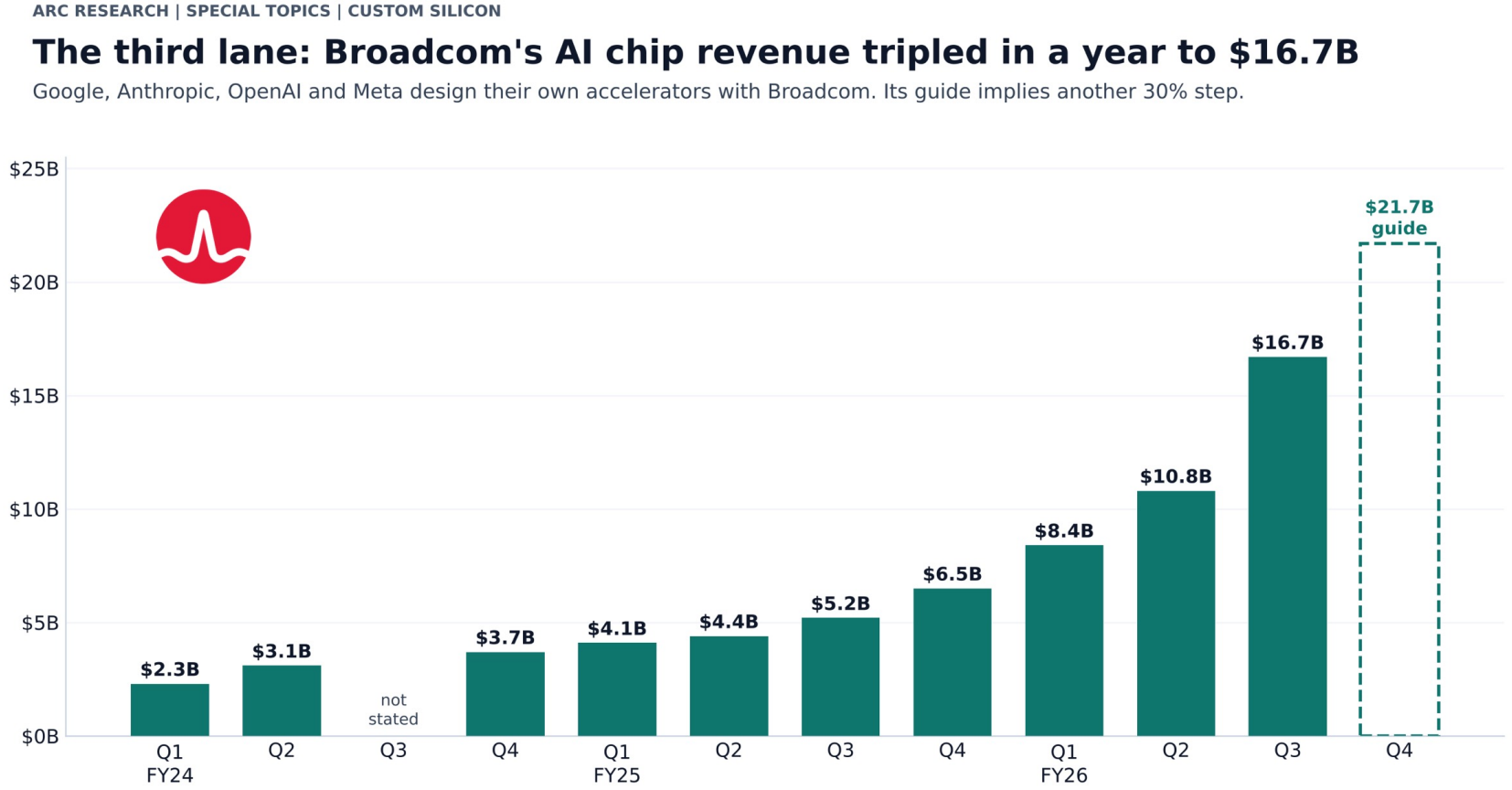
AWS custom-chip business, annual run rate (Q2 2026)

**2027**

Maia becomes Microsoft's primary inference system

Sources: Google Cloud blogs and release notes; AWS and About Amazon (Jassy letter 9 Apr 2026, Q2 2026 call); Microsoft blogs (Maia 100/Cobalt Nov 2023, Maia 200 26 Jan 2026, Maia 300 12 Oct 2026); Meta AI blog. Dates are first announcement; TPU v1 date is its public disclosure (deployed internally 2015).

# The labs are building their own chips



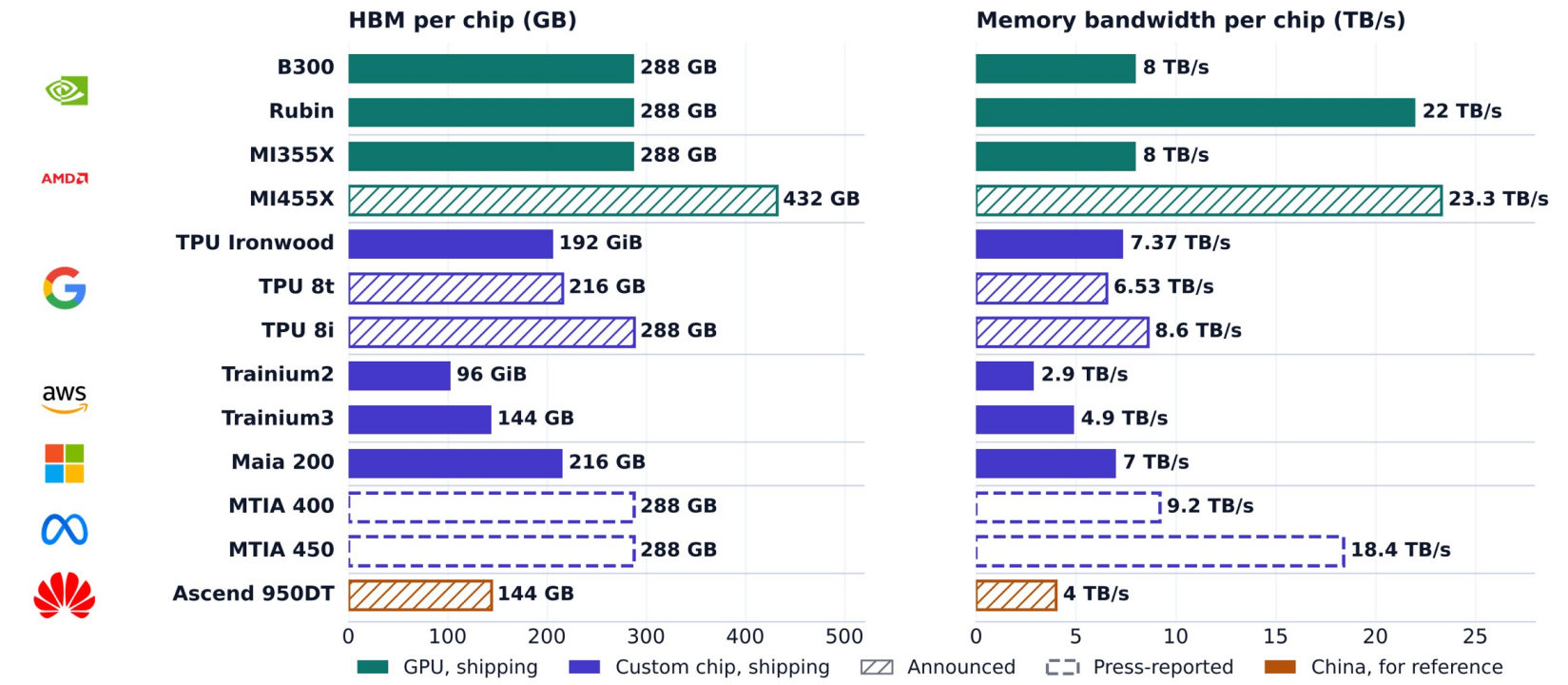

## SO WHAT

Broadcom AI: $16.7B, 3.2x

Google, Anthropic, OpenAI, Meta

Cheaper per token, less flexible

# Every big buyer now has its own chip




Source: Nvidia, AMD, Google Cloud, AWS, Microsoft (26 Jan 2026) and Huawei vendor pages; Meta MTIA per Tom's Hardware (12 Mar 2026), dashed. GiB figures (Ironwood, Trainium2) drawn in GB (x1.074, ours).
Per chip, as published. Compute is not compared: vendors publish it at different precisions and sparsity, so the numbers are not like for like.

## SO WHAT

Six designers, one memory race

New parts: ~3x bandwidth

Compute specs not like for like

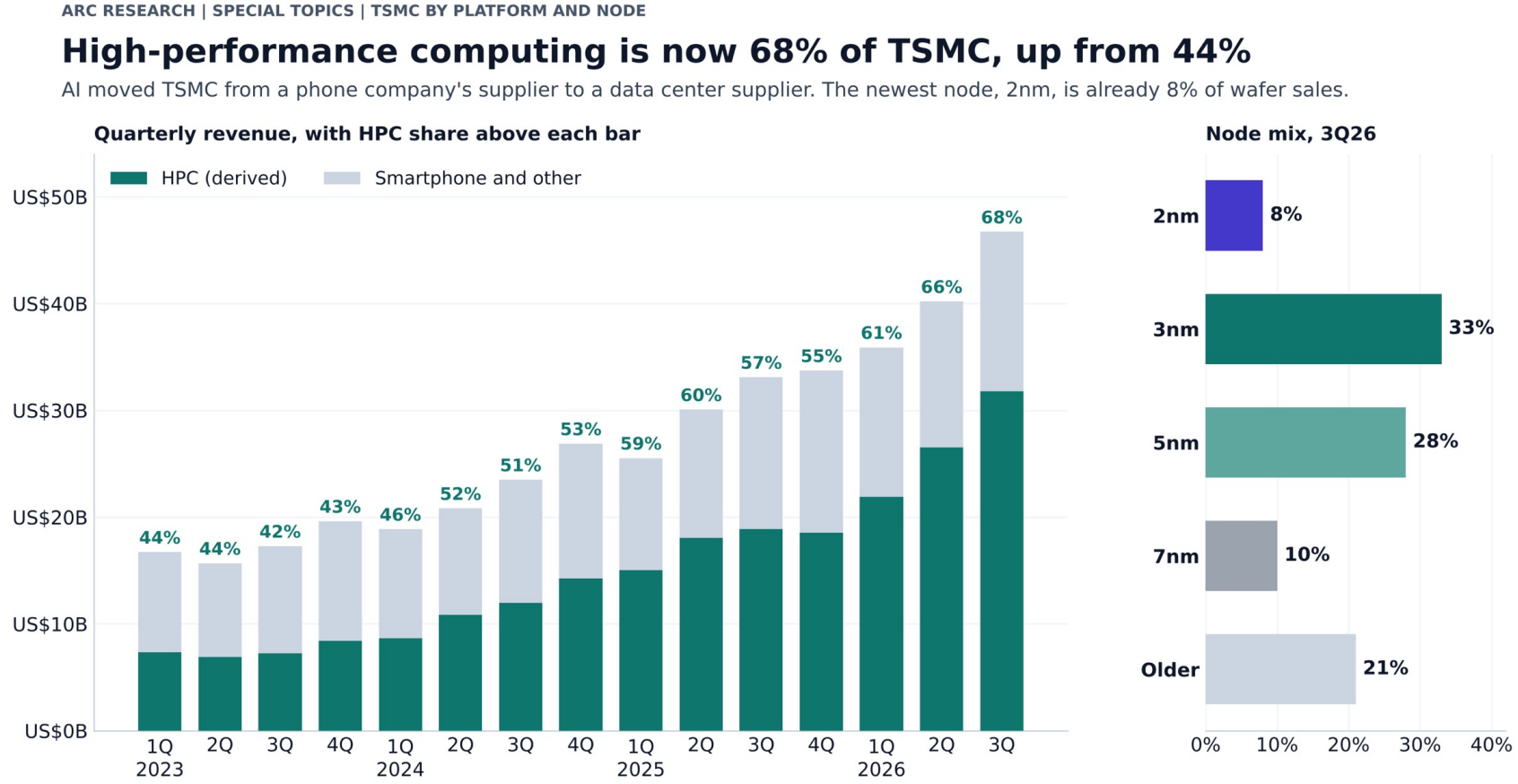
# Seven chip programs, two models: buy a GPU or build your own

| Nvidia | AMD | Google | Amazon | Microsoft | Meta | OpenAI |
|---|---|---|---|---|---|---|
| GPU | GPU | custom chip | custom chip | custom chip | custom chip | custom chip |
| Rubin | MI455X / Helios | Ironwood, TPU 8 | Trainium3 | Maia 200 | MTIA 300 to 500 | with Broadcom |
| shipping since Aug | in production | Ironwood available | shipping 2026 | deployed, Iowa | new chip each year | racks from H2 2026 |
| **$89B** | **432 GB** | **~100** | **>$25B** | **3 nm** | **3 GW** | **10 GW** |
| data center revenue, one quarter (Jul 2026) | HBM4 per chip, most of any GPU | customers on TPUs; Anthropic up to 1M | custom-chip run rate (incl. Graviton CPUs) | 216 GB HBM3e, 750 W | line of sight via Broadcom, to 2028 | agreed, 2027 to 2029; 1.3 GW in 2027 |

**AI servers shipped in 2026, forecast (TrendForce)**

Sources: Nvidia Q2 FY2027 CFO commentary; AMD MI455X page; Alphabet Q3 2026 call (28 Oct 2026) and Anthropic-Google announcement (Oct 2025); Amazon Q2 2026 call (30 Jul 2026), run rate covers all Amazon custom chips; Microsoft blog (26 Jan 2026); Broadcom Q3 FY2026 call (2 Sep 2026) for Meta and OpenAI gigawatts; OpenAI (13 Oct 2025); TrendForce (20 Jan 2026), remaining 2.5% other. Dashed status = announced or not yet confirmed shipping.

# AI turned a phone supplier into a data center supplier



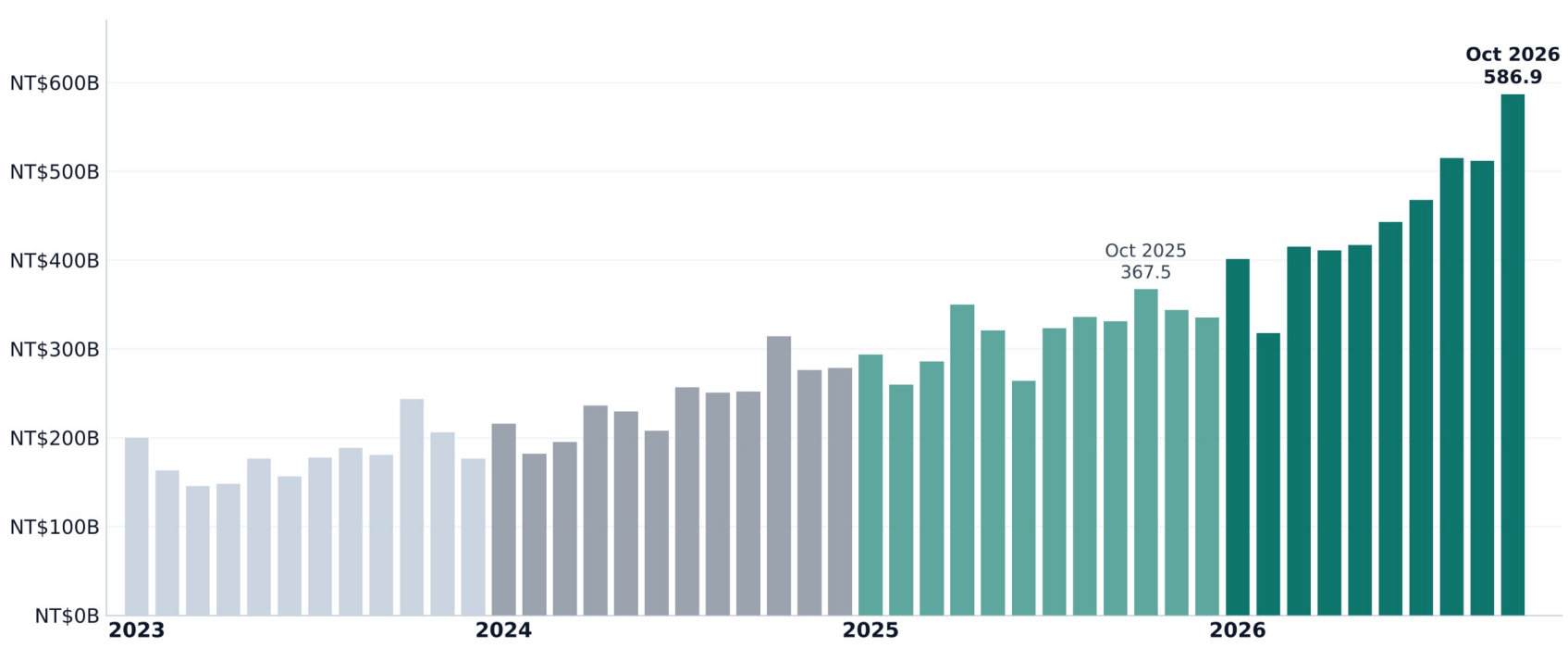

## SO WHAT

HPC share: 44% to 68%

Revenue 2.8x since early 2023

2nm already 8% of wafers

# The freshest hard data says demand is intact



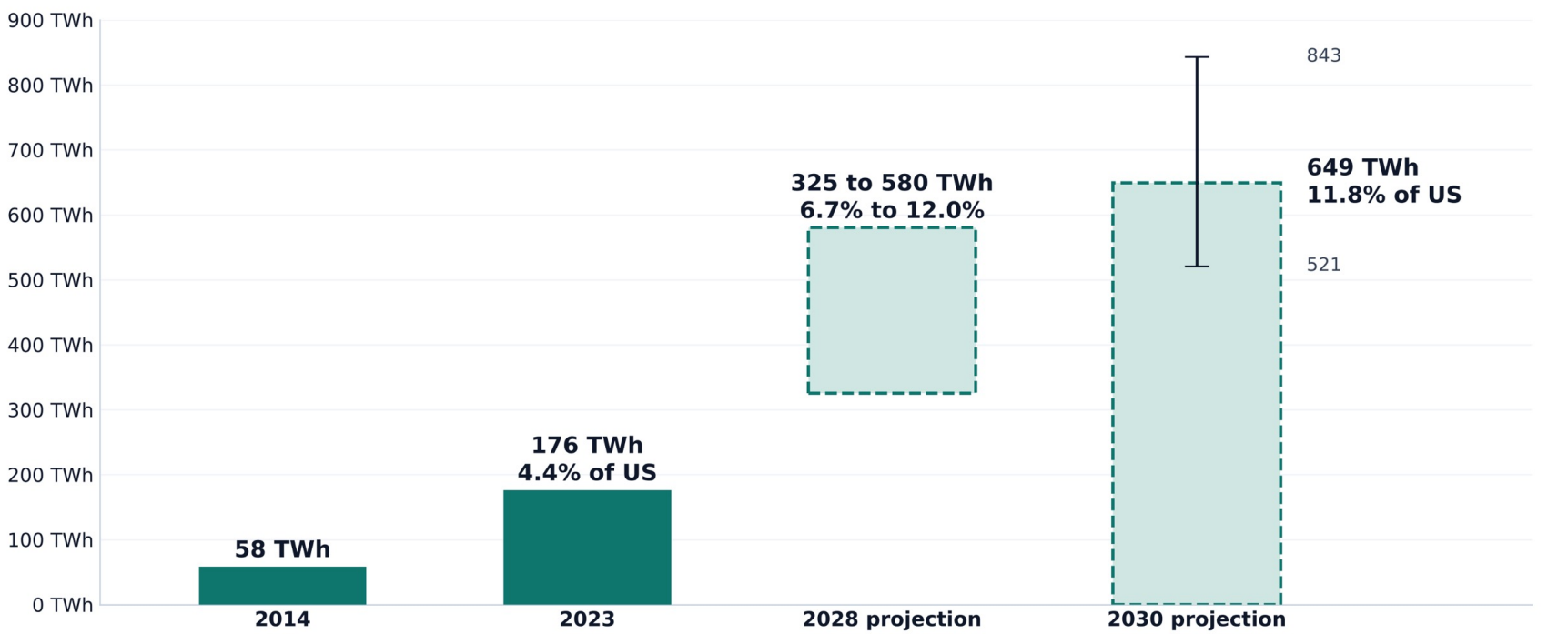

## SO WHAT

October: record, +60% on year

Reported during the selloff

2026 capex raised to $60–64B

# Data centers could use an eighth of US power by 2030



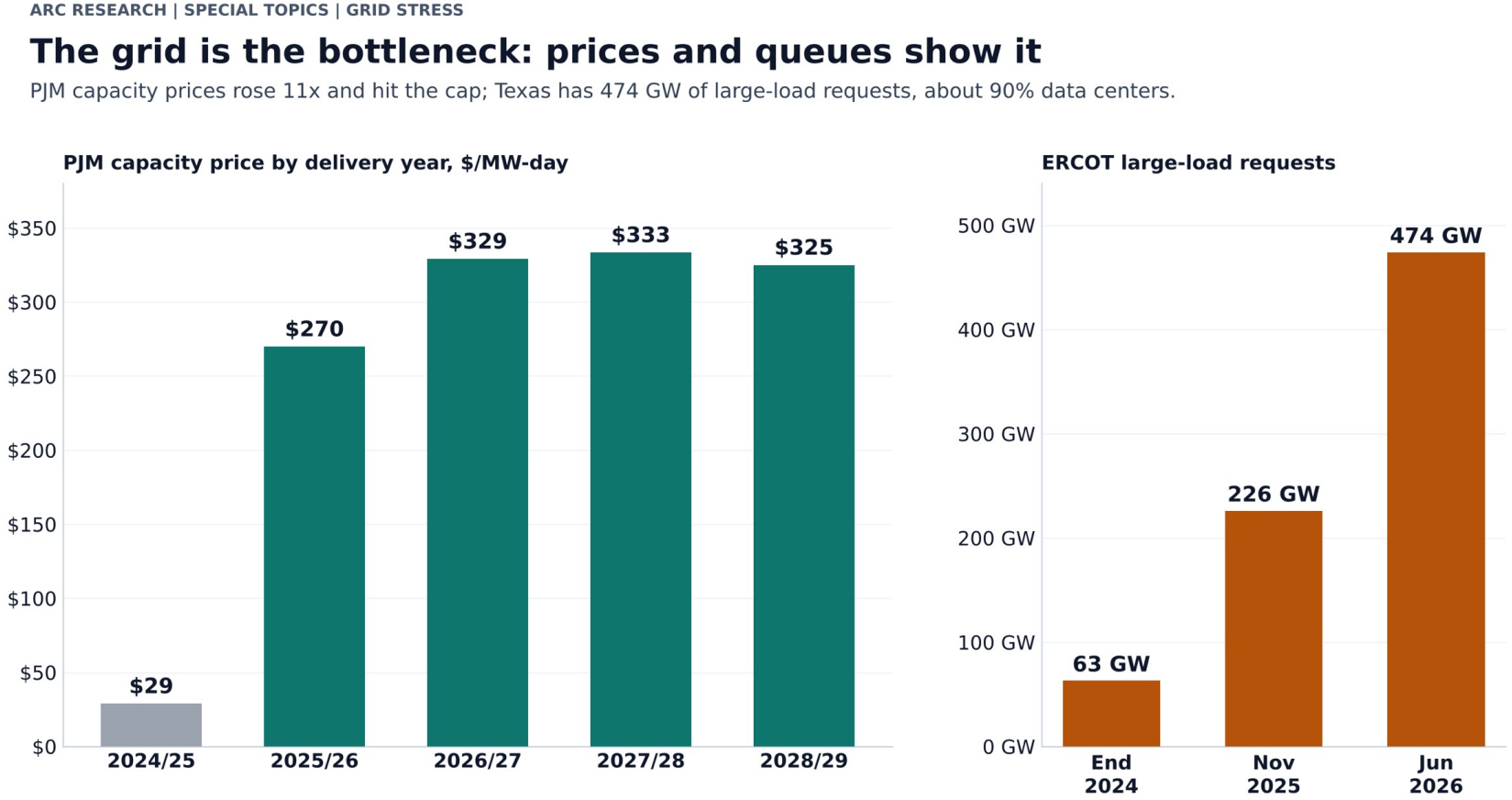

## SO WHAT

176 TWh in 2023: 4.4% of US

649 TWh projected for 2030

US demand flat for 15 years before

# The grid is already short




## SO WHAT

PJM capacity price: 11x

Texas queue: 474 GW of requests

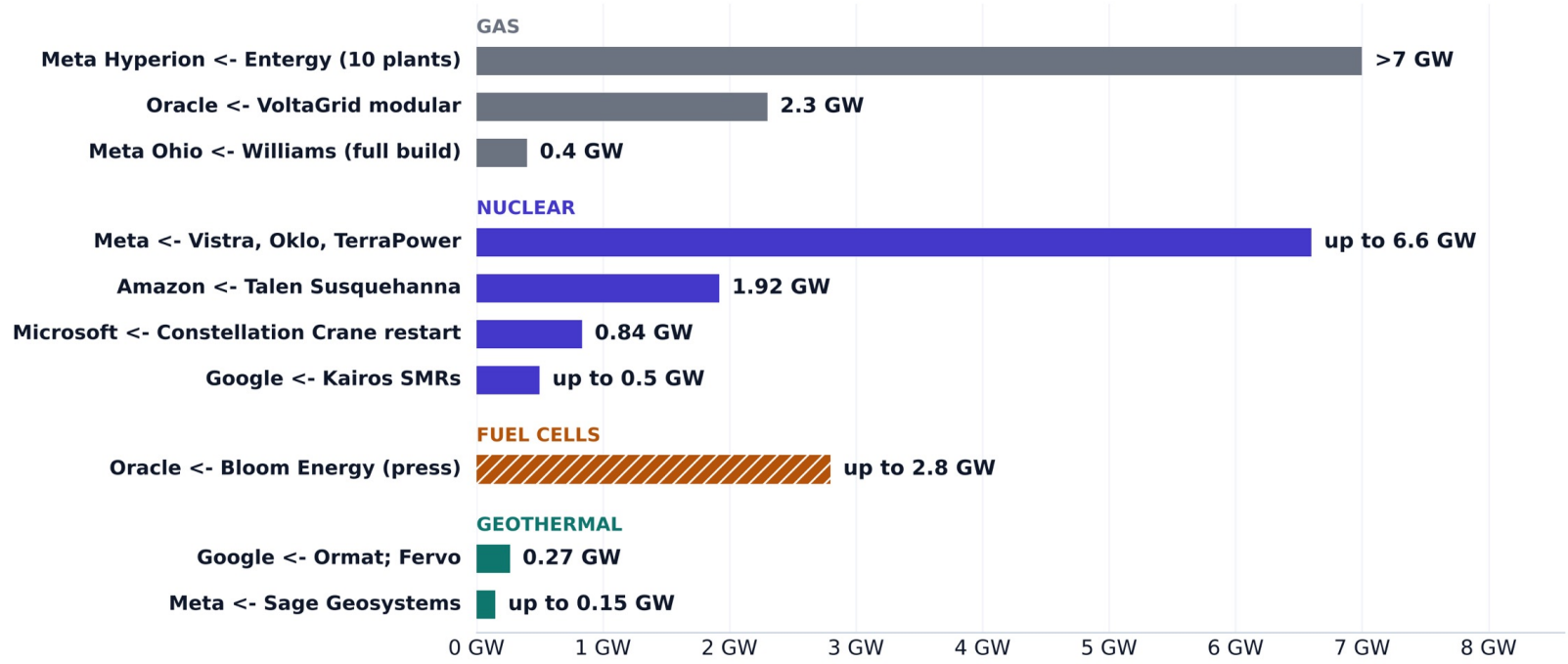
So builders bring their own power

# No single source is enough, so they are buying all of them

ARC RESEARCH | SPECIAL TOPICS | ENERGY DIVERSIFICATION

## Data centers are buying every kind of firm power they can find

Gas is the fast answer, nuclear the long one. In these named deals the two are almost level at about 10 GW each.




Source: company announcements via Utility Dive, ESG Dive, POWER Magazine, Power Engineering, Google blog, The Register (2024-2026); Meta Hyperion via press (20 Jul 2026). Selected named deals, not a complete list; sizes are contracted or 'up to' figures, not operating capacity.
Hatched = press-reported only (Bloom-Oracle, IWR Online, 23 Oct 2026). Totals: gas 9.7 GW, nuclear 9.9 GW (ours). GE Vernova gas backlog plus reservations: 122 GW at 30 Sep 2026.

### SO WHAT

Gas now, nuclear for the 2030s

Turbines: 122 GW on order

New names join the trade

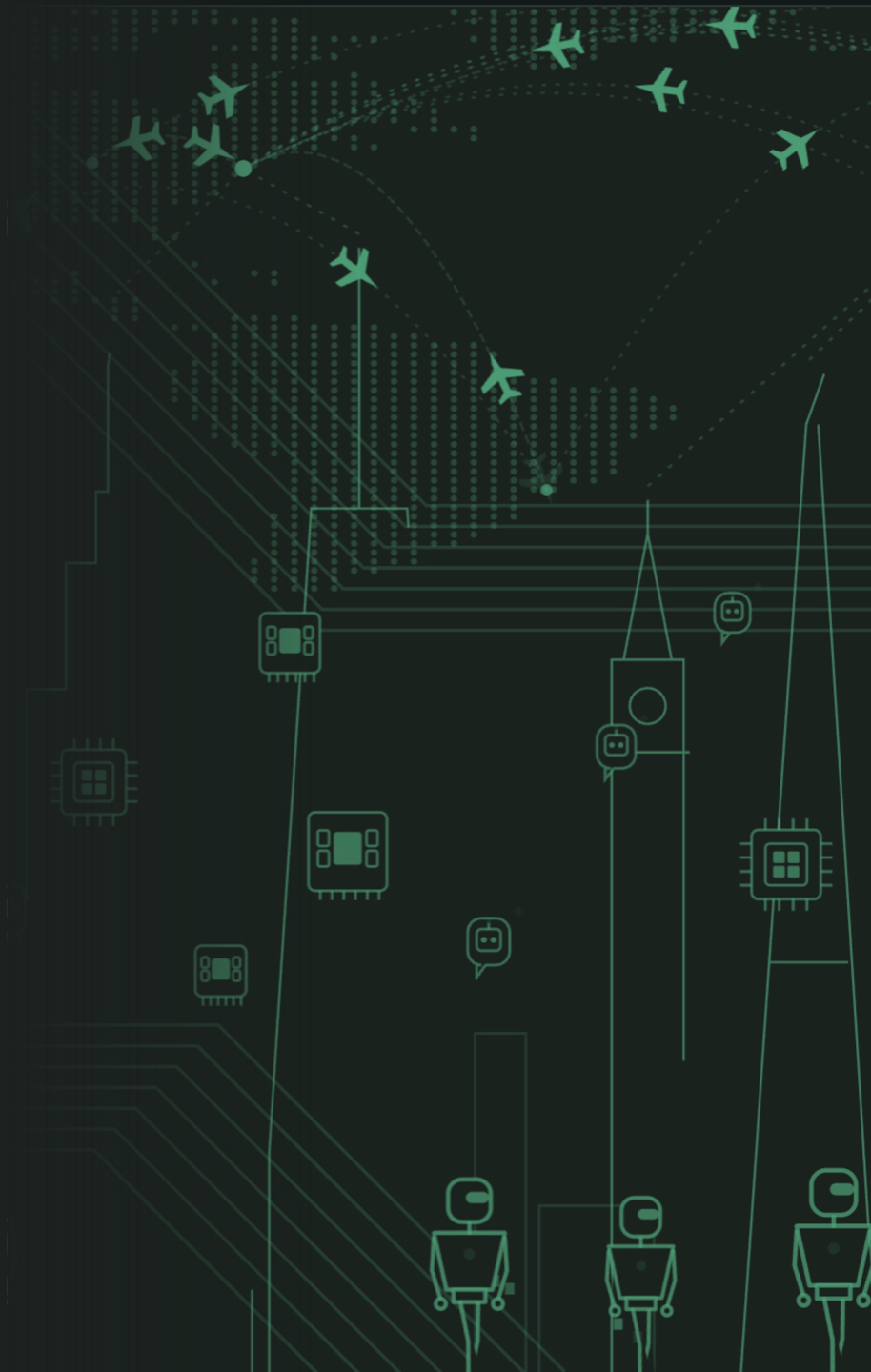
CHAPTER 4 OF 8

# 2025: agents and the memory squeeze

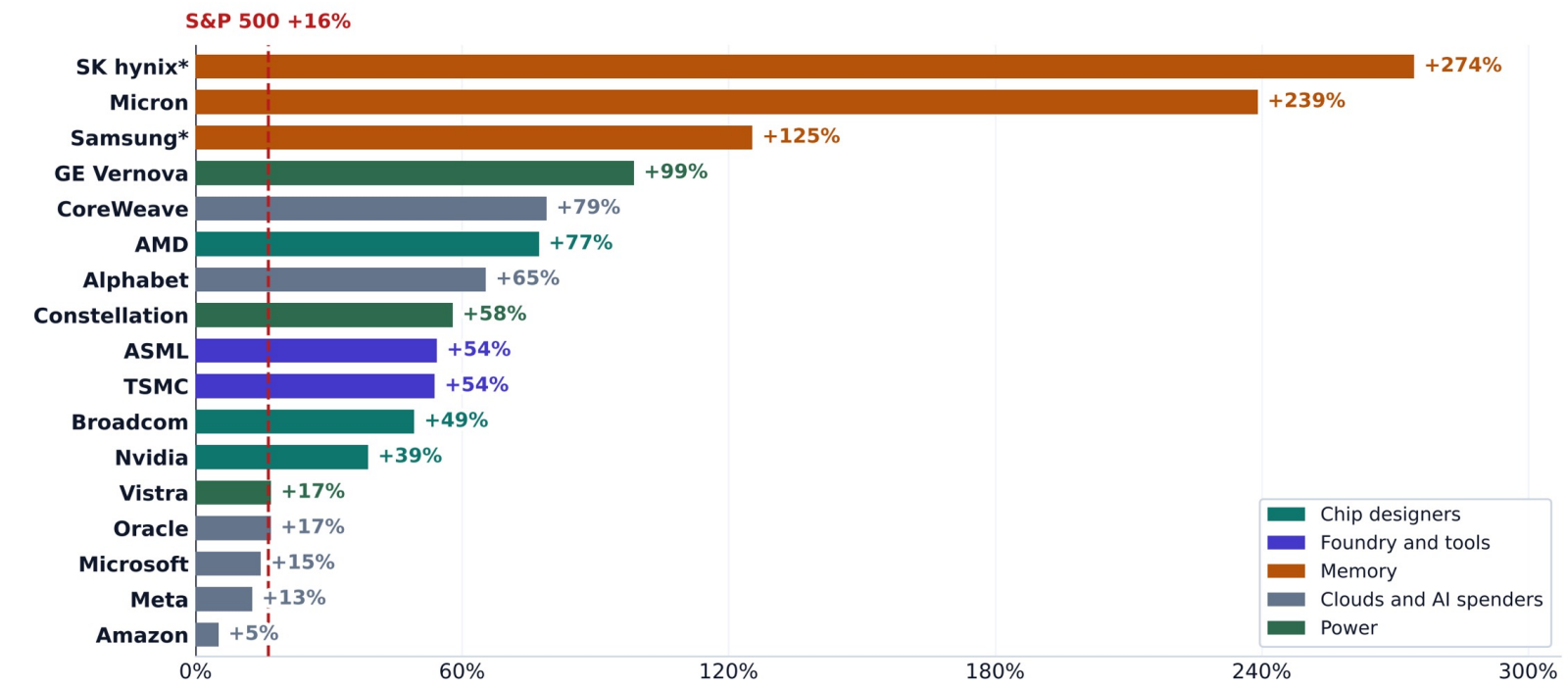
Agents and long context land on memory

COMING UP


| | | | |
|---|---|---|---|
| 2025 stock returns | 28 | The memory makers' windfall | 34 |
| Usage: tokens | 29 | SK hynix, Samsung and their stocks | 35 |
| From chat to agents | 30 | HBM backlog and valuations | 36 |
| The growing KV cache | 31 | Korea's chip exports | 37 |
| Three kinds of memory | 32 | DRAM prices | 38 |
| Memory per chip | 33 | | |

# 2025: memory takes the lead



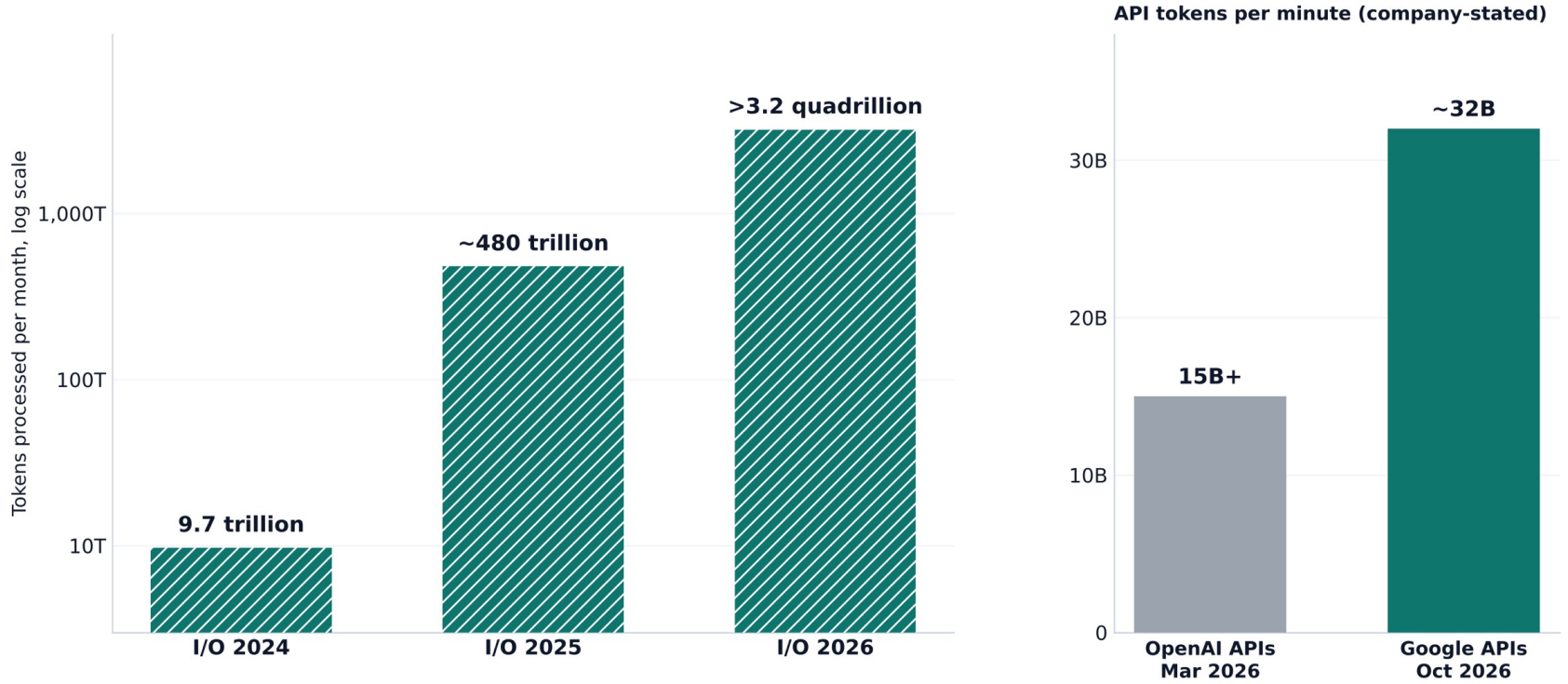

## SO WHAT

SK hynix +274%

Survived DeepSeek and tariffs

Nvidia only +39%

# Usage is growing faster than the models




## SO WHAT

Tokens are the unit of demand

About 330x in two years

Google API: 4x in a year

# Agents change the hardware mix

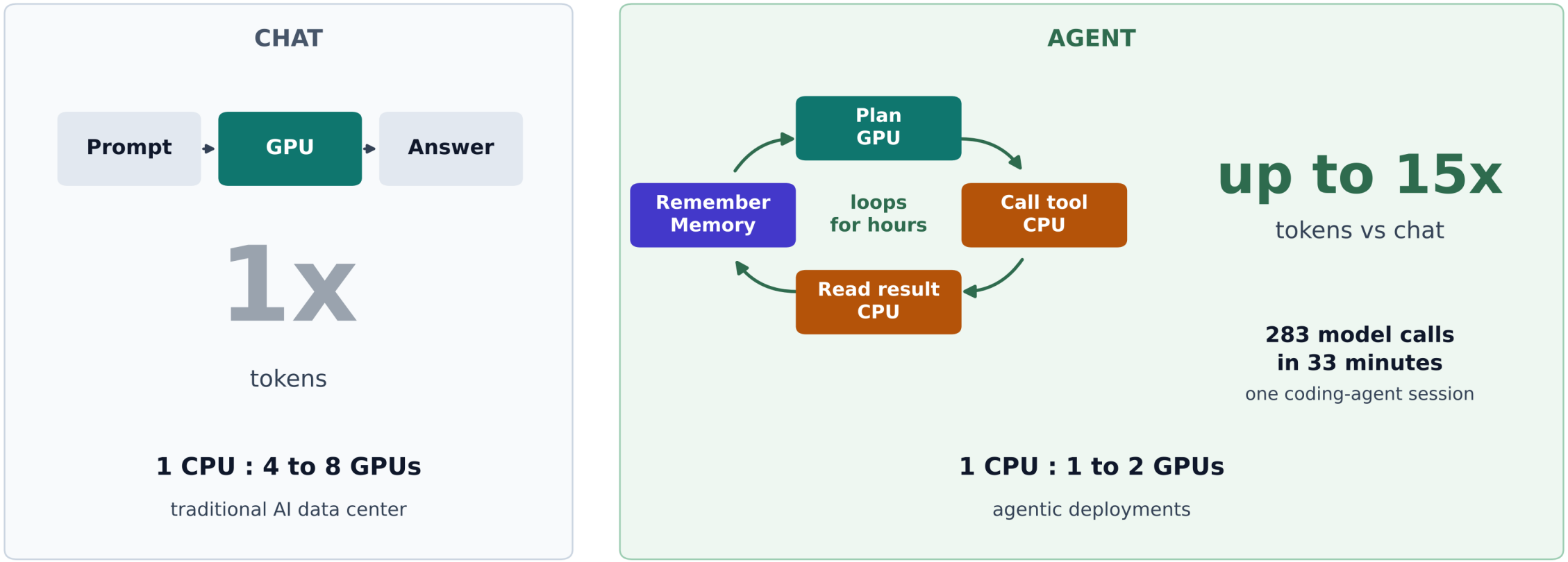



Illustration. 15x: Anthropic estimate for multi-agent systems, cited by NVIDIA (developer blog, 5 May 2026); 283 calls: session measured in the same blog. CPU:GPU ratios: TrendForce, 14 Apr 2026.

# Longer context means a bigger KV cache

## The cache grows with the context

KV cache size vs a 15,000-token turn, same model

**Start of an agent session**

**1x**

15,000 tokens

**Peak of the same session**

**10.4x**

156,000 tokens

**Full 2026 context window**

**66.7x**

1,000,000 tokens

A key and a value per token, per layer: memory rises with context.

## Where the cache lives

NVIDIA's memory tiers for inference context

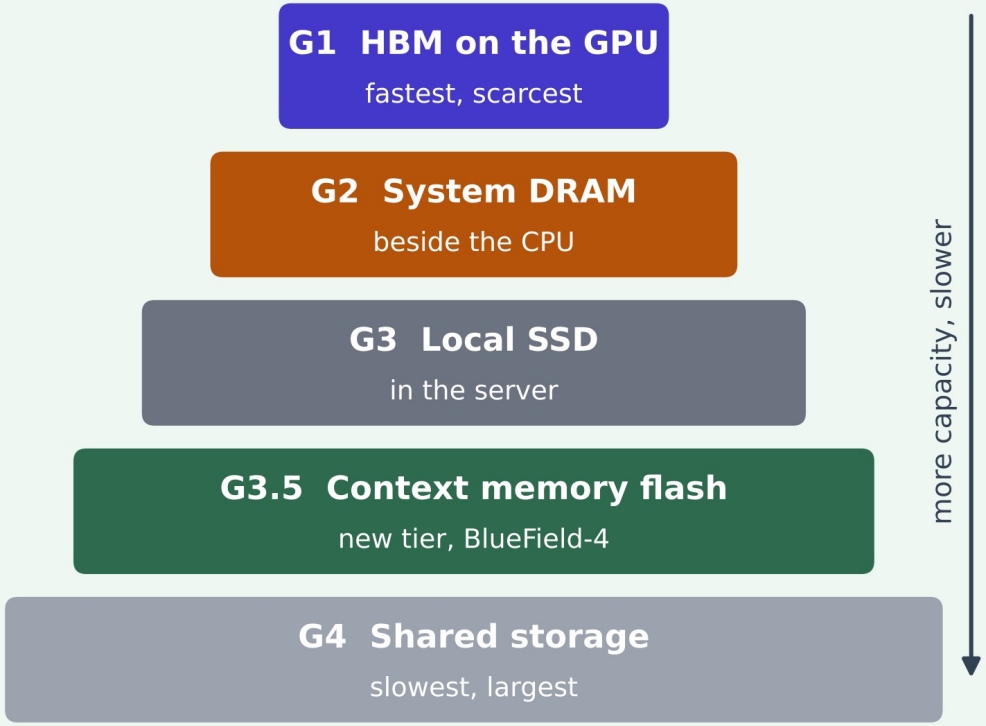
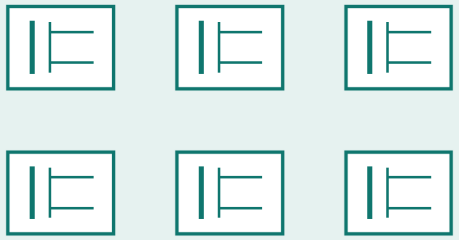



Illustration. Multiples = context tokens / 15,000 (ours); KV cache grows linearly with sequence length per NVIDIA. 15,000 and 156,000: coding-agent session in NVIDIA developer blog (5 May 2026). 1,000,000: 2026 flagship context windows (lab model pages). Tiers: NVIDIA blog on BlueField-4 Inference Context Memory Storage (Jan 2026).

# Three kinds of memory, three jobs

## SRAM

on the chip itself



**6 transistors per bit**

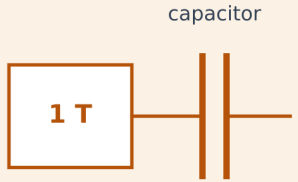
fastest, no refresh, least dense

**44 GB**

on one Cerebras WSE-3 wafer

## DRAM

beside the CPU




**1 transistor + 1 capacitor**

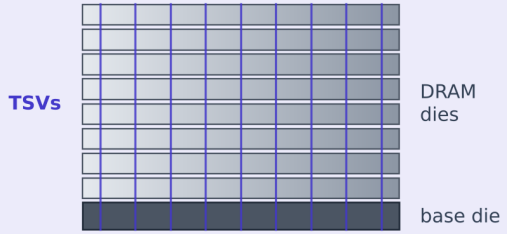
dense and cheap, must be refreshed

**54 TB**

LPDDR5X in one Vera Rubin rack (up to)

## HBM

stacked beside the GPU




**DRAM dies stacked, wired through**

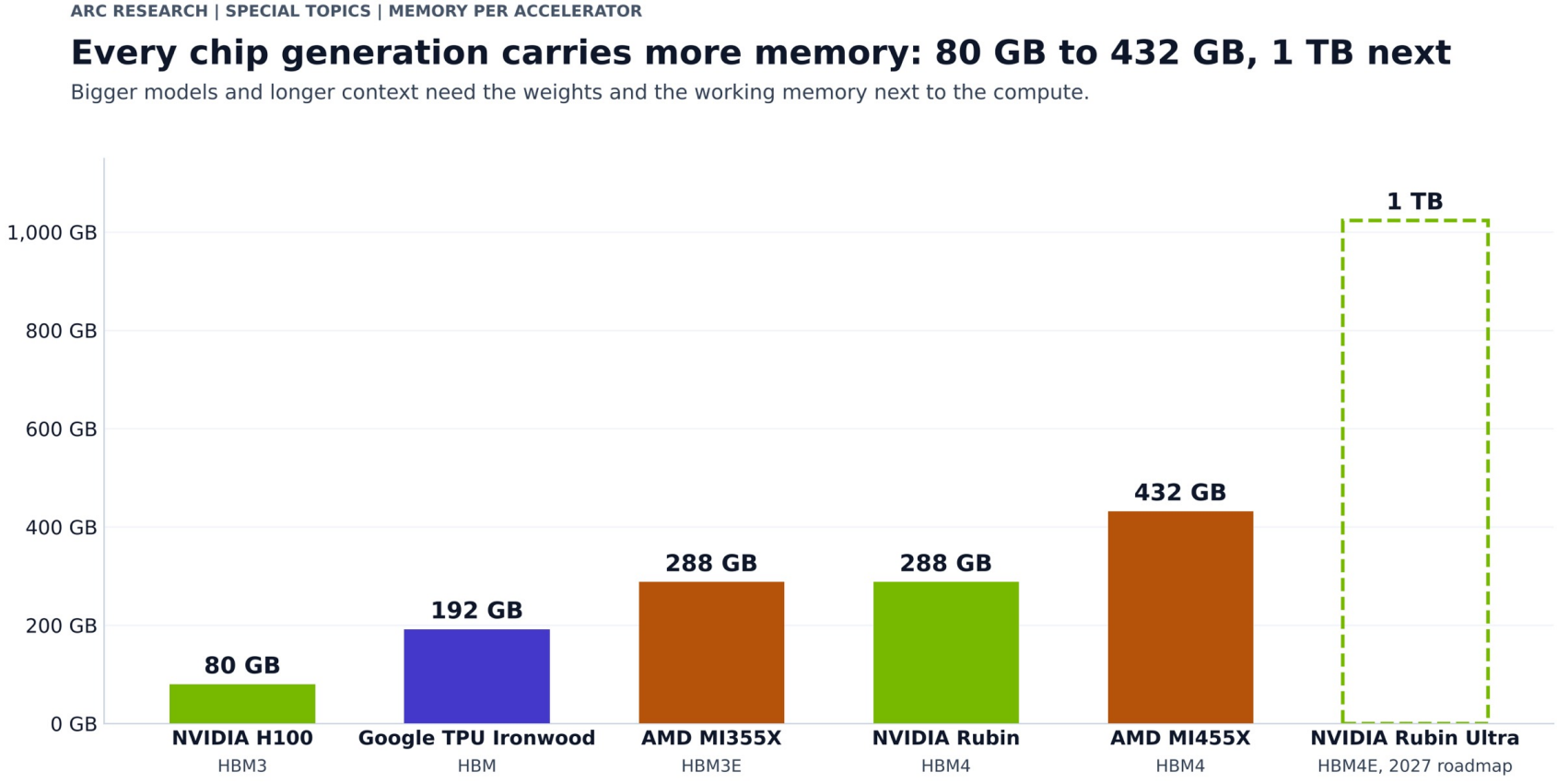
2,048 I/O per HBM4 stack, up to 64 GB

**288 GB**

HBM4 on one Rubin GPU

Illustration, not to scale; the stack shows 8 dies for clarity (layer count varies by product). Cell structures are standard. Figures: Cerebras WSE-3 via ServeTheHome (13 Mar 2024); NVIDIA Vera Rubin NVL72 page; JEDEC HBM4 (2,048 I/O, up to 64 GB).

# Each chip generation carries more memory




**SO WHAT**

80 GB to 432 GB per chip

HBM uses 3x the wafer area

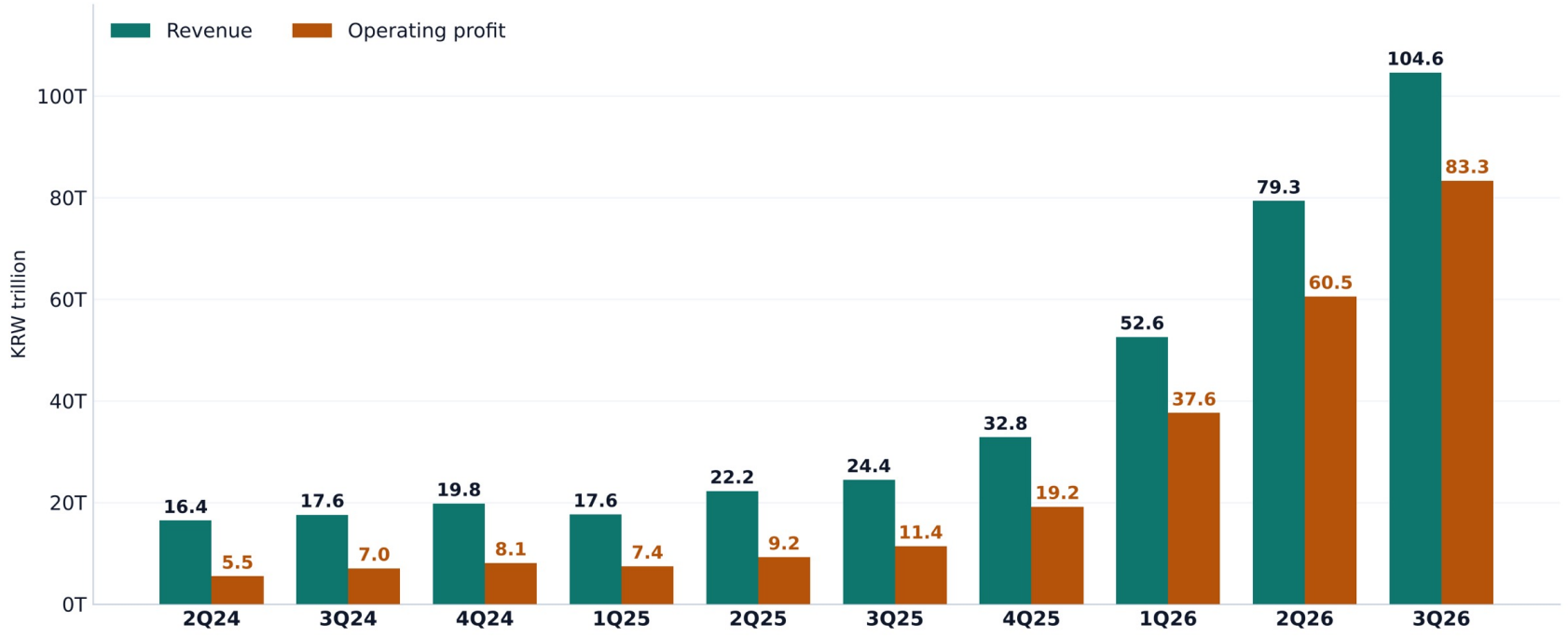
About 20% of DRAM wafers

# Memory makers are earning chip-designer margins

ARC RESEARCH | SPECIAL TOPICS | MEMORY SUPPLIER RESULTS

## SK hynix: revenue up 4.3x in a year, with an 80% operating margin

Memory is where the AI trade's pricing power is most visible right now. Samsung and Micron show the same shape.




Source: SK hynix quarterly results releases (K-IFRS consolidated; 3Q26 preliminary, 28 Oct 2026), via SK hynix newsroom and release reprints (TechNode, Digitimes, TechPowerUp).
Multiples = 104.6 / 24.4 and 83.3 / 104.6 (ours). KRW trillion.

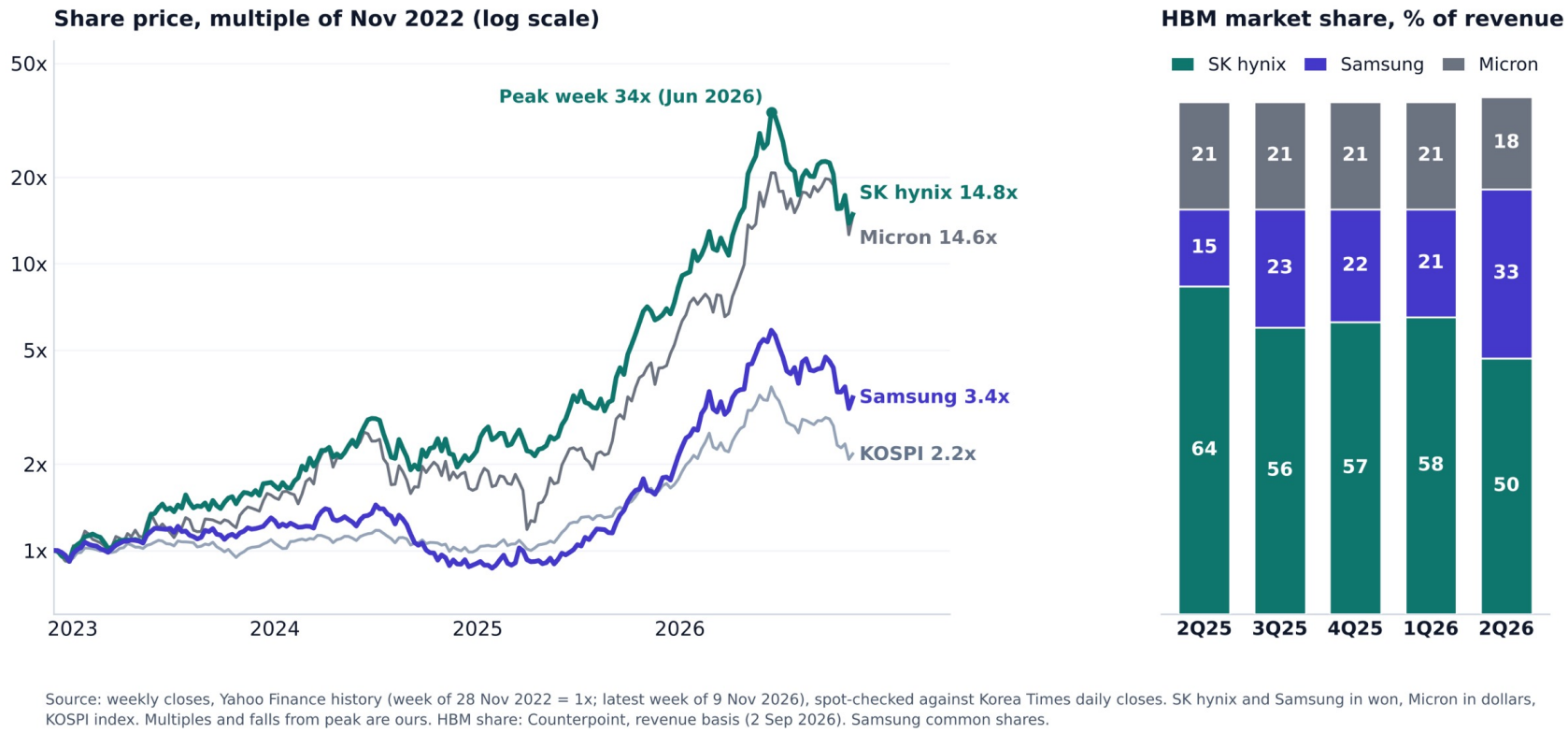
### SO WHAT

SK hynix: 80% operating margin

Micron: $54B quarter, 4.8x

Historically a cyclical industry

# SK hynix and Samsung make the HBM, and their stocks rode it




## SO WHAT

Two Korean firms: ~83% of HBM

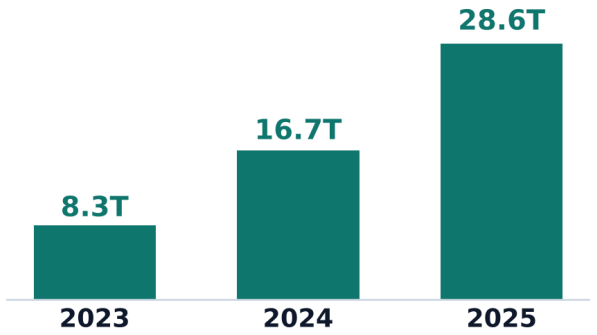
SK hynix: 34x, then -56%

Samsung winning back share

# Sold out into 2027, spending more than ever, worth $1 trillion each

SK hynix

**Capex, KRW trillion**




**Market value**

**$41B** → **$1T**

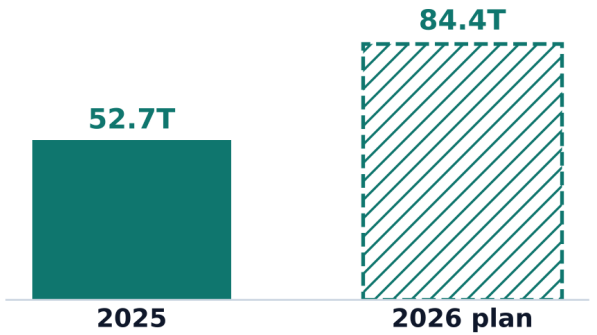
end-2022 — first crossed, 27 May 2026

**Order book**

**Long-term deals with 14 customers cover >50% of 2027-28 capacity**

SAMSUNG

**Capex, KRW trillion, whole company**




**Market value**

**$293B** → **$1T**

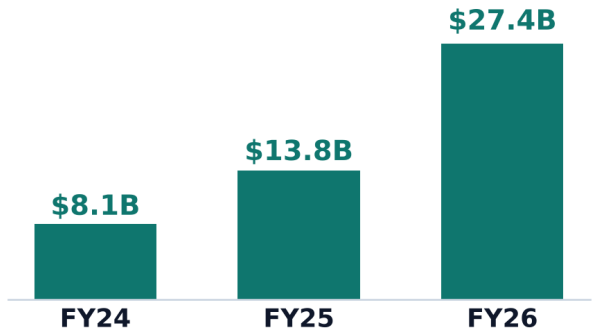
end-2022 — first crossed, 6 May 2026

**Order book**

**2027 HBM supply 'already sold out'**

micron

**Capex, USD billion, net**




**Market value**

**n/a** → **$1T**

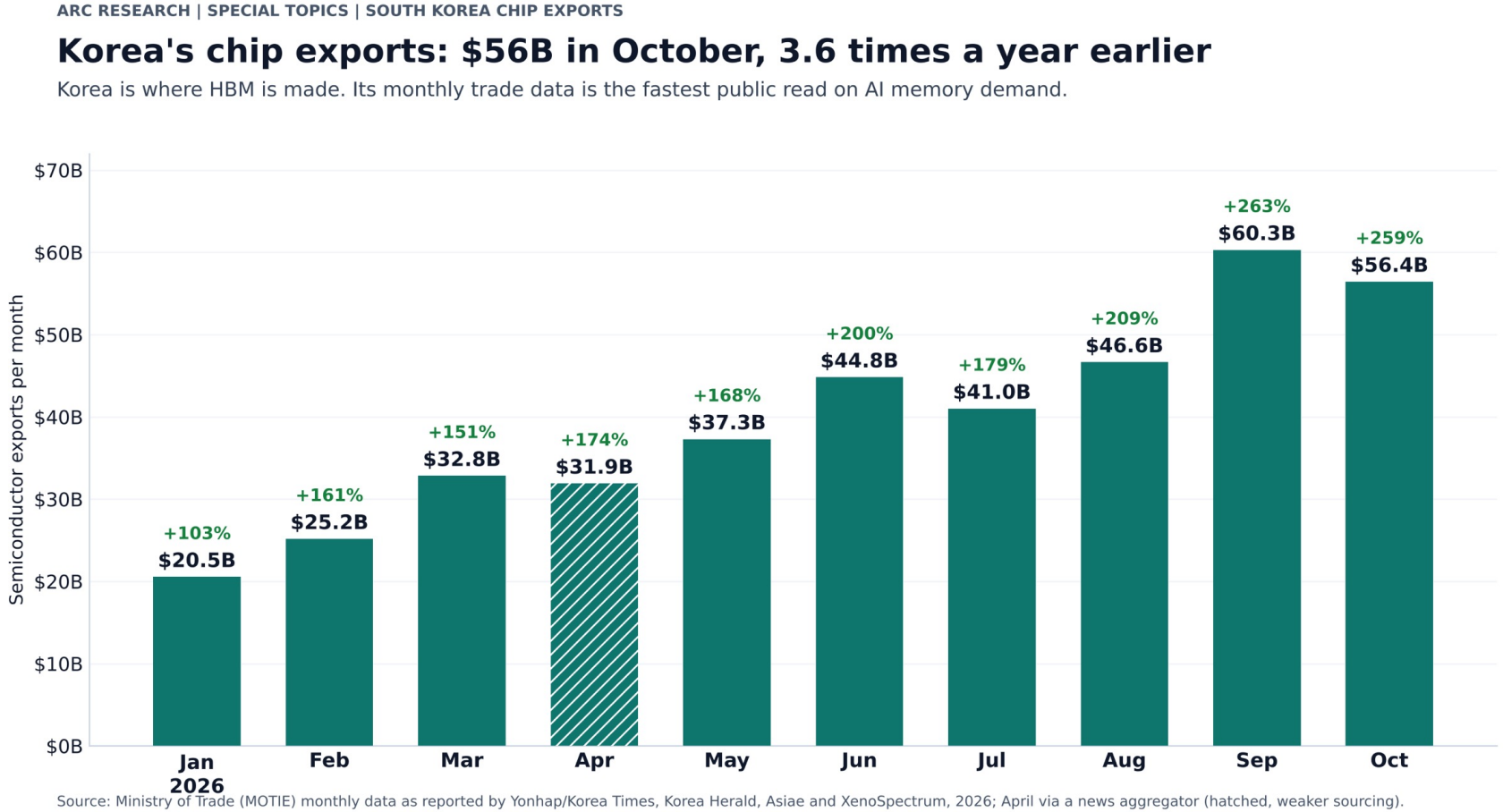
end-2022 not found — first crossed, late May 2026

**Order book**

**Deals for most 2027 HBM bits; $12.8B customer deposits, FY26**

Sources: capex: SK hynix via tickerleague (cash-flow basis); Samsung 2025 IR disclosure and 2026 plan via Korea Herald (dashed, a plan); Micron 8-K, FY ends early Sept, net of government incentives (FY26 gross $30.7B). Market value: companiesmarketcap (end-2022), Korea Herald ($1T dates). Order book: SK hynix Q3 2026 release (28 Oct 2026); Samsung Q3 call via Korea Herald; Micron FQ4 2026 call via The Next Platform and The Register, deposits from its 8-K.

# Korea’s exports show the memory boom in real time




## SO WHAT

$56B in October, +259%

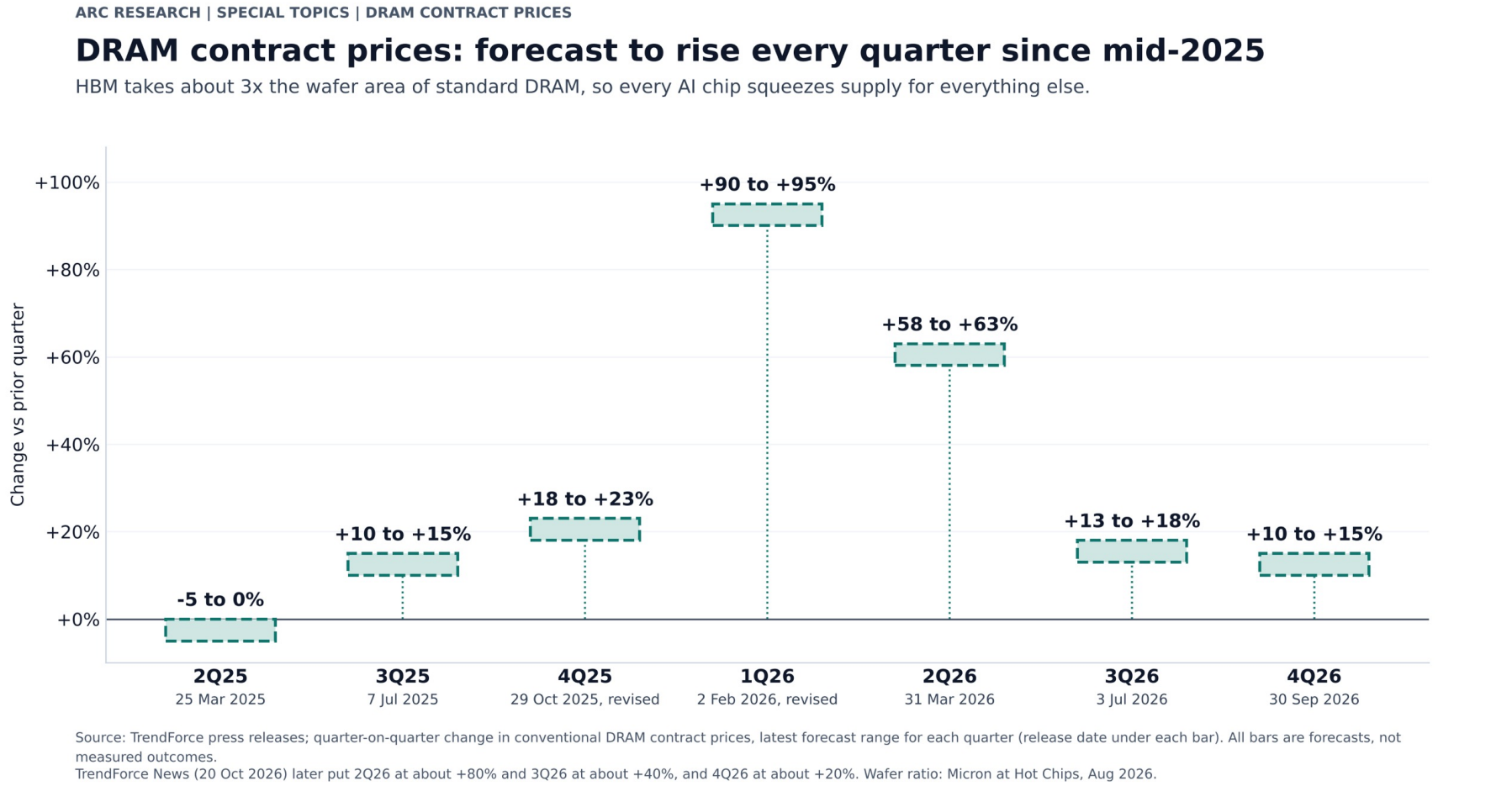
Fastest public read on HBM

Exports to China are restricted

SK hynix

SAMSUNG

# Everyone else pays for AI’s memory




## SO WHAT

Six straight quarters of rises

Peak forecast: +90% in a quarter

PCs and phones bear the cost

SK hynix

SAMSUNG

micron

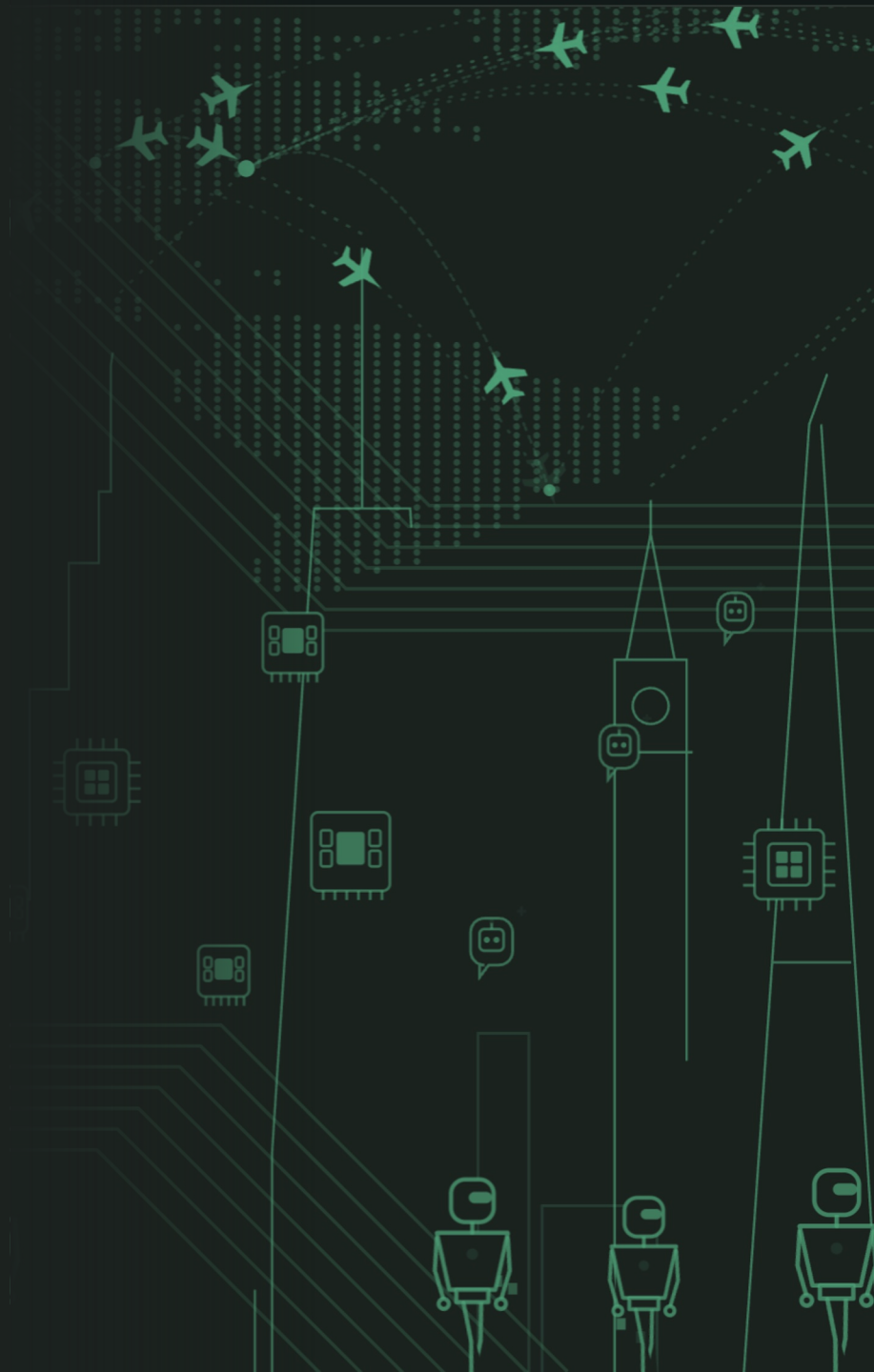
CHAPTER 5 OF 8

# 2025: DeepSeek and the China split

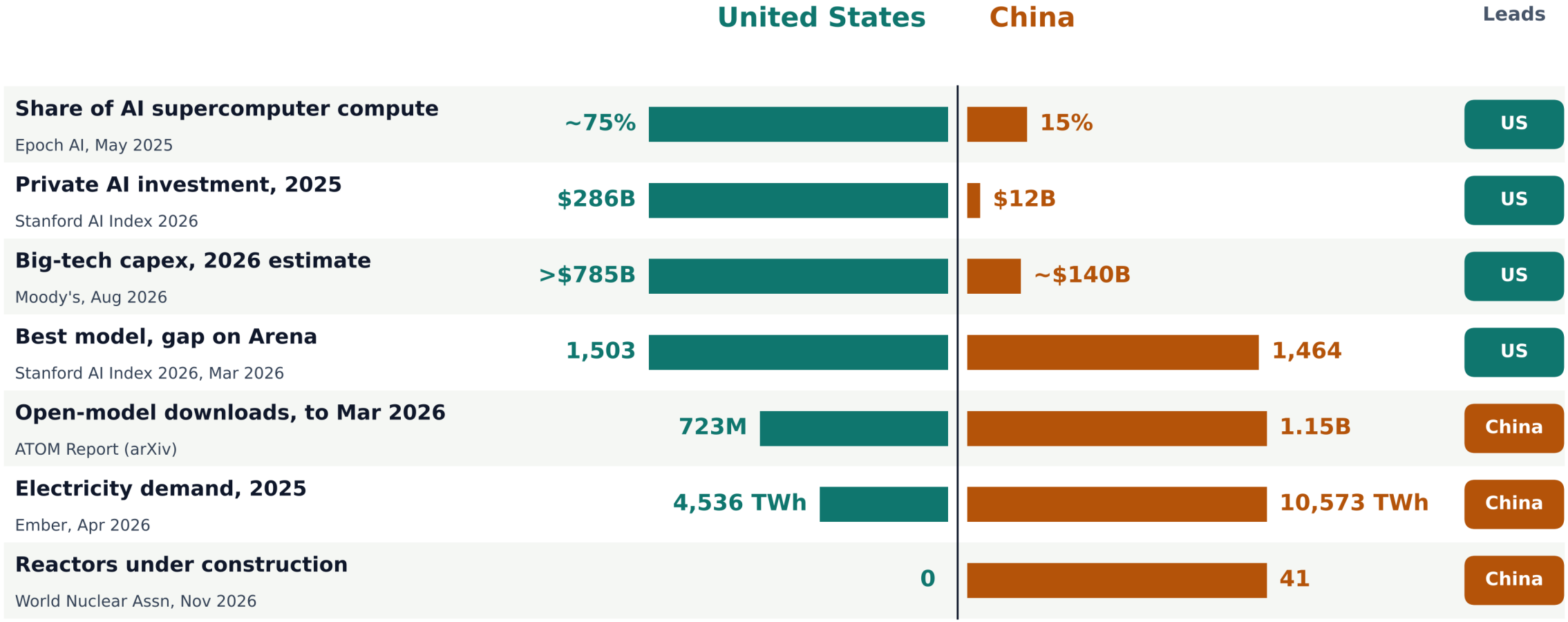
Who leads where, and how the two chains separated

COMING UP


| | | | |
|---|---|---|---|
| Who leads where | 40 | Nvidia’s China sales | 45 |
| Open vs closed models | 41 | China’s parallel chain | 46 |
| Capex, US vs China | 42 | Huawei | 47 |
| Power, US vs China | 43 | Huawei vs Nvidia roadmaps | 48 |
| Controls and counter-moves | 44 | CXMT memory | 49 |

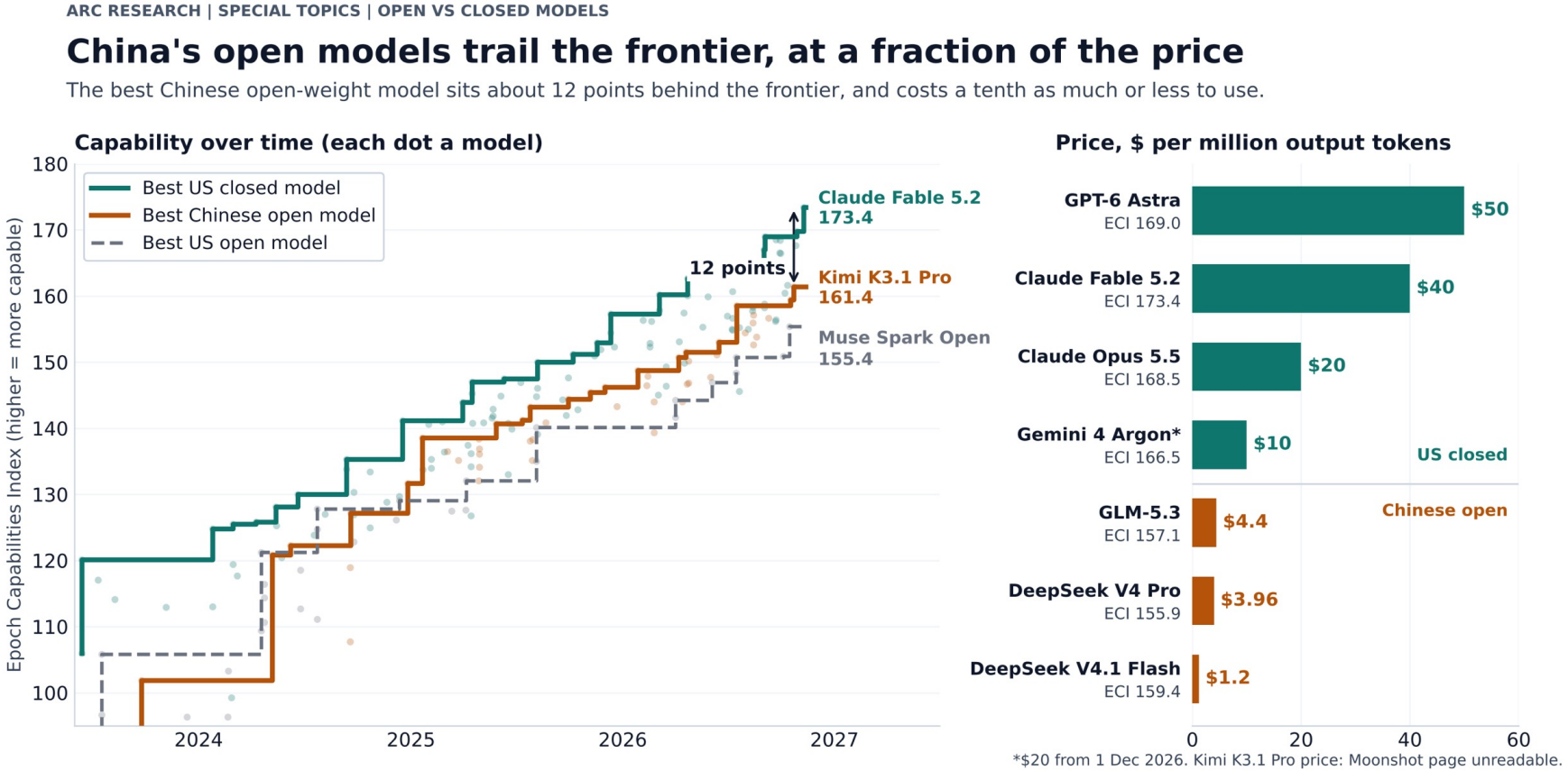



# US vs China: who leads where




Bars scaled within each row. Arena row: Elo scores of the best US model (Claude Opus 4.6) and best Chinese model (ByteDance Dola-Seed-2.0) in March 2026, a 2.7% gap. Epoch AI says Chinese models trail the US frontier by about 7 months on average (Jan 2026). Compute share: Epoch AI, 'about three-quarters' US, May 2025 data. Capex: six US hyperscalers vs leading Chinese tech companies. Stanford figures via IEEE Spectrum and TNW.

# China gives its models away; the US sells the best ones



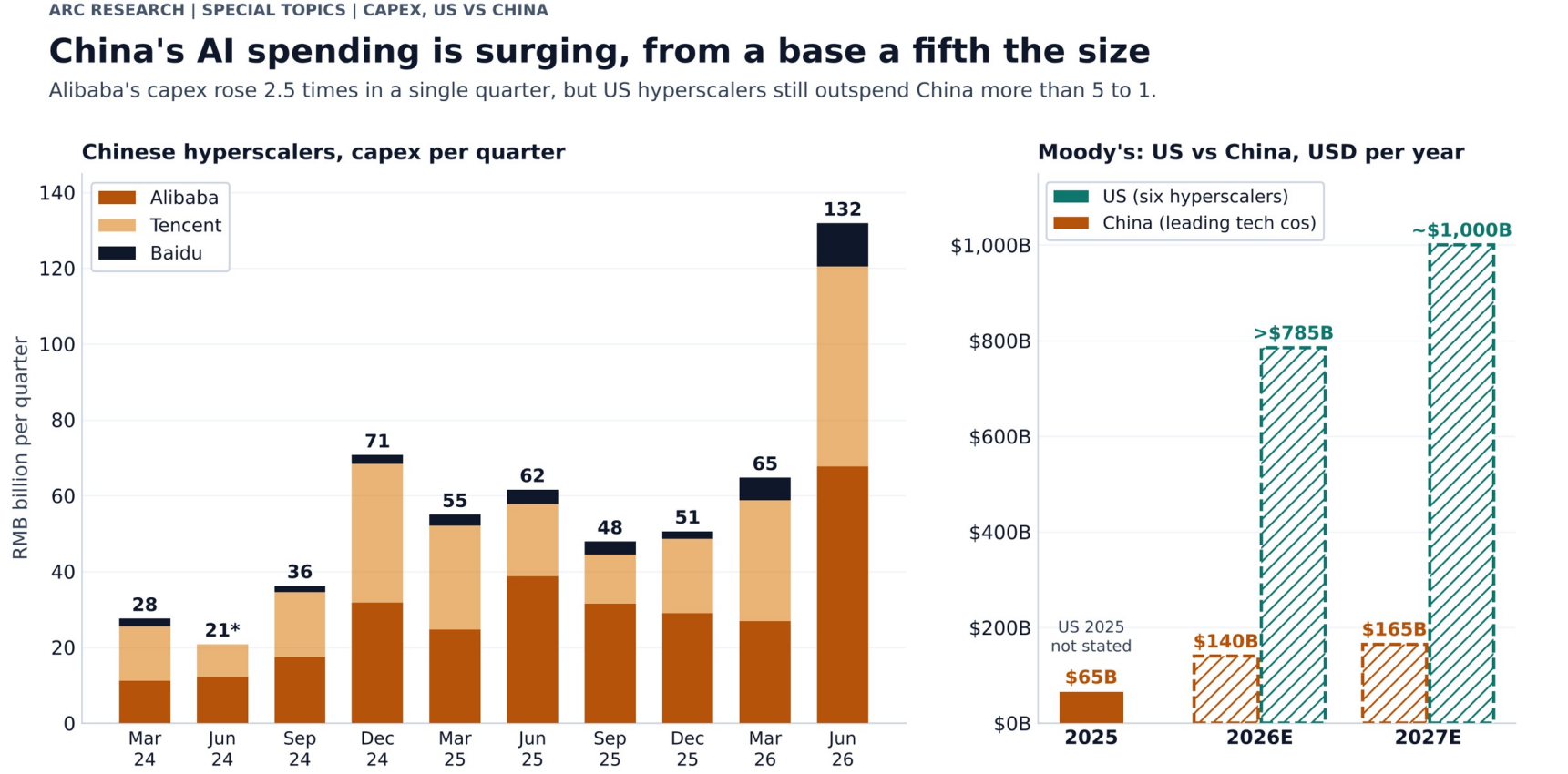

## SO WHAT

~12 points behind

Leads US open models

A tenth of the price

# China is spending fast, from a much smaller base



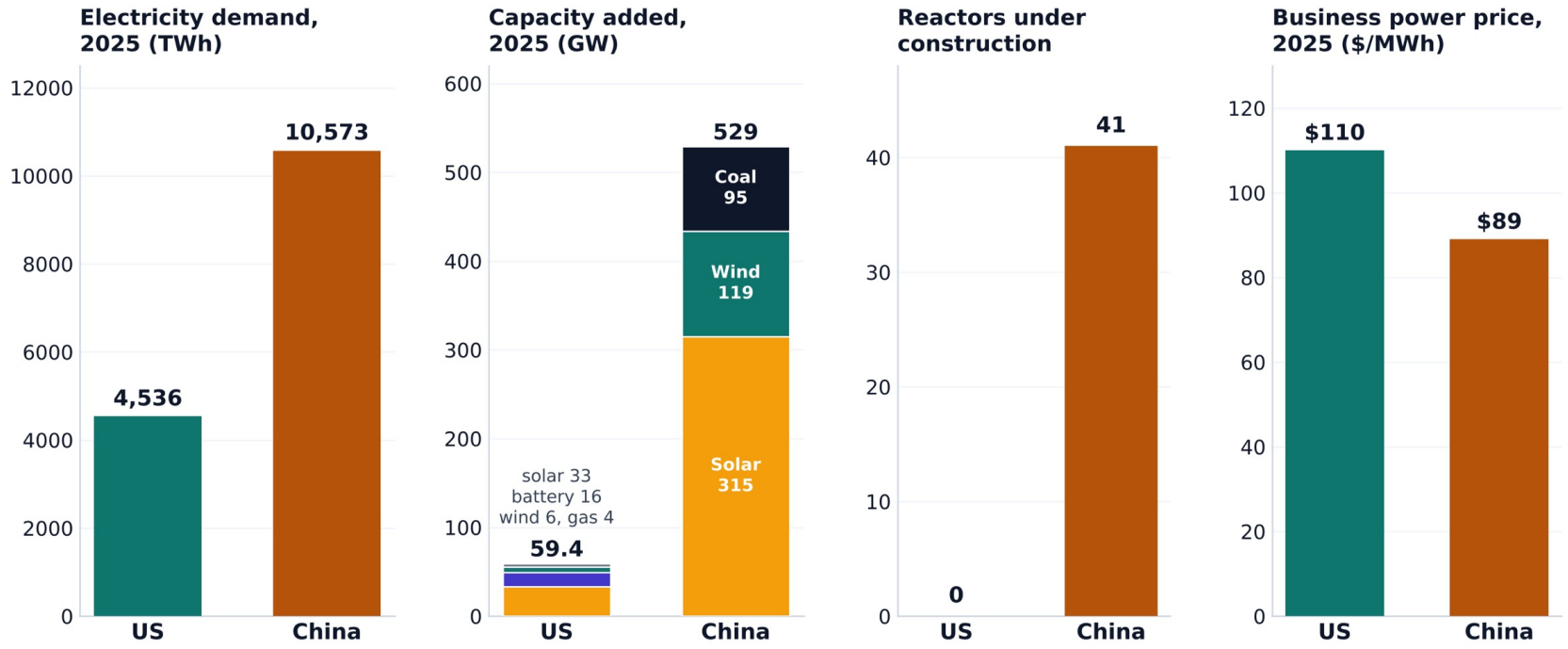

## SO WHAT

RMB 132B in one quarter

US: >5x China in 2026

Capped by chips, not cash

# China has the power the US is short of



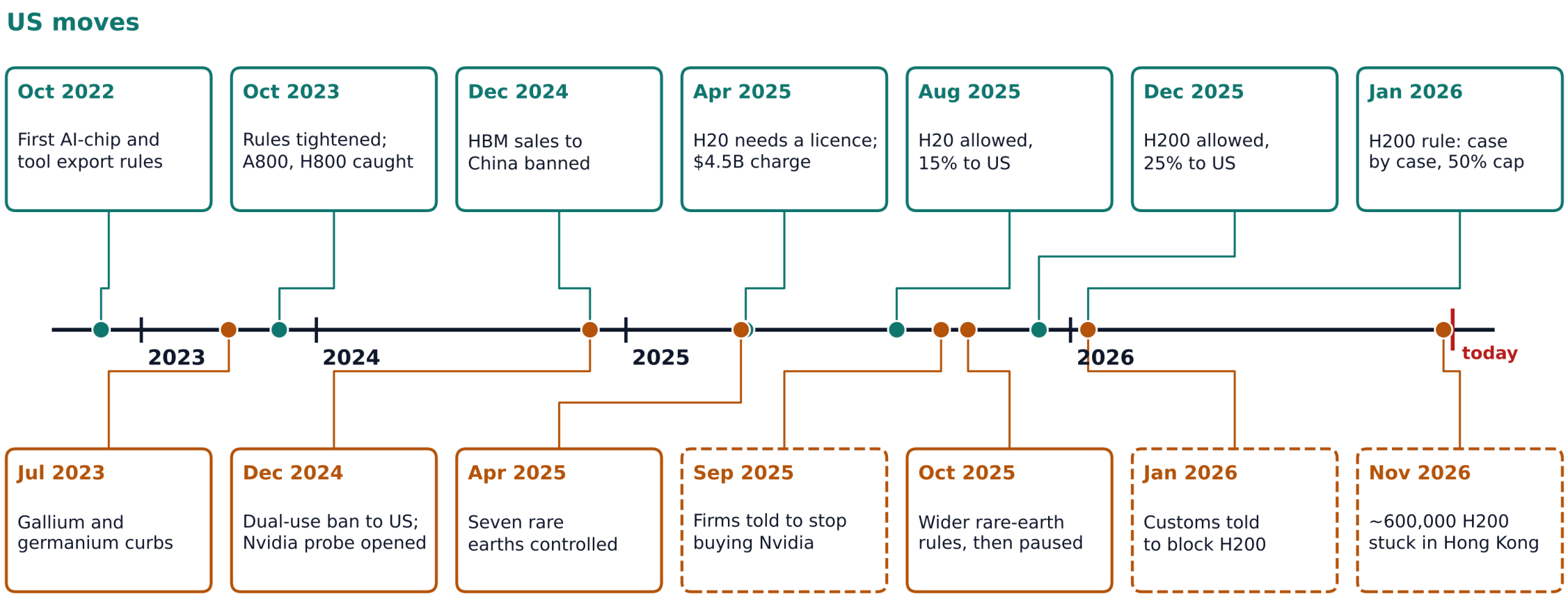

## SO WHAT

2.3x the electricity

~9x the new capacity

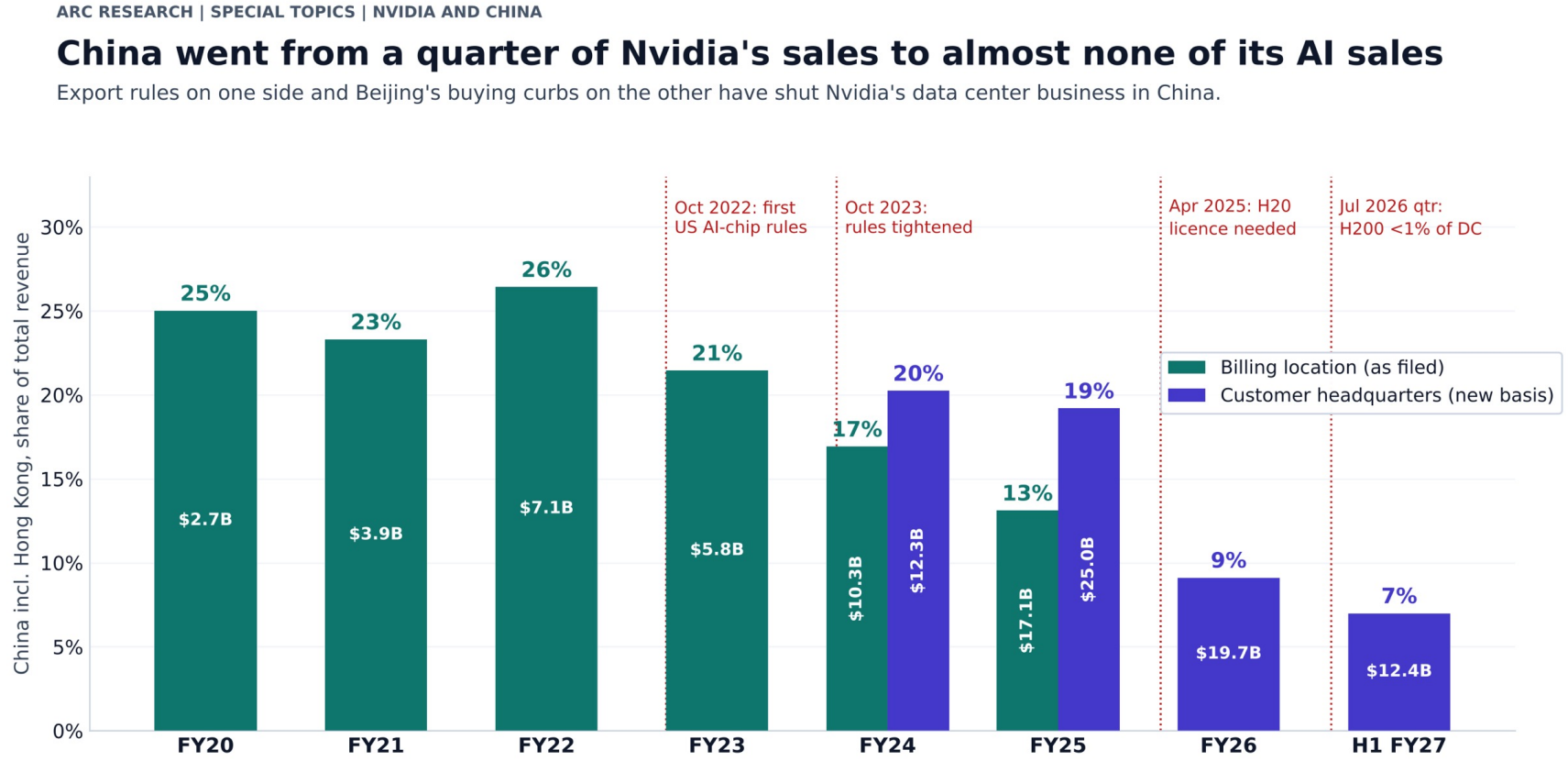
41 reactors to none

# Four years of moves and counter-moves




Sources: BIS rules (7 Oct 2022, 17 Oct 2023, 2 Dec 2024, H200 rule effective 15 Jan 2026); Nvidia 8-K (9 Apr 2025) and Q1 FY2026 results; White House statements (11 Aug and 8 Dec 2025); MOFCOM announcements (3 Jul 2023, No. 46 Dec 2024, No. 18 Apr 2025, Nos. 55-62 Oct 2025; suspension 7 Nov 2025); SAMR (15 Sep 2025). Dashed = press-reported: FT (17 Sep 2025), Reuters (14 Jan and 5 Nov 2026).

# China went from a quarter of Nvidia’s sales to almost none of its AI sales




## SO WHAT

25% to 7% of revenue

Guidance assumes zero China

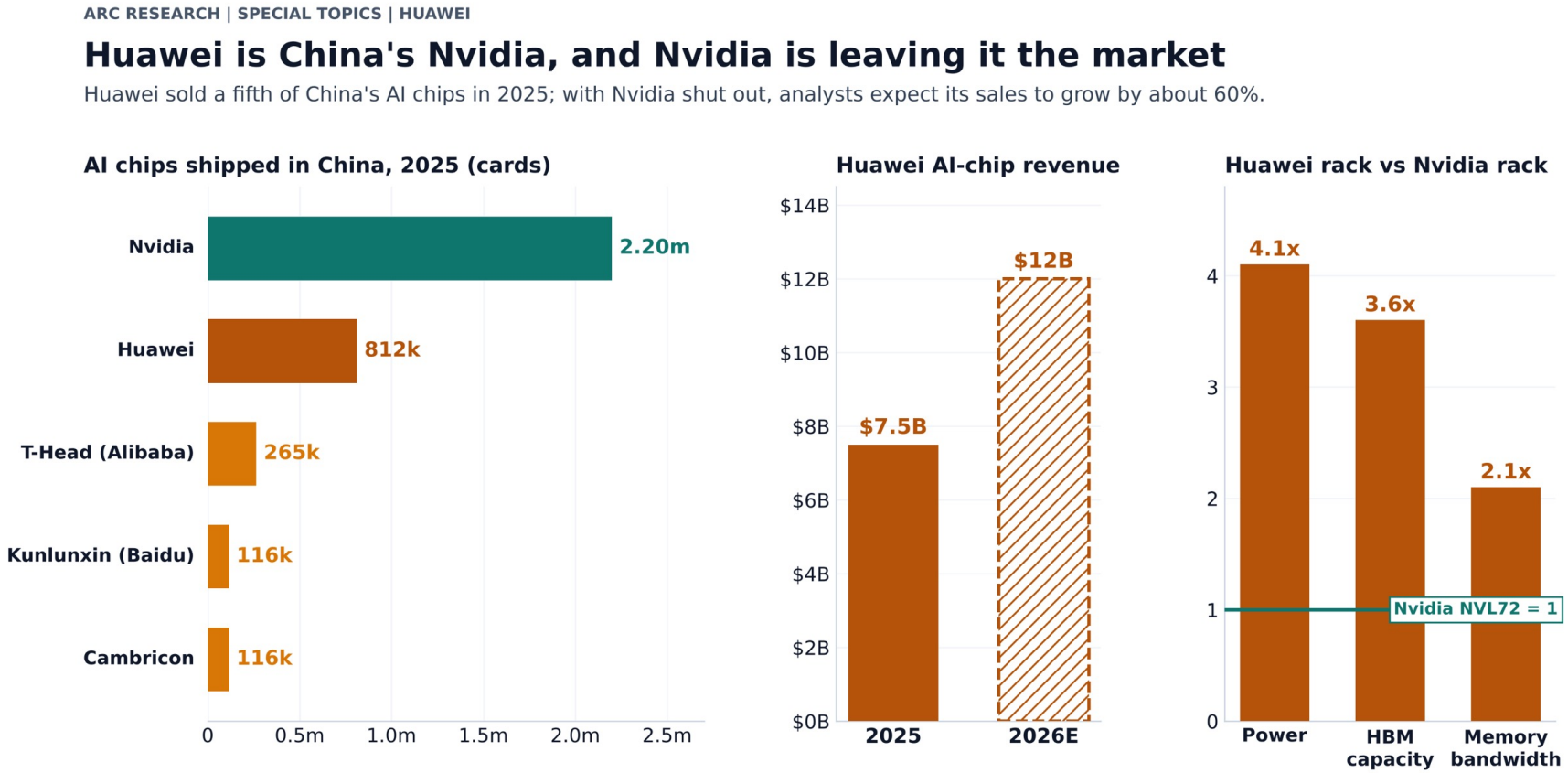
Upside if Beijing relents

# China is building a parallel chain, layer by layer

| LAYER | US-LED CHAIN | CHINA'S PARALLEL CHAIN | THE SIGNAL |
|---|---|---|---|
| AI chips | Nvidia, AMD, custom ASICs | Huawei Ascend 950, Cambricon | **Cambricon revenue +104% y/y (Q3)** |
| HBM | SK hynix, Samsung, Micron: HBM4 | Huawei's own HBM; CXMT | **CXMT HBM3 yield ~25% (modeled)** |
| Foundry | TSMC: 2nm, 3nm; $46.7B a quarter | SMIC: 7nm-class; $3.0B a quarter | **about 1/15th of TSMC revenue** |
| Equipment | ASML, AMAT, Lam, TEL | China bought $49.3B in 2025 | **#1 buyer of chip tools** |
| Models | US frontier labs | DeepSeek, Qwen, Kimi | **~7 months behind (Epoch AI)** |
| Capex | $202B a quarter (4 US firms) | ~$140B in 2026e (Moody's) | **a fraction of US spending** |

ARC Research schematic. Huawei roadmap (Huawei, Sep 2025 and Sep 2026); Cambricon Q3 2026 filing via TechNews; CXMT yield: SemiAnalysis model; SMIC Q2 2026 results via XenoSpectrum; SEMI (7 Apr 2026); Epoch AI (9 Jan 2026, via Trending Topics); Moody's via TechNode (28 Aug 2026); US capex: SEC filings, ours. TSMC 3Q26 vs SMIC 2Q26 (latest each; 3.0 / 46.7, ours). SMIC 7nm-class: press-reported.

# Huawei: China’s answer to Nvidia




Source: left: IDC via Reuters (2 Apr 2026), AI accelerator cards shipped in China in 2025 (Nvidia about 2.2m, about 55%; Chinese vendors 1.65m). Middle: Bernstein via Tom's Hardware (5 May 2026), Huawei AI-chip revenue, 2026 an estimate (dashed). Right: SemiAnalysis (16 Apr 2025), Huawei CloudMatrix 384 relative to Nvidia GB200 NVL72; it also delivers 'almost double' the compute.

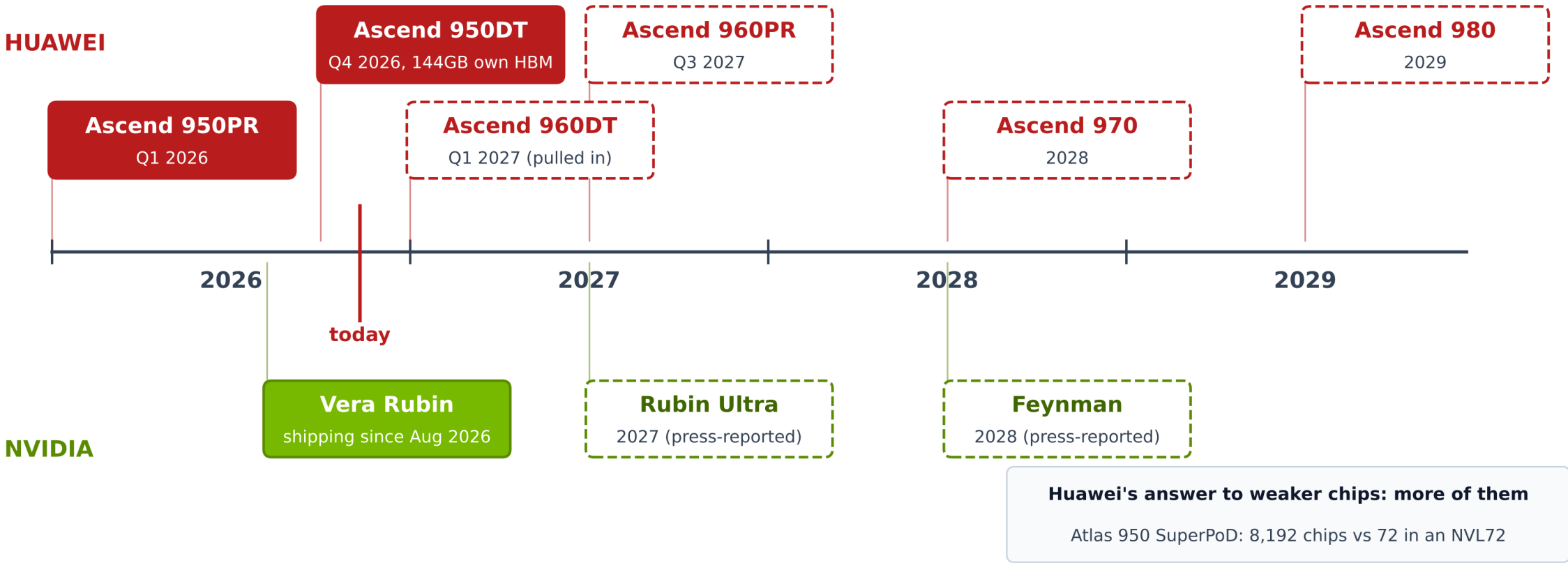
## SO WHAT

#1 Chinese AI chip

+60% sales expected

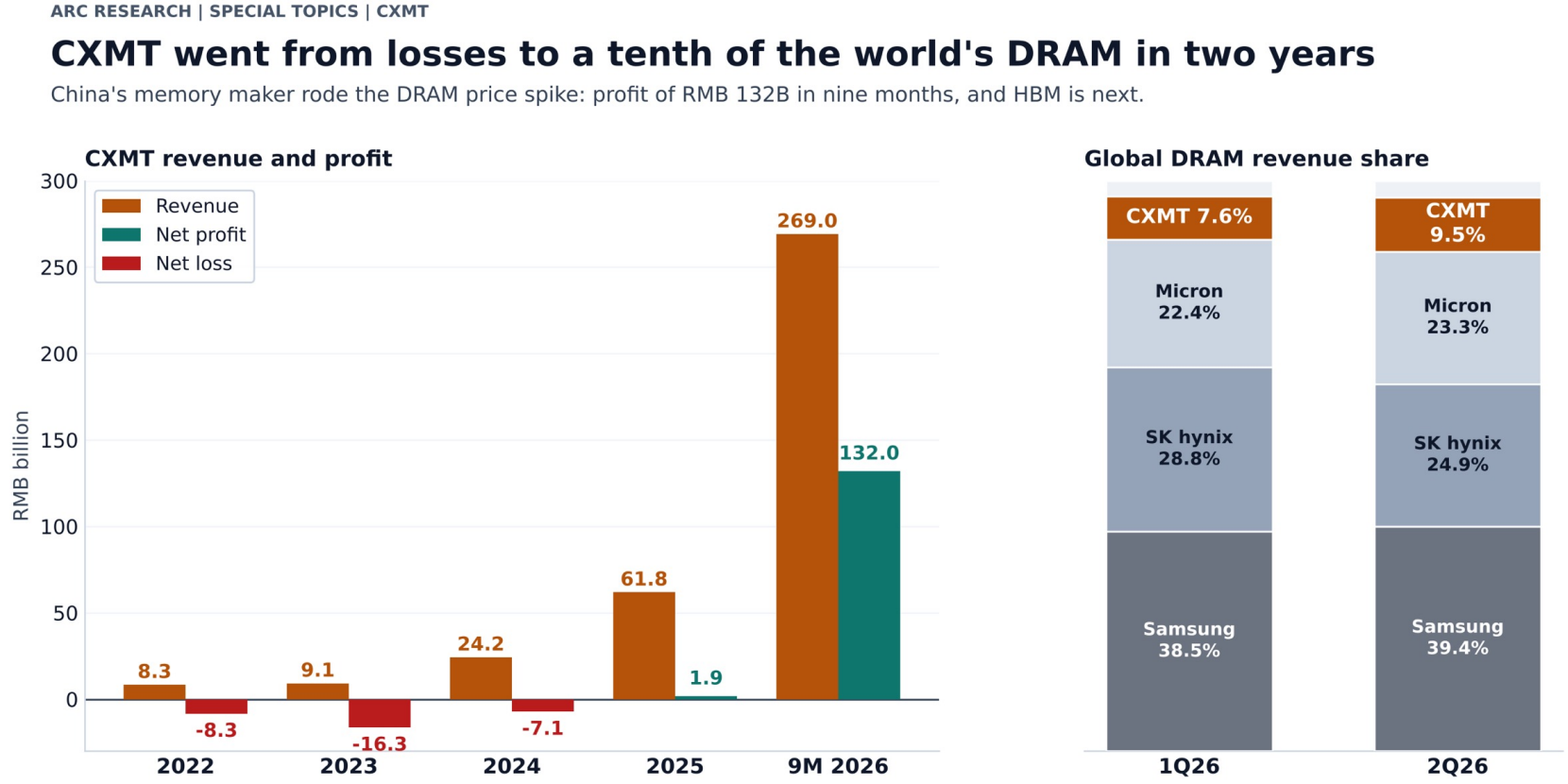
More chips, more power

# Huawei answers weaker chips with more of them




Solid = launched or shipping; dashed = announced roadmap. Huawei: Huawei Connect keynotes (18 Sep 2025; 17 Sep 2026, which pulled the 960 forward). NVIDIA: Vera Rubin shipments per Q2 FY2027 call; Rubin Ultra and Feynman per press coverage of GTC 2026 (Jon Peddie). Chip counts are per scale-up system, not per chip performance.

# CXMT: China’s memory maker is now a global player




Source: left: CXMT prospectus figures via TrendForce (2023-2025) and Gelonghui (2022); 9M 2026 from the company's Q3 report (30 Oct 2026) via NeoTeo; net profit attributable to parent.
Right: TrendForce DRAM revenue ranking, 1Q26 and 2Q26 (CXMT $7.34B and $14.62B); 'other' is the remainder (ours). Omdia's 2025 figures use a different method and are not shown.

## SO WHAT

9.5% of world DRAM

RMB 132B profit, 9 months

HBM yields still low

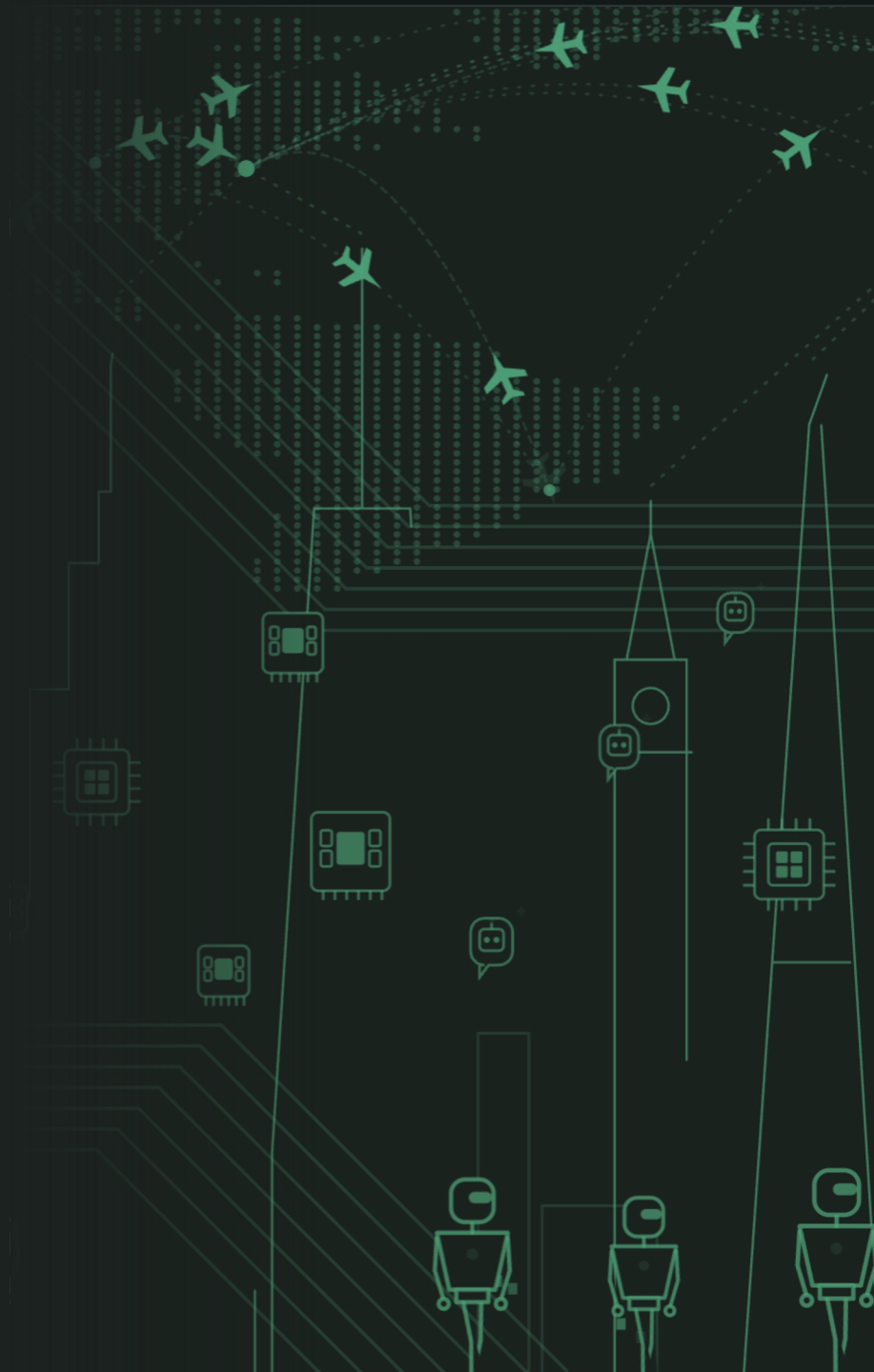
CHAPTER 6 OF 8

# 2025: gigawatts and circular money

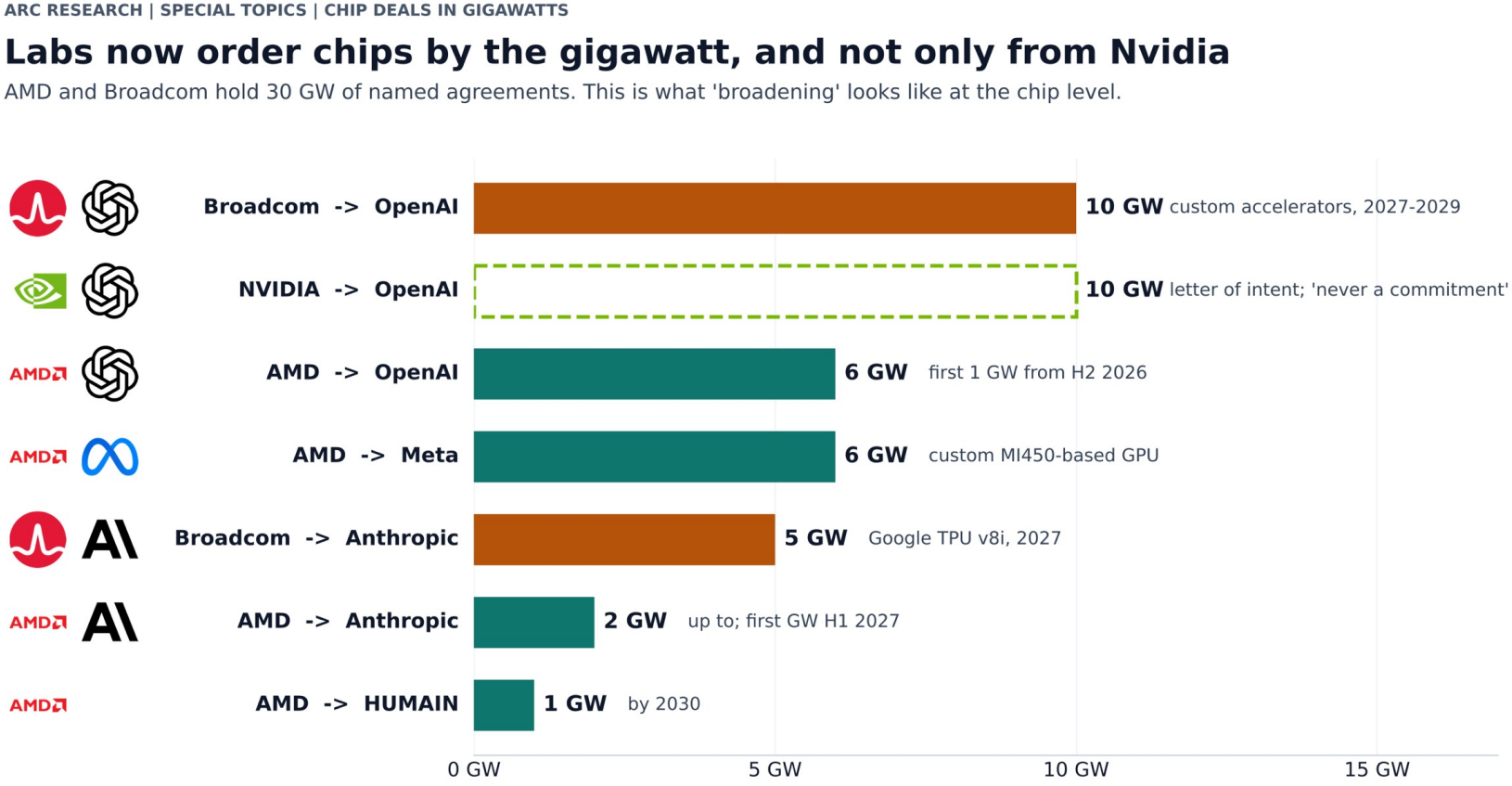
Deals sized in power, and money that loops back

## COMING UP


| | | | |
|---|---|---|---|
| Deals in gigawatts | 51 | The neolabs | 55 |
| Nvidia as financier | 52 | ASML to TSMC to Nvidia | 56 |
| Neoclouds and Nvidia | 53 | Gigawatt campuses | 57 |
| The money loop | 54 | AI and US growth | 58 |

# Chips are now ordered by the gigawatt



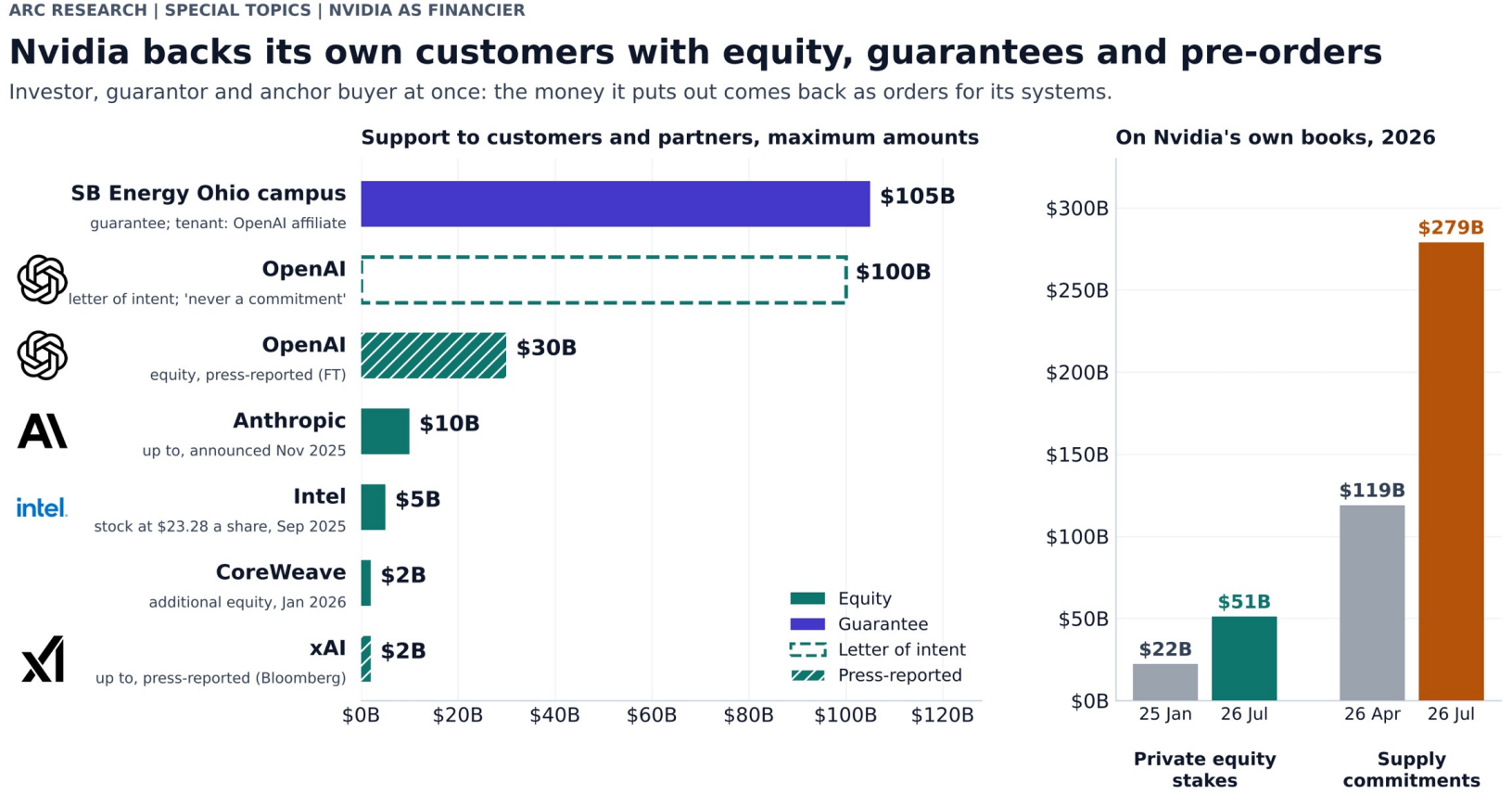

## SO WHAT

30 GW outside Nvidia

Agreements, not shipments

Power is the unit that binds

# Nvidia is acting like the central bank of AI




Source: NVIDIA 10-K, 10-Q and CFO commentaries (guarantee, non-marketable equity securities, supply commitments); NVIDIA, Anthropic and Intel releases; CoreWeave via SiliconANGLE.
Solid = company-stated. Dashed = letter of intent the CEO later said was never a commitment. Hatched = press-reported, amount unconfirmed. Maximum amounts, not cash paid; do not sum.

## SO WHAT

Investor, guarantor, pre-buyer

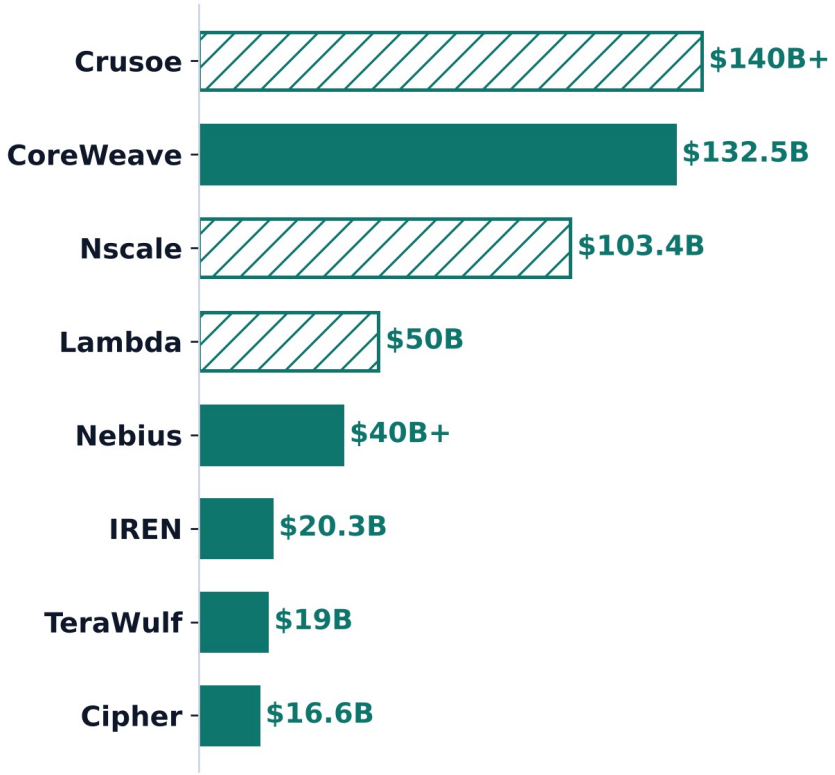
The money returns as chip orders

Stakes: $22B to $51B in six months

# Neoclouds: Nvidia's customers beyond the hyperscalers

## Neocloud contracted backlog

Latest, 2026. Hatched = private or press-reported




## How Nvidia backs them

| | |
|---|---|
| **CoreWeave** | $2B stake (~13%) + $6.3B pledge to buy unsold capacity to 2032 |
| **Nebius** | $2B, ~9% stake incl. warrant (Mar 2026) |
| **Lambda** | Rents back ~18,000 of its own GPUs, $1.5B over 4 years |
| **IREN** | Up to $2.1B; also a $3.4B customer |
| **Nscale** | $1B of its convertible notes (Sep 2026) |
| **All clouds** | $29B multi-year cloud deals + $3.5B of lease guarantees (Nvidia 10-Q, Jul 2026) |

## Why Nvidia does it

**44%**

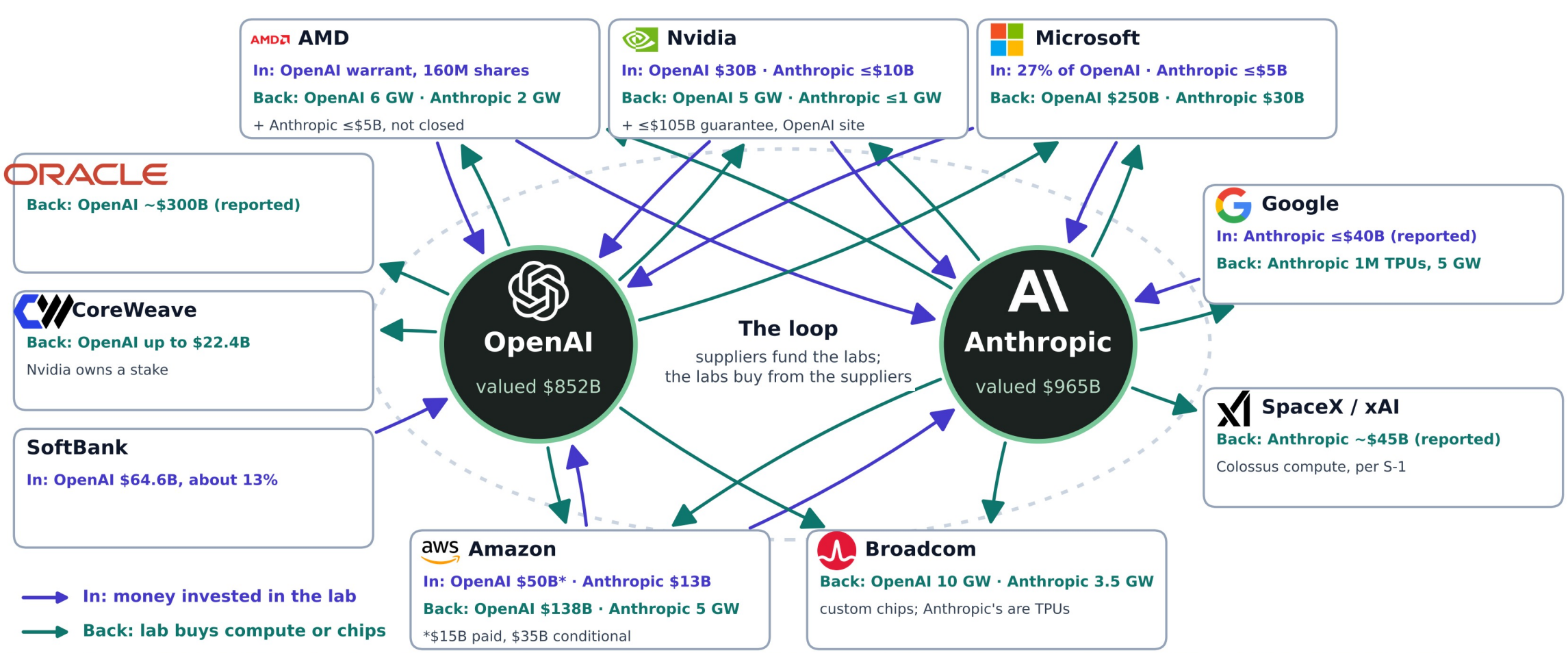
of Nvidia revenue came from three customers (1H FY27). The hyperscalers also design their own chips. Neoclouds run mostly on Nvidia.

**The risk**

CoreWeave debt $46.9B; some bonds yield ~15%. Chanos: Nvidia is “putting money into money-losing companies” to buy its chips.

Sources: company releases and SEC filings (CoreWeave, Nebius 6-K, IREN, TeraWulf, Cipher); Nvidia 10-Q (Jul 2026) and 10-Q customer concentration; press: Next Platform, TechSpot, TNW/Reuters (Lambda), WSJ (Lambda backlog), Crusoe and Nscale reports, Reuters (CoreWeave bonds), Chanos (Nov 2025). Backlog definitions differ by company.

# It all comes down to OpenAI and Anthropic

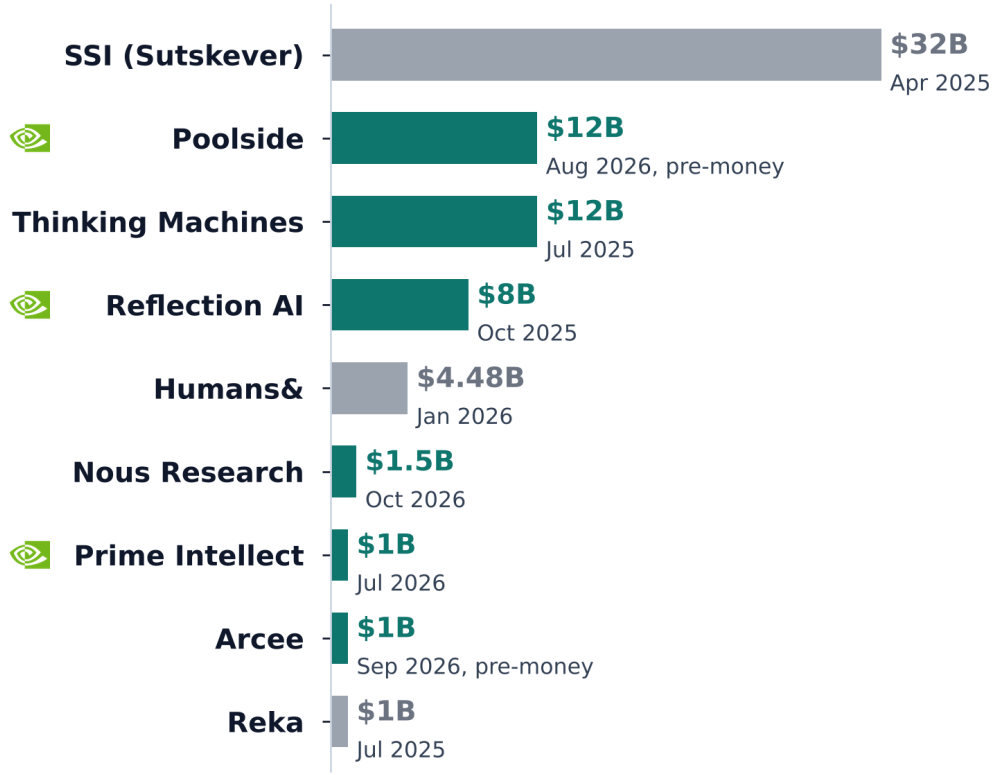



Sources: OpenAI (27 Feb 2026), Nvidia, Microsoft (28 Oct 2025), Amazon (27 Feb, 20 Apr 2026), Anthropic (23 Oct, 18 Nov 2025; 28 May 2026), AMD (6 Oct 2025; 22 Jul 2026), OpenAI-Broadcom (13 Oct 2025), CoreWeave, SoftBank. Reported: Oracle (WSJ), Google (Bloomberg), Nvidia guarantee (Implicator), Broadcom 3.5 GW (Tom's Hardware), SpaceX (HotHardware), OpenAI value (investingLive). Ceilings and commitments over different years; do not add.

# The neolabs: new labs, and America's open-model push

## Neolabs: valuation at latest confirmed round

Teal = releases open weights. Nvidia logo = Nvidia invested in that round or deal.




## America's open-weight push

| | | |
|---|---|---|
| **Thinking Machines** | Inkling, 975B parameters | Jul 2026 |
| **Reflection AI** | Beam, 501B, Apache 2.0 | Oct 2026 |
| **Arcee** | Trinity Large, 400B | 2026 |
| **Nvidia** | Nemotron 3 Ultra open models | Jun 2026 |
| **Ai2** | Olmo 3, fully open | Nov 2025 |

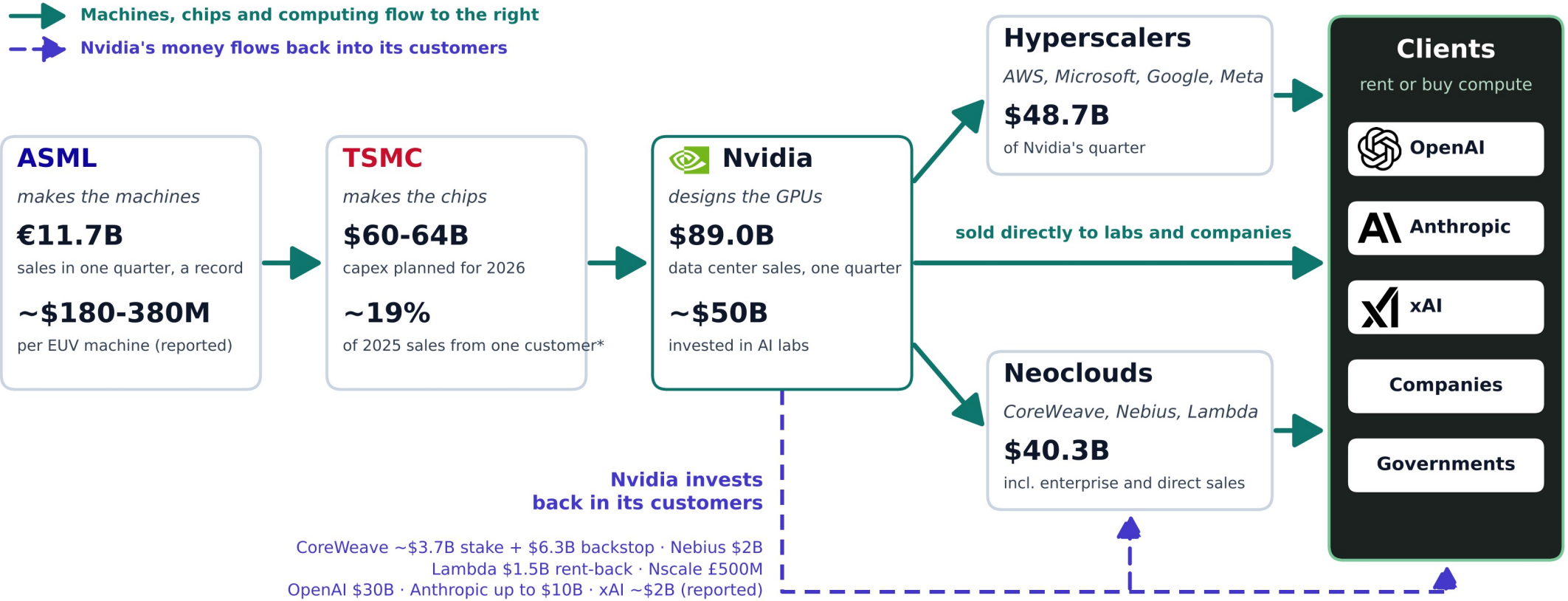
### Why it matters

"Since about April 2025, Chinese AI companies have been the clear leader in open-weight models."
— Nathan Lambert, ATOM Project, Sep 2026

US AI Action Plan (Jul 2025): "America has leading open models founded on American values"

Sources: company announcements and press (TNW, SiliconANGLE, Decrypt, Forbes, Dealroom); Reflection and Thinking Machines model posts; Nvidia GTC 2026; Ai2; ATOM Project; FedScoop (AI Action Plan). Rounds in talks (Thinking Machines ~$40B, Periodic Labs) and Mistral (€21B+, Sep 2026) are not charted.

# From ASML's machines to your chatbot, and Nvidia's money back again



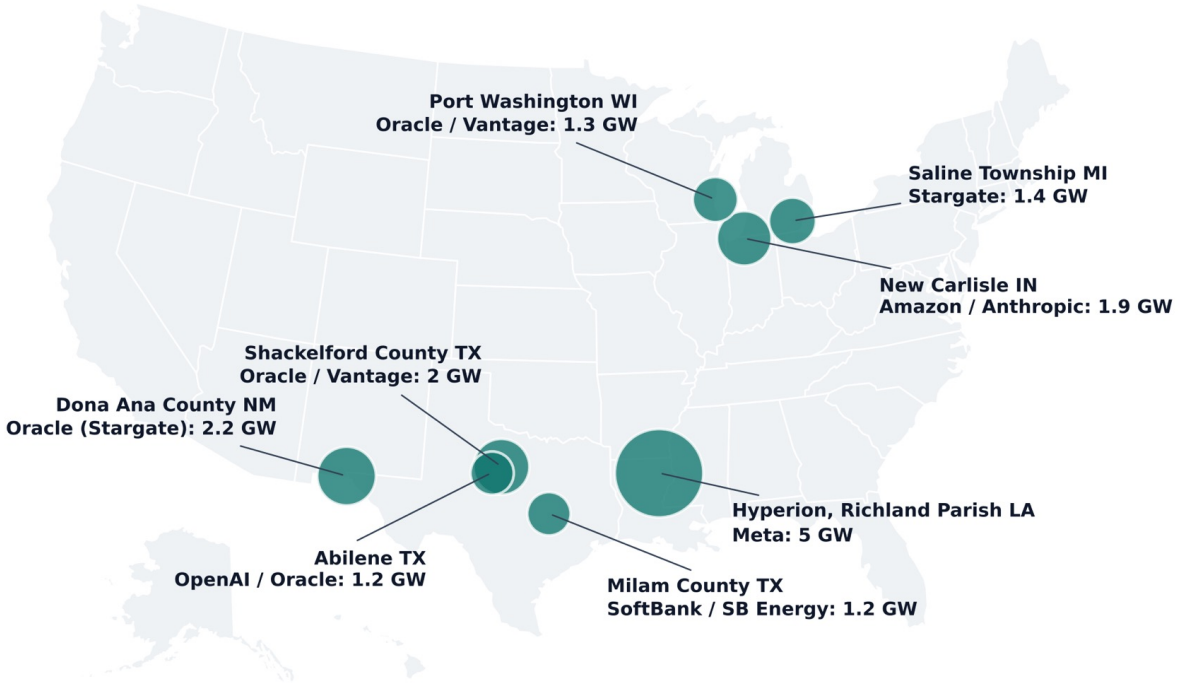

Sources: ASML Q3 2026 results (14 Oct 2026); EUV prices press-estimated (Tom's Hardware). TSMC Q3 2026 call (via Futurum). *TSMC's unnamed 'Customer A', 19% of 2025 revenue, widely taken to be Nvidia (Commercial Times via TrendForce). Nvidia Q2 FY2027 CFO commentary, quarter to 26 Jul 2026: Hyperscale $48.7B; AI clouds, industrial and enterprise $40.3B; 'nearly $50 billion' in frontier labs. Stakes: 13F, CoreWeave 8-K, press. Illustration.

# The new campuses are the size of power plants

ARC RESEARCH | SPECIAL TOPICS | GIGAWATT CAMPUSES

## Where the gigawatts are going: eight named campuses, 16 GW planned

Texas and the Midwest dominate. One Louisiana site alone is planned at 5 GW, backed by more than 7 GW of new gas plants.




Source: Epoch AI (Stargate site review, 17 Apr 2026; AI data center directory, 2 Aug 2026) for Stargate and New Carlisle; Meta confirmation of Hyperion (13 Jul 2026) via press. Planned capacity, not operating; Shackelford is Epoch's estimate.
Bubble area = planned GW. Points placed at the named town or county (approximate, ours). Sum = 16.2 GW (ours). Not a complete list. Geometry: Highcharts Map Collection (Natural Earth).

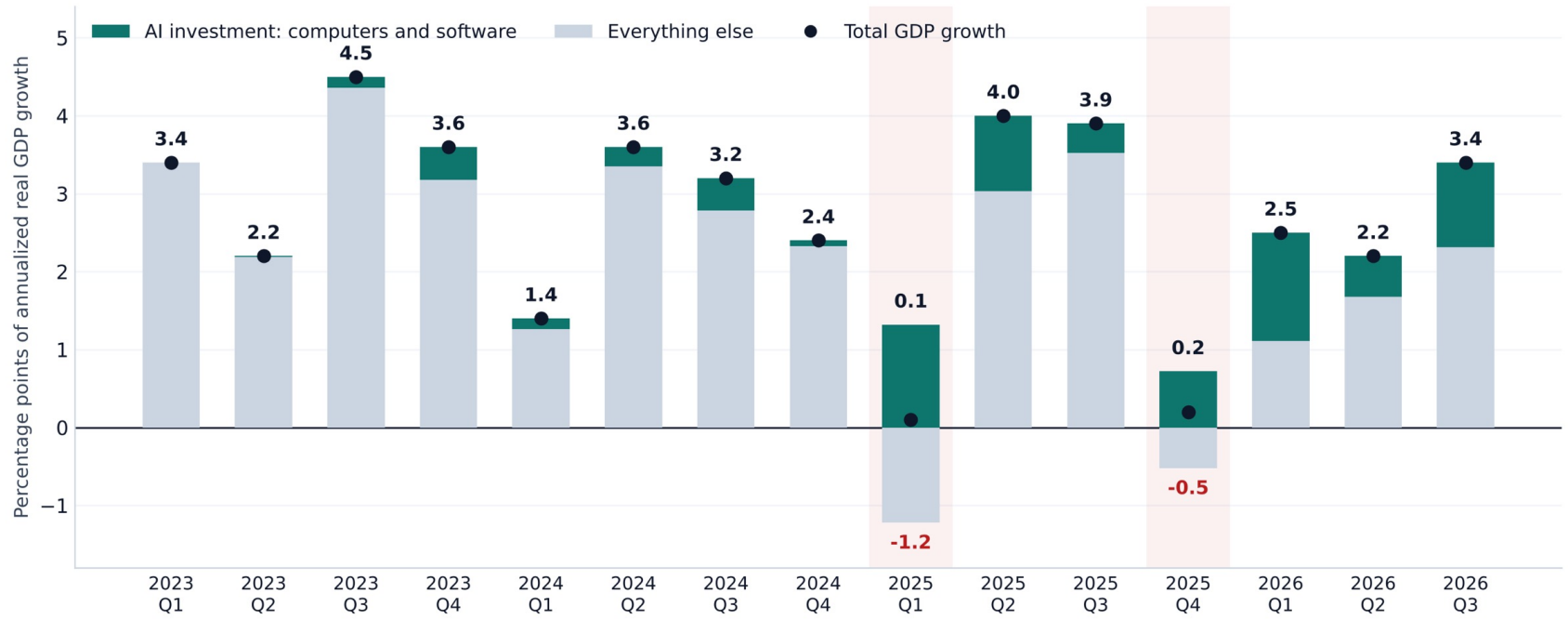
**SO WHAT**

8 sites: 16 GW planned

Sited where power is, not people

$35–60B per gigawatt

# AI spending kept US growth positive in 2025




Source: BEA, contributions to percent change in real GDP (seasonally adjusted annual rates), via FRED, vintage 29 Oct 2026; 3Q26 is the advance estimate. AI investment = information processing equipment + software (a proxy; includes non-AI spending). 'Everything else' = GDP growth minus that (ours). Gross of imports: much AI equipment is imported, which subtracts from GDP elsewhere.

## SO WHAT

~1 point a quarter since 2025

Two quarters: rest of economy shrank

Net of imports: much smaller

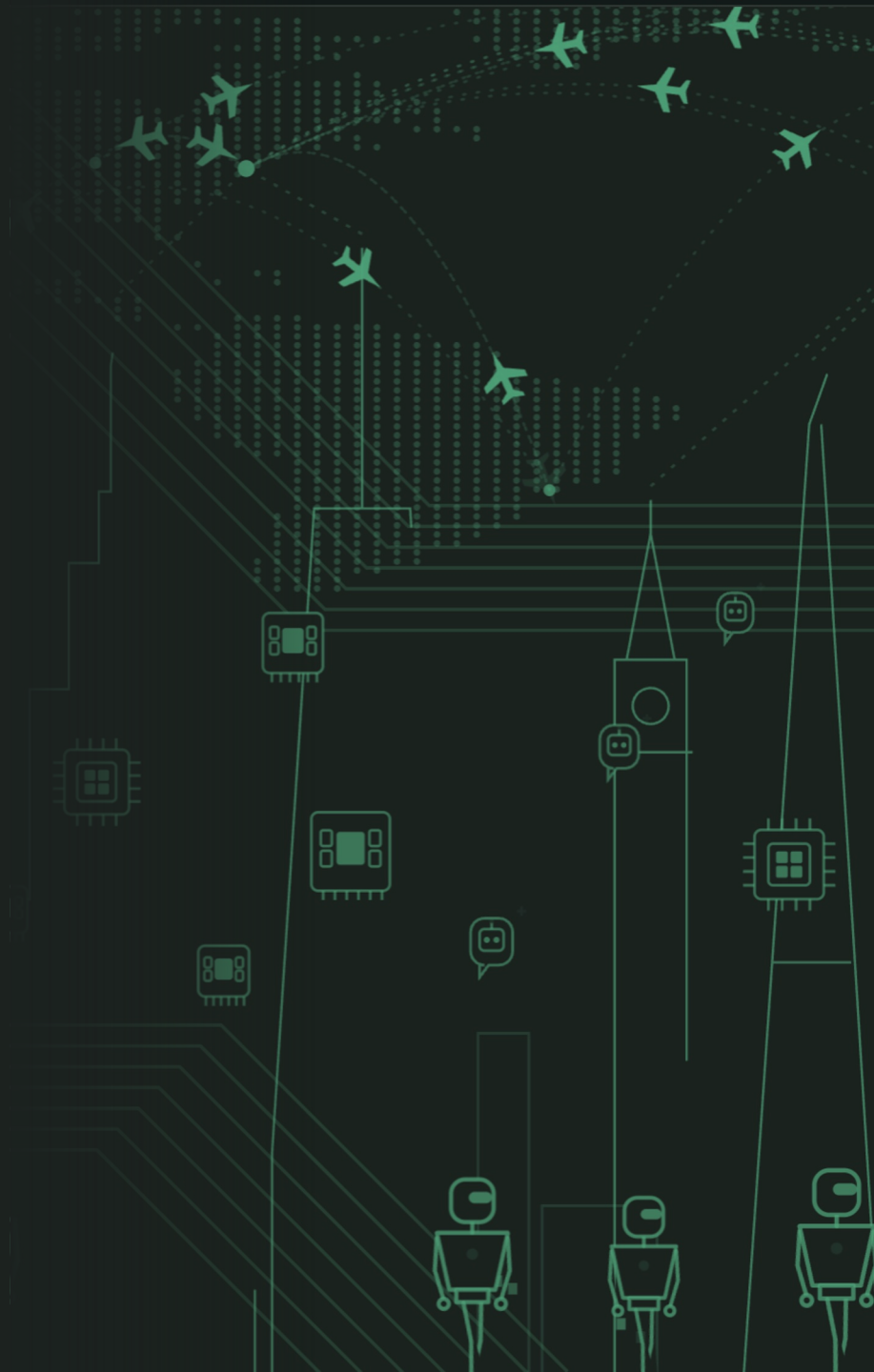
CHAPTER 7 OF 8

# 2026: bigger chips, bigger bets

The rally widens, models reach trillions, capex hits records

COMING UP


| | | | |
|---|---|---|---|
| 2026 returns so far | 60 | Nvidia’s revenue | 68 |
| The rally widens | 61 | CPUs are back | 69 |
| The frontier today | 62 | Cerebras wafer-scale chips | 70 |
| Agents and RSI (METR) | 63 | What Nvidia sells | 71 |
| Model size | 64 | Inside the chip | 72 |
| 2026 plans and beyond | 65 | Marvell buys Lumentum | 73 |
| Cloud revenue | 66 | Sites running and being built | 74 |
| Old GPUs cost more | 67 | | |

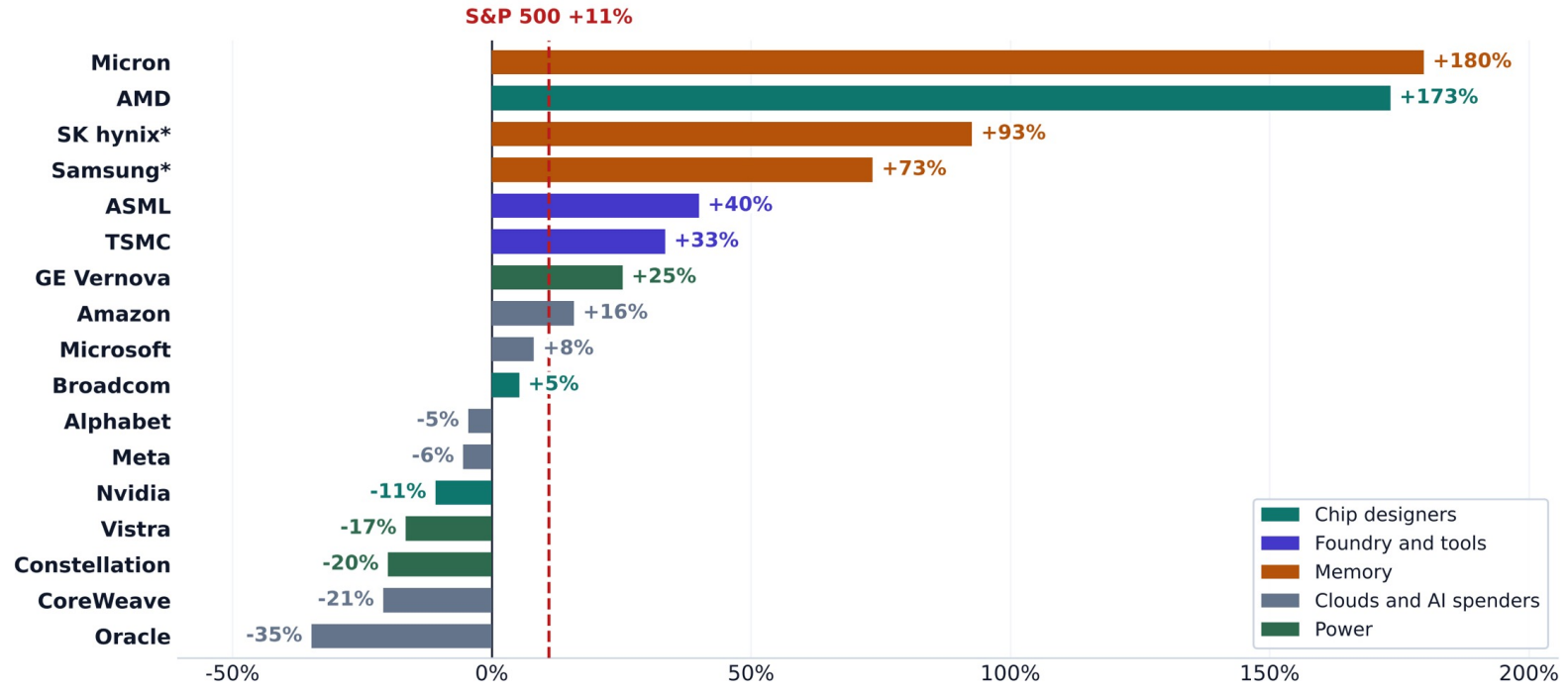



# 2026 so far: memory and AMD up, Nvidia down

ARC RESEARCH | SPECIAL TOPICS | THE AI TRADE, YEAR BY YEAR

## 2026 so far: memory and AMD on top, Nvidia and the spenders behind

Micron and AMD nearly tripled before November's selloff; Nvidia, Oracle and the power names are down for the year.




Source: price returns computed by us from split-adjusted closes (digrin.com; stockanalysis.com for 12 Nov 2026), cross-checked with macrotrends; not dividend-adjusted.
*In Korean won. S&P 500 via SPY. CoreWeave 2025 from its $40 IPO price (28 Mar). Names missing in a year had not listed or lack a verified start price.

### SO WHAT

Micron +180%, AMD +173%

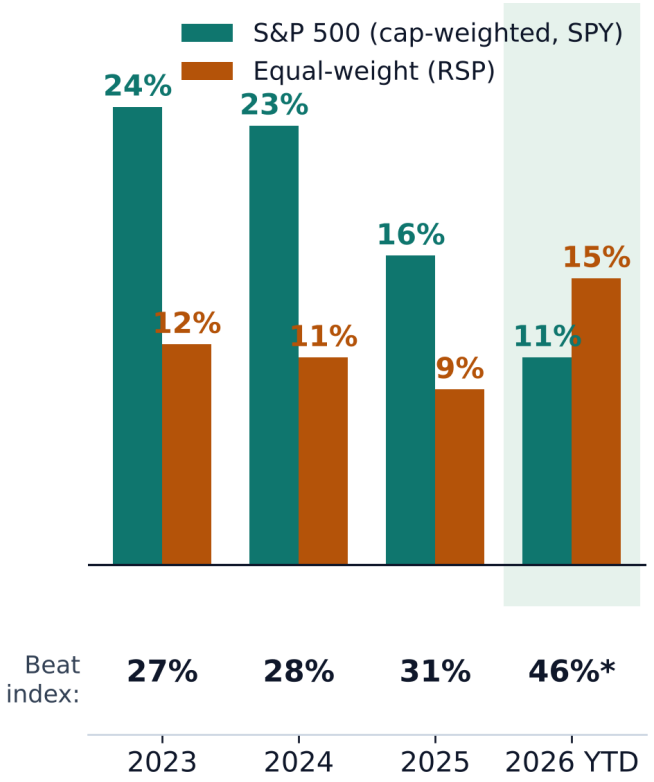
Nvidia -11% after Nov selloff

Debt-funded names hit hardest

# 2026: the rally widens beyond AI, and AI costs more to own

## The average stock wins in 2026

Equal-weight leads, first time since ChatGPT




## Sector returns, % (sorted by 2026)

Energy, health care and staples lead this year

| | 2023 | 2024 | 2025 | 2026 YTD |
|---|---|---|---|---|
| Energy | -4 | +2 | +4 | +44 |
| Health care | +0 | +1 | +12 | +21 |
| Technology | +55 | +21 | +24 | +21 |
| Staples | -3 | +9 | -1 | +17 |
| Materials | +10 | -2 | +8 | +15 |
| Real estate | +8 | +2 | -1 | +10 |
| Industrials | +16 | +16 | +18 | +9 |
| Financials | +10 | +28 | +13 | +6 |
| Utilities | -10 | +20 | +13 | +3 |
| Cons. discretionary | +38 | +26 | +6 | +1 |
| Communication | +51 | +33 | +22 | -9 |

## Paying less for AI

Valuations came down

**22.5x → 18.7x**

S&P 500 forward P/E

Dec 2025 → 30 Oct 2026 (FactSet)

**>30% → ~10%**

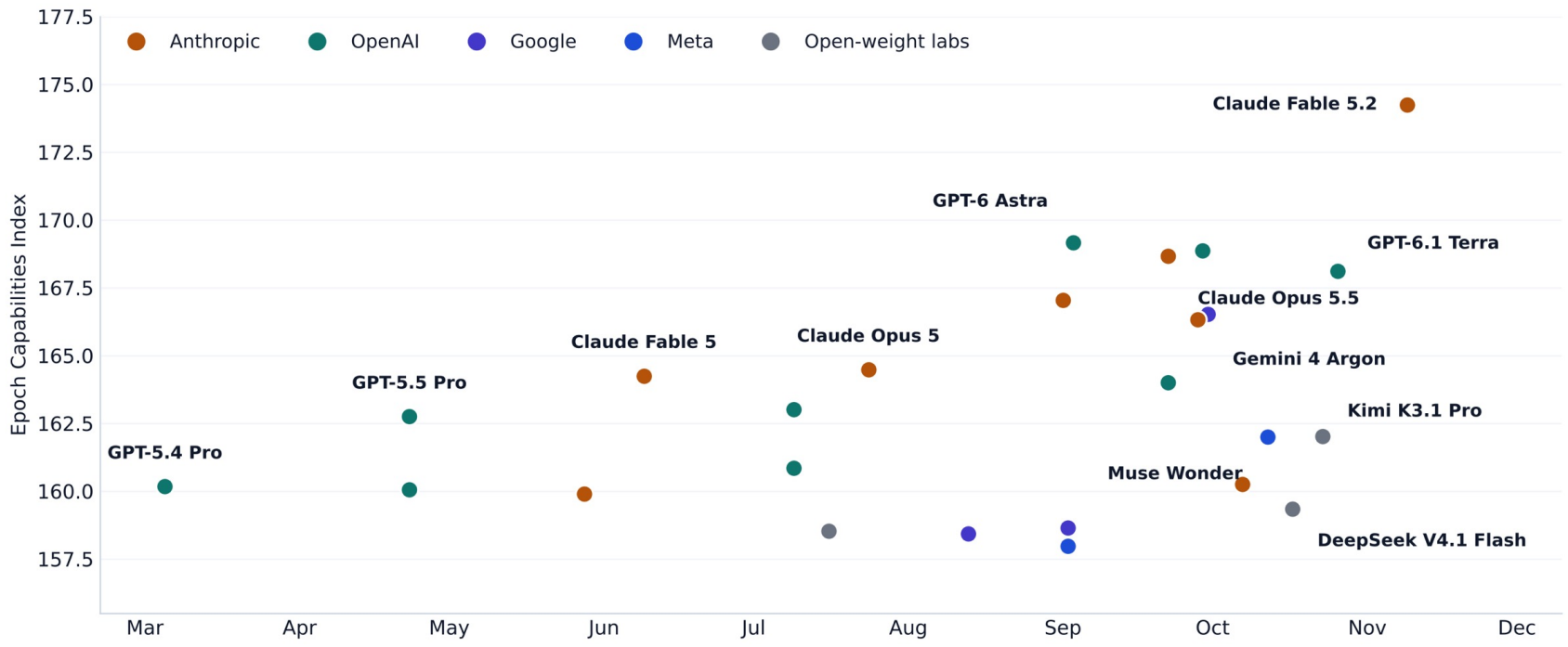
Mag 7 P/E premium

2020s → Jul 2026 (Morgan Stanley)

Spenders punished for capex: Meta and Alphabet fell on 2026 spending guides

Price returns computed by us, SPY, RSP and Select Sector SPDRs, year-end closes (digrin.com) to 12 Nov 2026 (stockanalysis.com, StatMuse); no dividends. *Members beating the index: First Trust (2026 = first half). P/E: FactSet; Morgan Stanley via Yahoo Finance. Capex reactions: Reuters, Trustnet.

# The frontier moves every few weeks




Source: Epoch AI, Epoch Capabilities Index (ECI), scores file updated 10 Nov 2026 (composite of 50+ benchmarks; no version number stated). Top 25 by ECI; release dates per Epoch.
Grey = open-weight labs (Moonshot, DeepSeek). Kimi K3.1 Pro release date is 23 Oct per Epoch (one press source says 5 Nov).

## SO WHAT

Top 25 all released in 2026

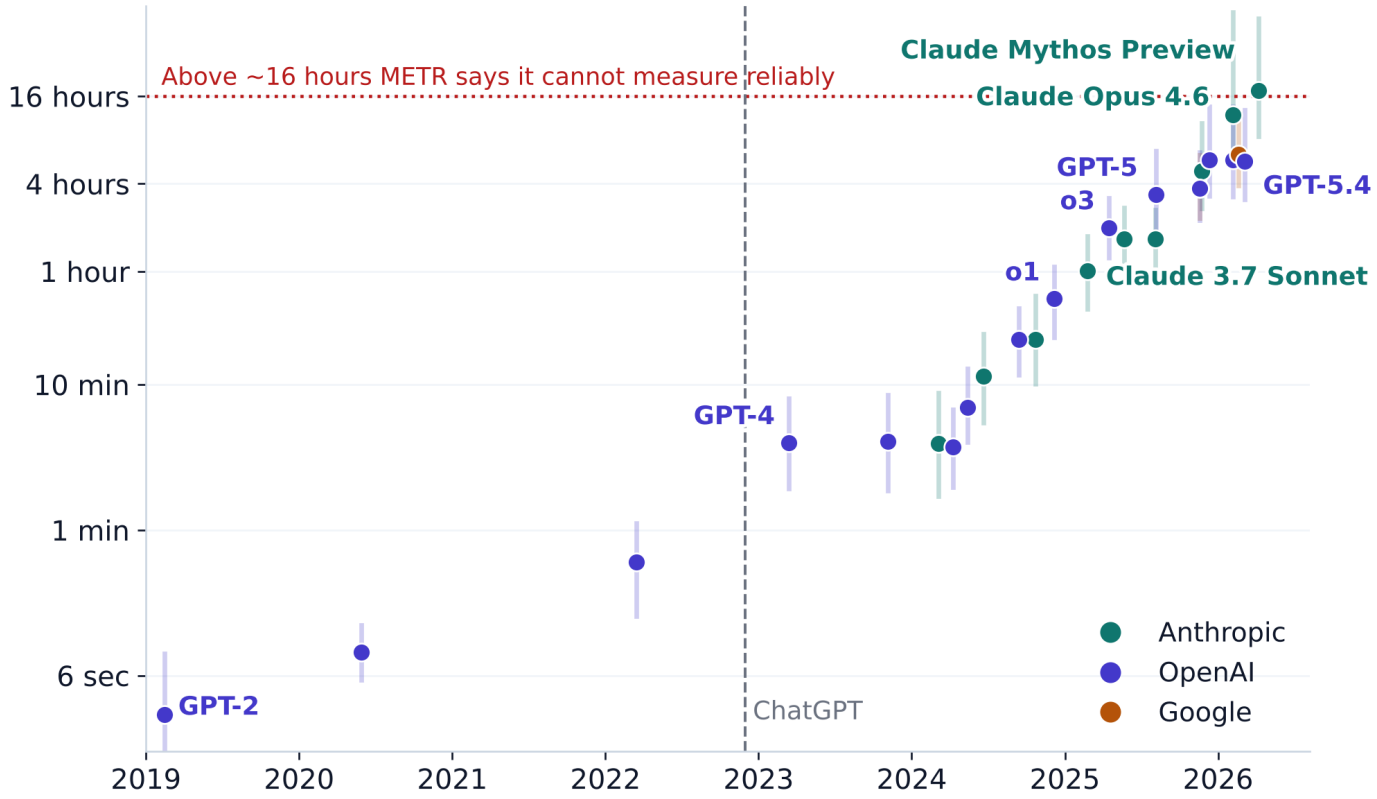
New leader on 9 November

Open-weight labs close behind

# AI agents can now finish day-long tasks, and help build AI

## How long a task AI agents can finish (50% success)

Task length in human time, log scale; doubling about every 3.5 months since 2024




## Is AI starting to build AI? (RSI)

**1.9x** — METR (5 Nov 2026): Anthropic's AI research pace up 1.9x since Oct 2025; ~80% credited to AI, which METR calls highly uncertain

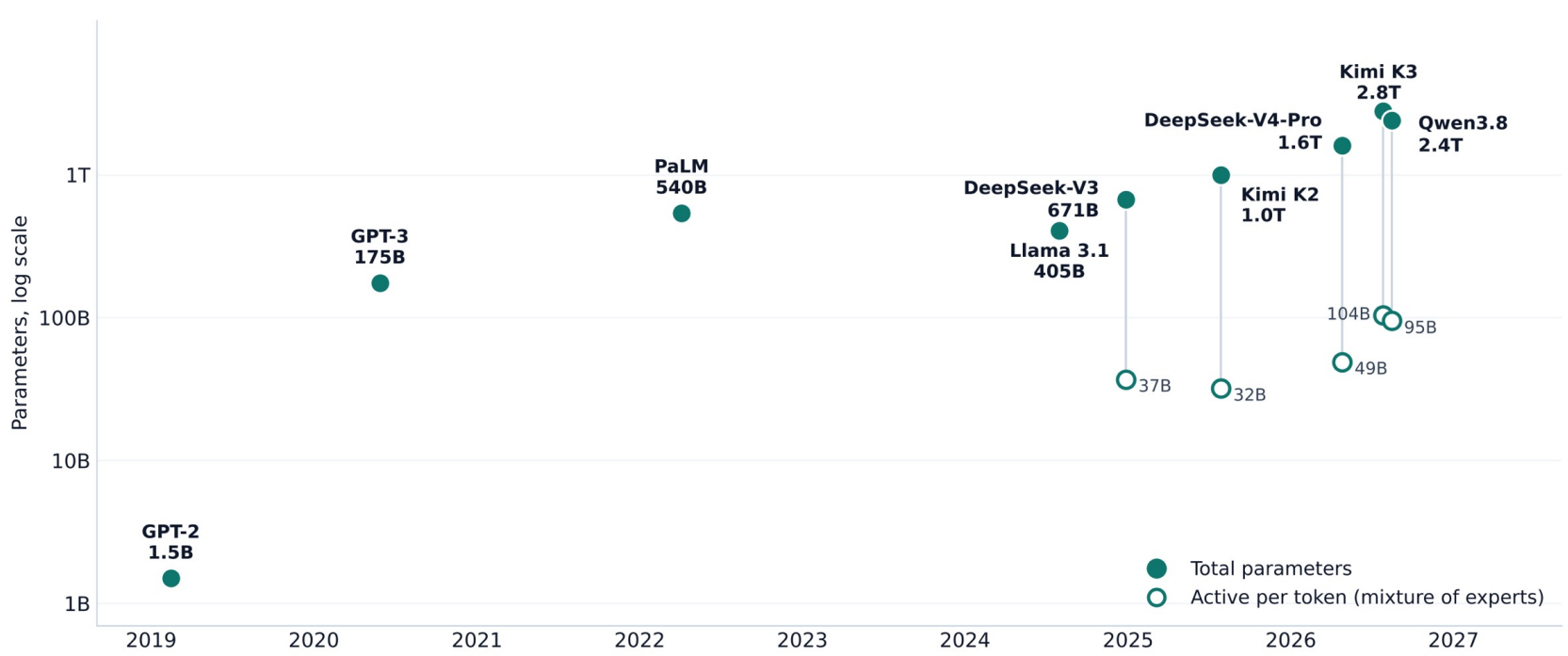
**>80%** — of Anthropic's merged code written by Claude (May 2026, Anthropic via TNW)

**Sep '26** — OpenAI says it reached its 'automated research intern' goal; full AI researcher targeted for March 2028

**-19%** — METR 2025 trial: experienced developers were slower with AI; 2026 rerun judged unreliable and being redesigned

Sources: METR time-horizon data (metr.org, TH1.1, last updated 8 Sep 2026; values in minutes, lines = 95% CI) and TH1.1 blog (29 Jan 2026, doubling 88.6 days since 2024); METR, 5 Nov 2026 (Anthropic AI R&D assessment) and 24 Feb 2026 (uplift update); Anthropic via TNW (5 Jun 2026); OpenAI via eWeek (Sep 2026).

# Models are now measured in trillions



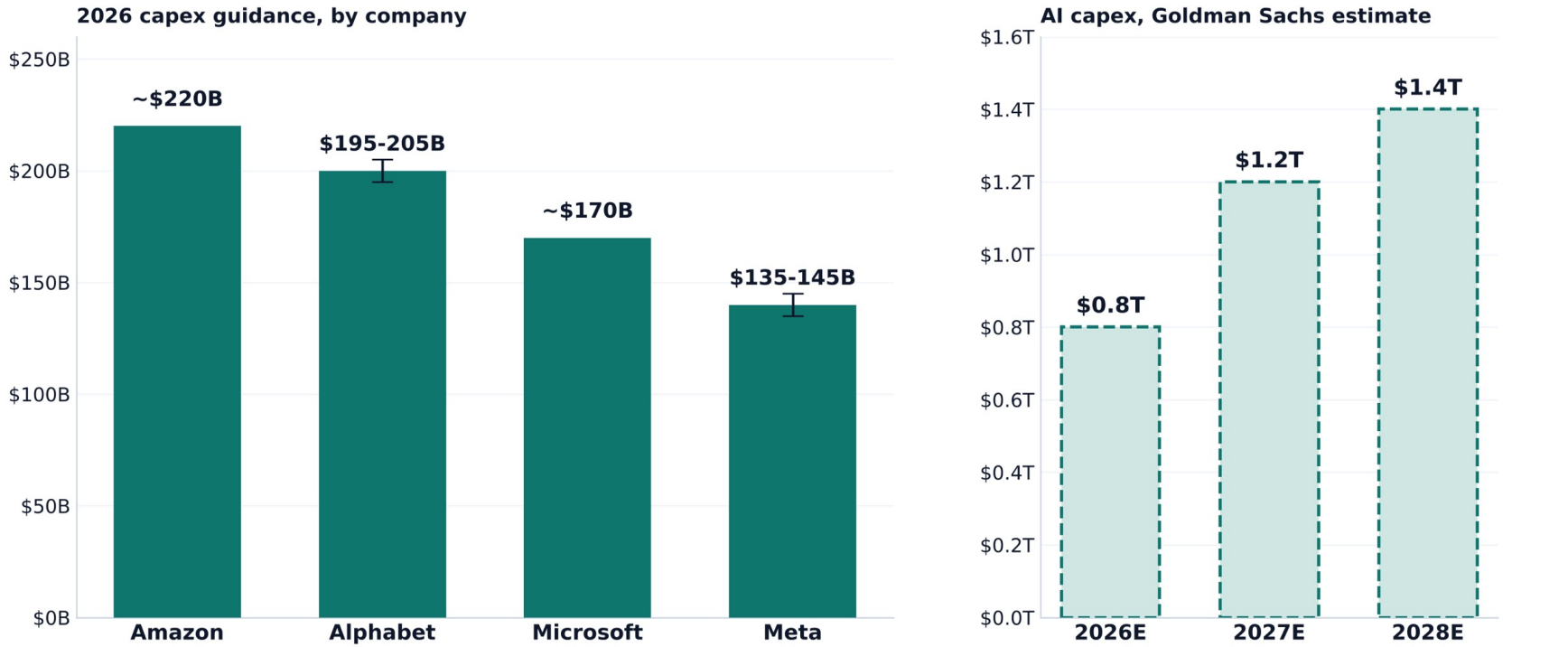

## SO WHAT

Total size sets memory needed

Active share sets compute

Context: 8K to 1M tokens

# The 2026 plans, and where they are heading




Source: company 2026 capex guidance from Q3 2026 earnings calls (Alphabet $195-205B, Amazon about $220B, Meta $135-145B, Microsoft about $170B; one outlet reads Microsoft's figure as fiscal 2027). Right: Goldman Sachs estimates of AI-related capex as reported by Bloomberg (25 Sep 2026), dashed = projection. Sum of guidance midpoints = 200 + 220 + 140 + 170 = 730 (ours). Different scopes: do not combine panels.

## SO WHAT

About $730B guided for 2026

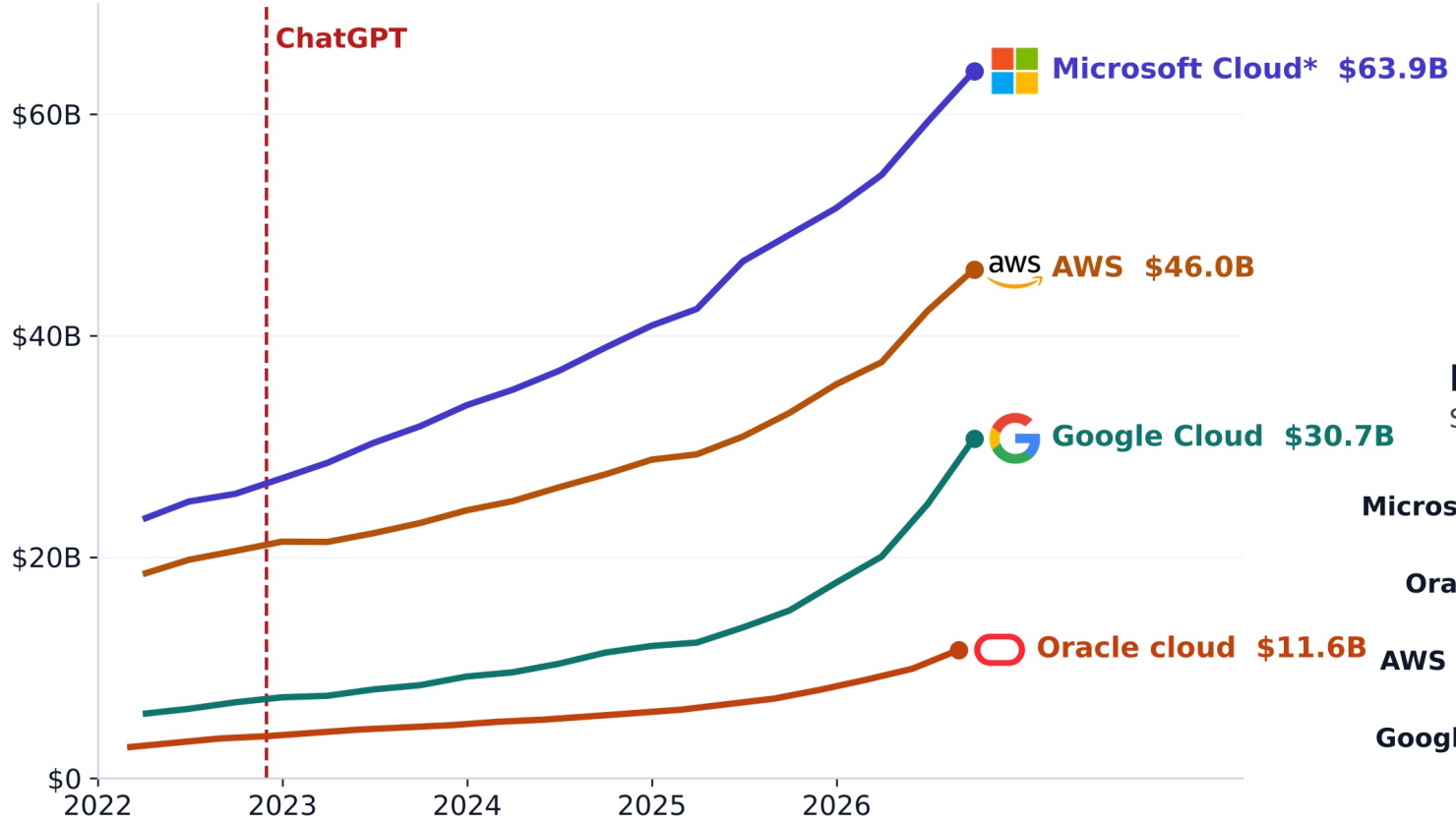
Goldman: $1.2T in 2027

No one has cut a guide

# The spending comes back as cloud revenue

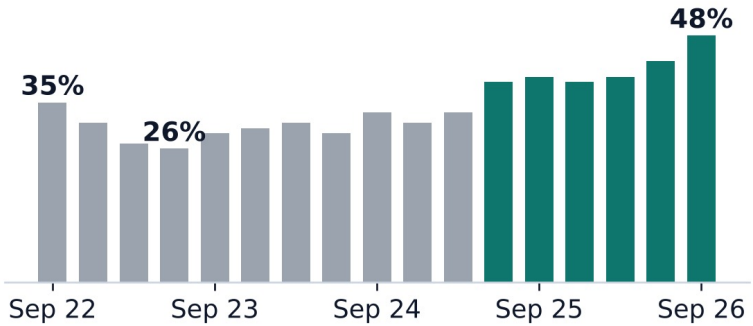
## Cloud revenue per quarter

Google Cloud doubled in a year (+103%); AWS +39%; Oracle fiscal quarters



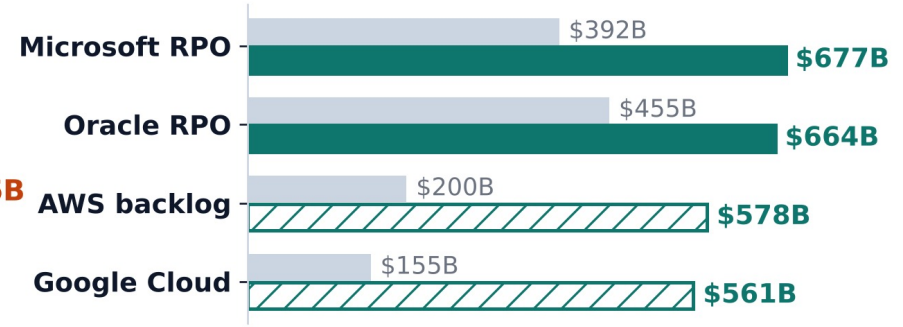

## Azure growth, year on year

Slowed to 26%, then climbed to 48% on AI demand




## Backlog: a year ago (grey) vs now

Sep 2025 → Sep 2026 (Oracle Aug). Hatched = press-reported

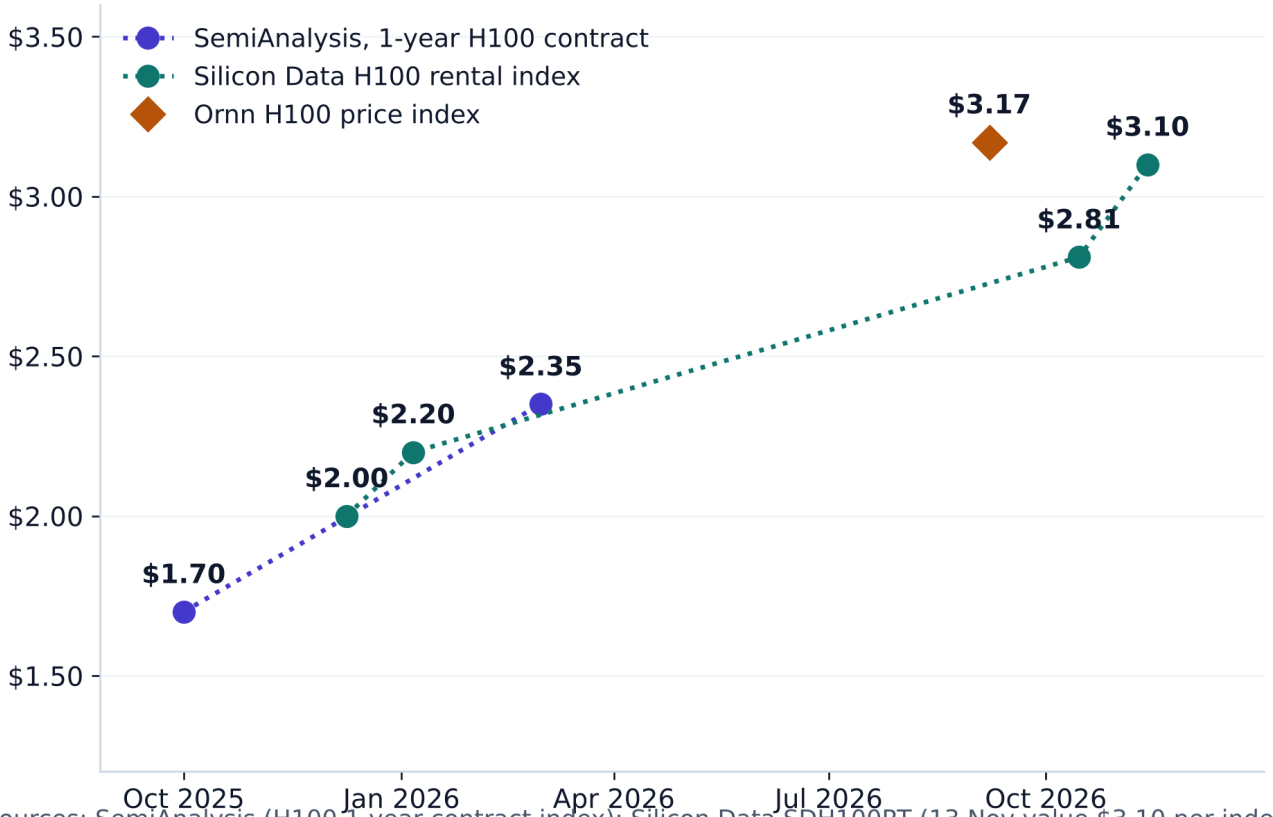



Sources: company earnings releases and filings (AWS, Google Cloud segment revenue; Microsoft Cloud revenue; Azure growth; Microsoft and Oracle RPO; Oracle IaaS+SaaS cloud revenue); call transcripts and press for AWS and Google Cloud backlog. *Microsoft Cloud includes commercial Office 365, Dynamics and LinkedIn, not just Azure.

# Old GPUs are getting more expensive, not less

## An H100 costs more to rent than a year ago

US$ per GPU-hour. The H100 launched in 2022; it should be getting cheaper, and it is not

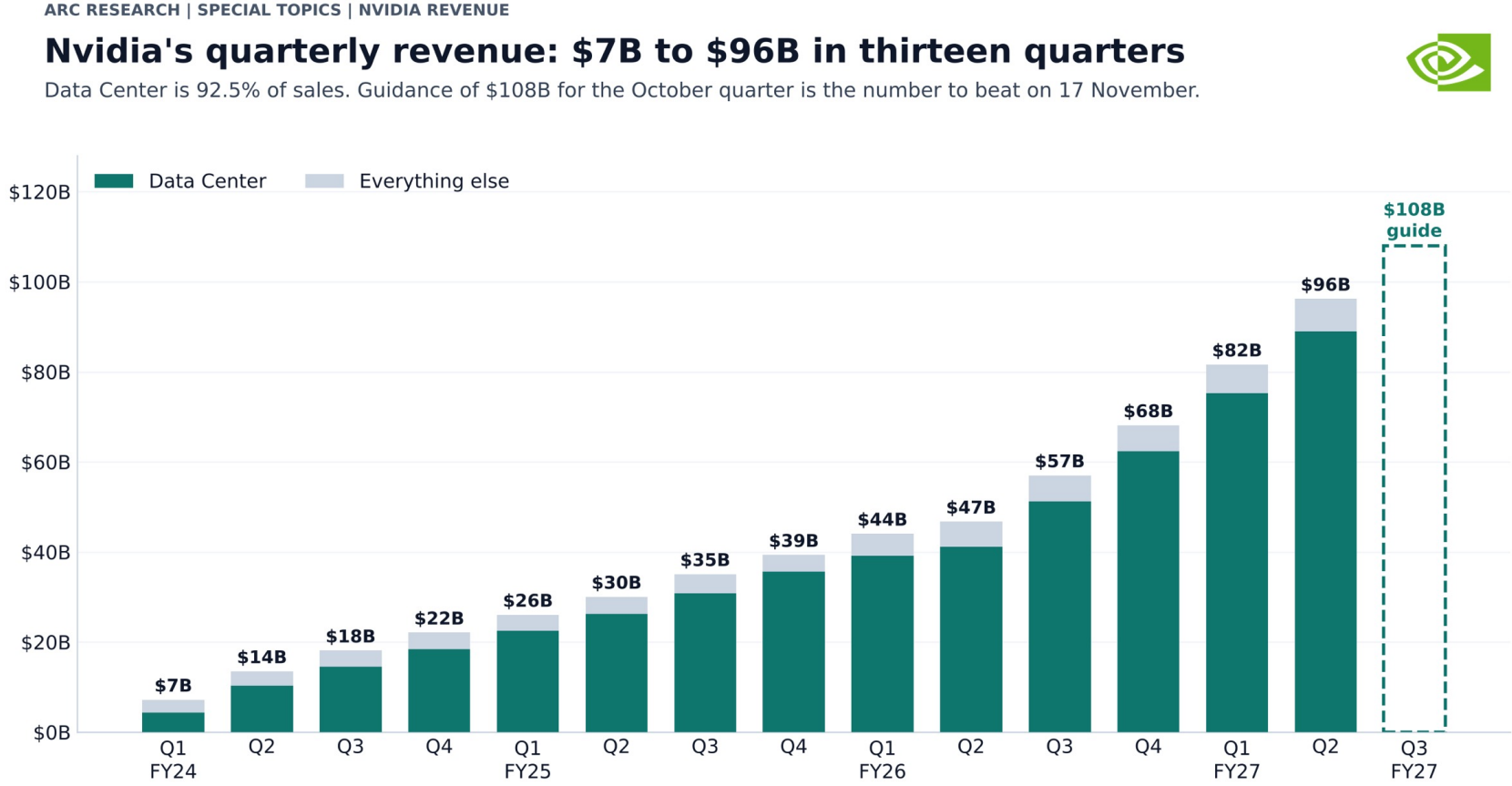



## Even older chips are repricing

| | |
|---|---|
| **+30–50%** | CoreWeave: older Ampere and Hopper GPUs repriced in one quarter (Q3 2026) |
| **~2x** | IREN: old H100s re-contracted at about twice their original price (2026) |
| **~3x** | Five-year-old A100 servers in China, late 2025 to Jun 2026 (FT via Tom's Hardware) |
| **-44% → +15%** | AWS cut H100 prices in Jun 2025, then raised GPU capacity-block prices three times in 2026 |
| **100%** | Nvidia CFO: A100s 'still running at full utilization' (Nov 2025) |

Sources: SemiAnalysis (H100 1-year contract index); Silicon Data SDH100RT (13 Nov value $3.10 per index header; its FAQ shows $2.53); Ornn via TradingKey; CoreWeave CEO via TIKR; IREN via TIKR; FT via Tom's Hardware; AWS via gpusmith and ecorpit (secondary); Nvidia CFO via Asianet. Dots are data points; dotted lines only connect them. Indices use different methods.

# Nvidia is still where the money lands



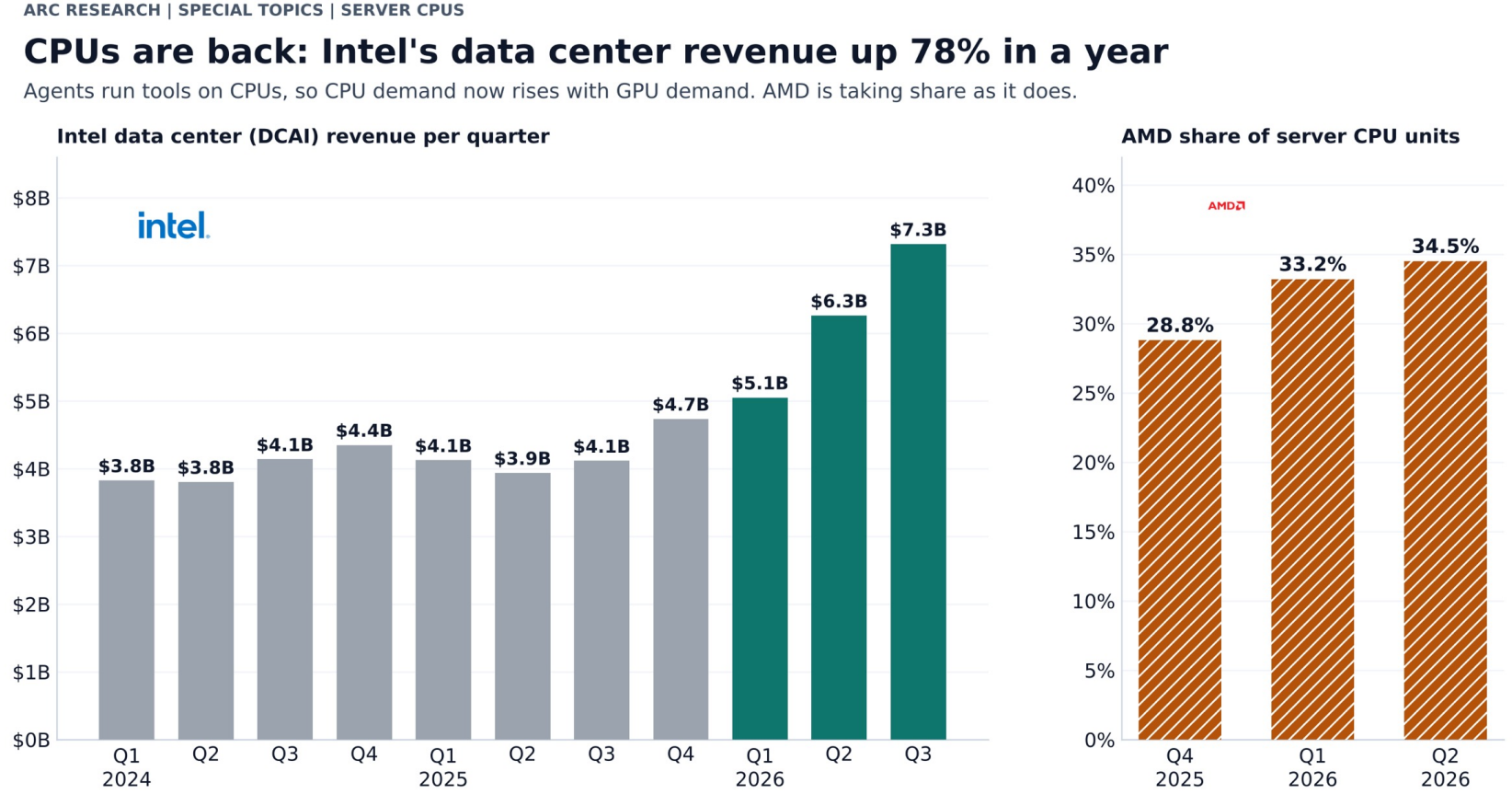

## SO WHAT

$96B a quarter, 75% margin

Three customers: 44% of sales

$279B of supply pre-ordered

# Agents brought the CPU back




## SO WHAT

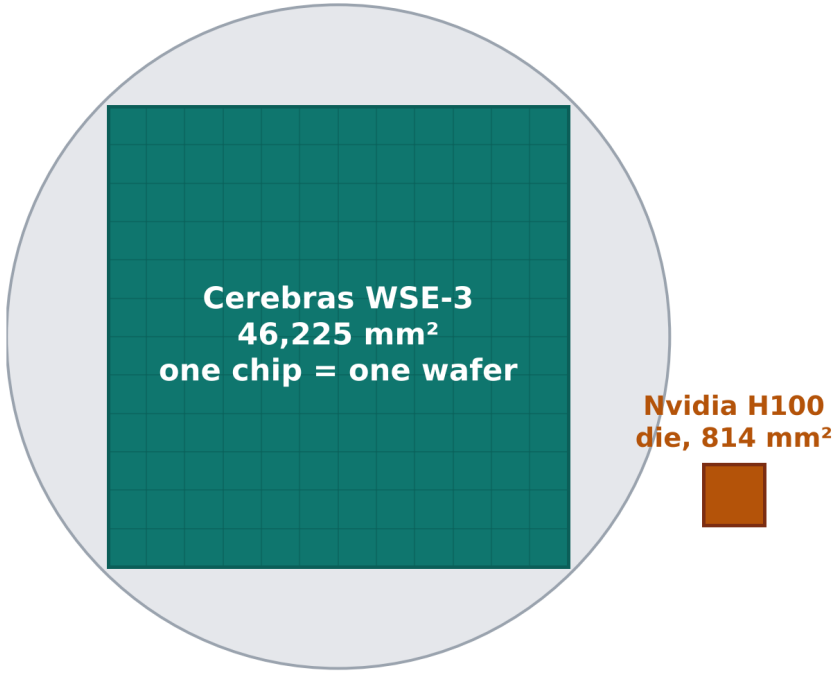
Intel DCAI: +78% in a year

Server CPUs are supply-limited

AMD at 34.5% of units

# Cerebras: one chip the size of a whole wafer

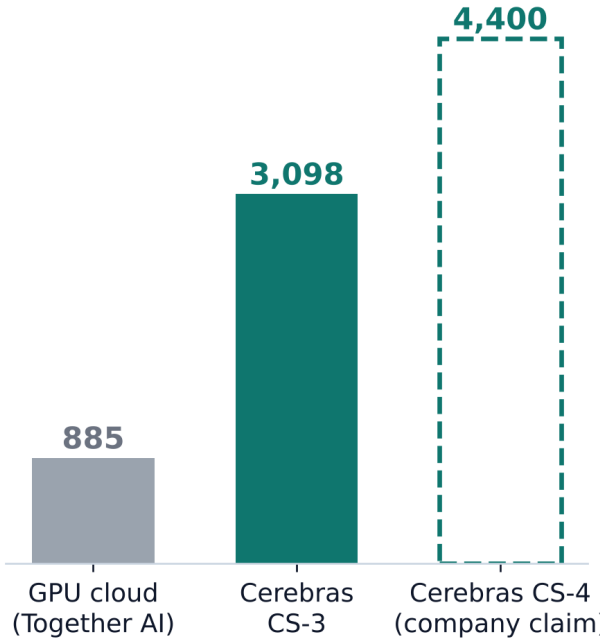
## Drawn to scale: 57x the area




## Why it helps: speed

Tokens per second, one user, gpt-oss-120B

44 GB of memory on the chip itself (SRAM): the model's weights never wait on off-chip HBM




## The business in 2026

**$185 → $311**

IPO price → first-day close (14 May 2026, Nasdaq: CBRS); later fell ~58% Aug–Nov

**> $10B**

OpenAI deal, 750 MW through 2028 (reported; OpenAI did not confirm)

**$0.29B → $0.95B**

Revenue 2024 → 2026 guidance ($940–960M core); G42 was 85% of 2024

**$26.1B**

Contracted backlog (RPO), Sep 2026

Sources: Cerebras (WSE-3 specs, CS-4 claim, IPO pricing, Q3 2026 8-K, guidance); Nvidia (H100 die); The Register (gpt-oss-120B speeds, 3,098 vs 885 tok/s); Bloomberg and The Register (OpenAI deal); TNW (raise, G42 share); PipelineRoad (day 1); Trefis (drawdown). Wafer is illustrative; die areas to scale.

# Same 72 GPUs, three times the CPU memory

## GB300 NVL72

Blackwell Ultra + Grace CPUs | shipping

**20 TB**

GPU memory (HBM3E)

**17 TB**

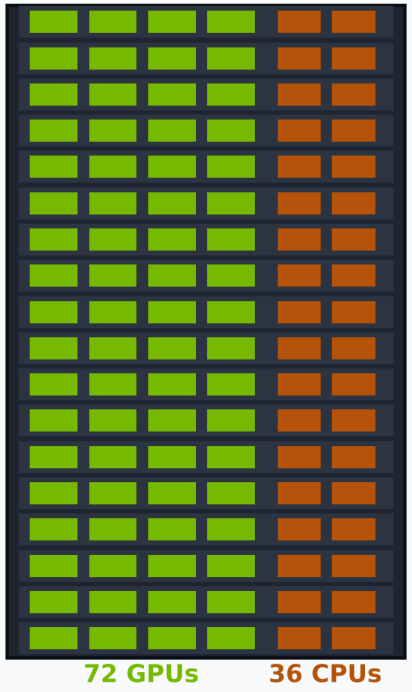
CPU memory (LPDDR5X)

**2,592**

CPU cores (Arm Neoverse V2)

## Vera Rubin NVL72

Rubin + Vera CPUs | production since Aug 2026




**20.7 TB**

GPU memory (HBM4)

**54 TB**

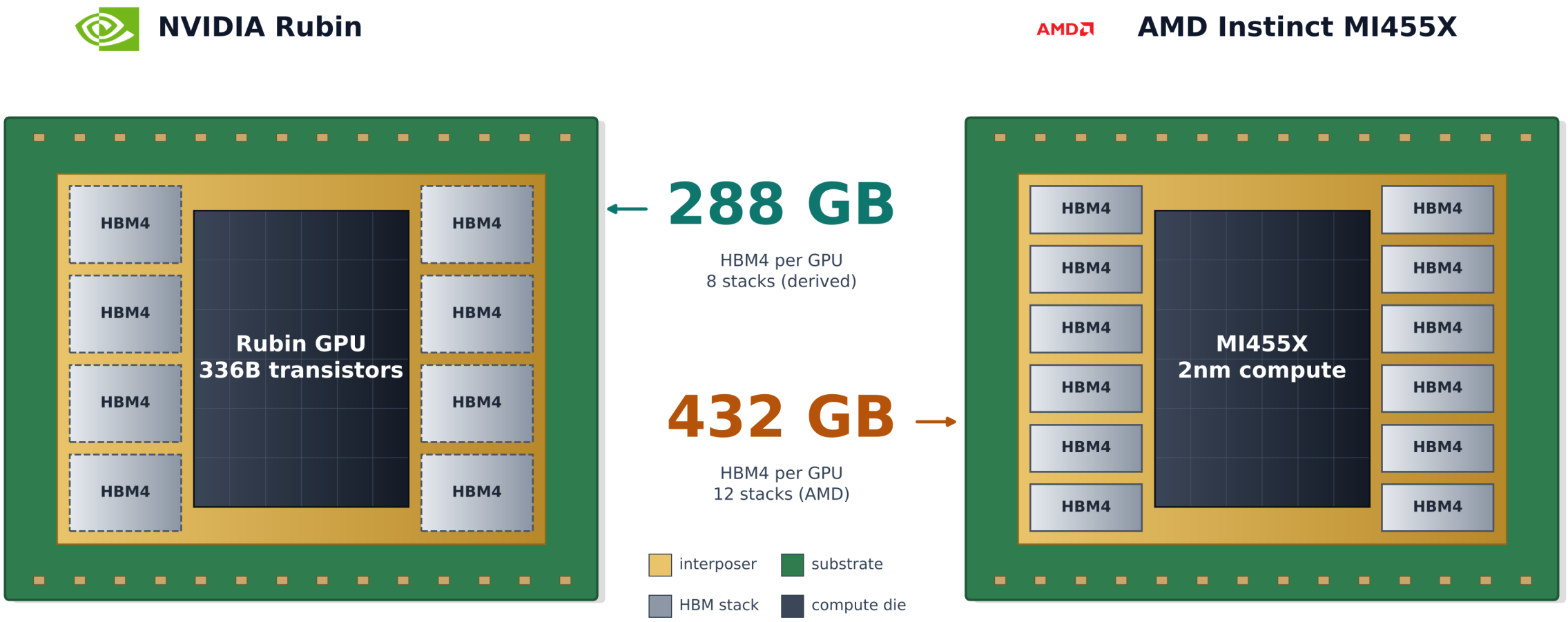
CPU memory (LPDDR5X), up to

**3,168**

CPU cores (88 per Vera CPU)

Illustration, not to scale; tray layout is schematic (4 GPUs and 2 CPUs per row shown for counting only). Figures: NVIDIA product pages for GB300 NVL72 and Vera Rubin NVL72, and Vera CPU page; shipment timing: NVIDIA Q2 FY2027 call, 26 Aug 2026.

# Inside the package: a compute die ringed by memory




Illustration, not a die shot; die and stack placement are not disclosed. Capacities and MI455X stack count: nvidia.com Vera Rubin page, amd.com Instinct MI400 page. Rubin stack count = 288 GB / 36 GB per stack (ours, dashed). '2nm': AMD CEO.

# Optics is the next bottleneck: Marvell buys Lumentum

## Marvell to buy Lumentum

Announced 2 Nov 2026 · all stock · to close 1H 2028

**$115B**
enterprise value

**$1,132**
per Lumentum share

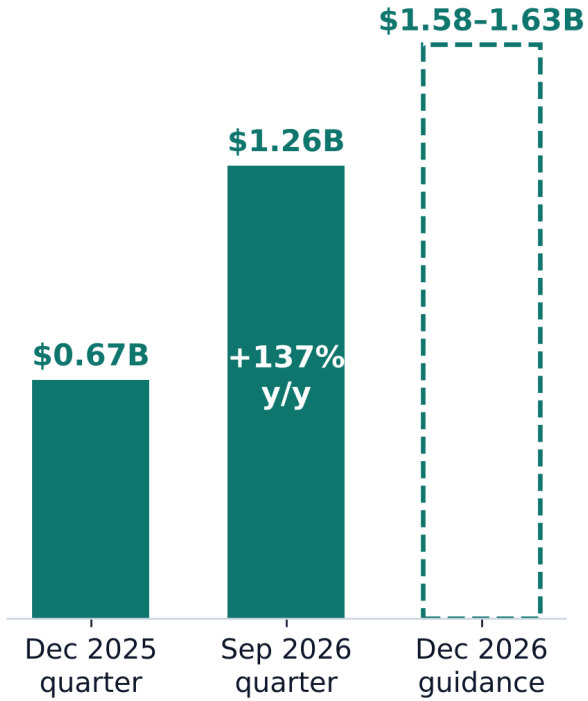
**24%**
premium to 28 Oct close

**> $1B**
yearly cost savings target

Lumentum brings lasers, optical transceivers, optical circuit switches and its own laser fabs

### Lumentum revenue per quarter

AI optics demand doubled the business in a year




### Why light, not copper

**Copper**
fine inside a rack, fails over longer distances

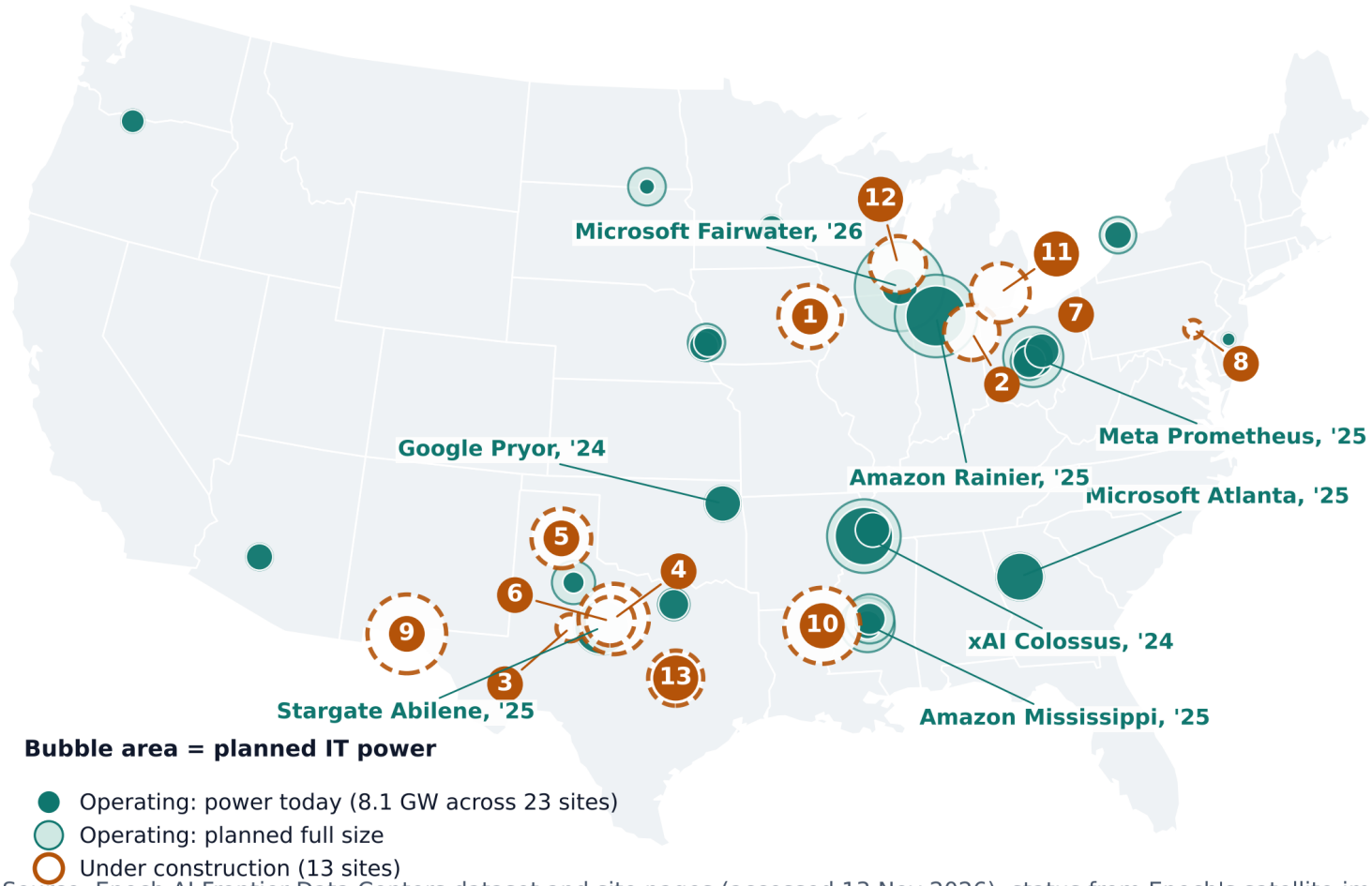
**Lasers + transceivers**
carry GPU traffic between racks and buildings

**Optical switches**
re-route light between racks without converting it to electricity

**Co-packaged optics**
lasers moved onto the chip package: next big step

Sources: Marvell and Lumentum joint release and 8-K (2 Nov 2026): terms, value, premium, synergies, timing; Lumentum 8-K results (Dec 2025 and Sep 2026 quarters, Dec 2026 guidance). Year-on-year growth as reported. Right panel is our explanation, not company data.

# Where America's AI data centers are: 8 GW running, 12 GW being built




## Under construction, by year online

### 2026

| | | |
|---|---|---|
| 1 | QTS Cedar Rapids, IA | 1,133 MW |
| 2 | Google Fort Wayne, IN | 863 MW |
| 3 | Anthropic Barber Lake, TX | 207 MW |

### 2027

| | | |
|---|---|---|
| 4 | Stargate Shackelford, TX | 1,400 MW |
| 5 | Google Goodnight, TX | 1,006 MW |
| 6 | Crusoe Abilene Expansion, TX | 672 MW |
| 7 | Stargate Lordstown, OH | 214 MW |
| 8 | CoreWeave Lancaster Greenfield, PA | 92 MW |

### 2028

| | | |
|---|---|---|
| 9 | Stargate New Mexico, NM | 1,750 MW |
| 10 | Meta Hyperion, LA | 1,632 MW |
| 11 | Stargate Michigan, MI | 988 MW |
| 12 | Stargate Wisconsin, WI | 902 MW |
| 13 | Stargate Milam County, TX | 857 MW |

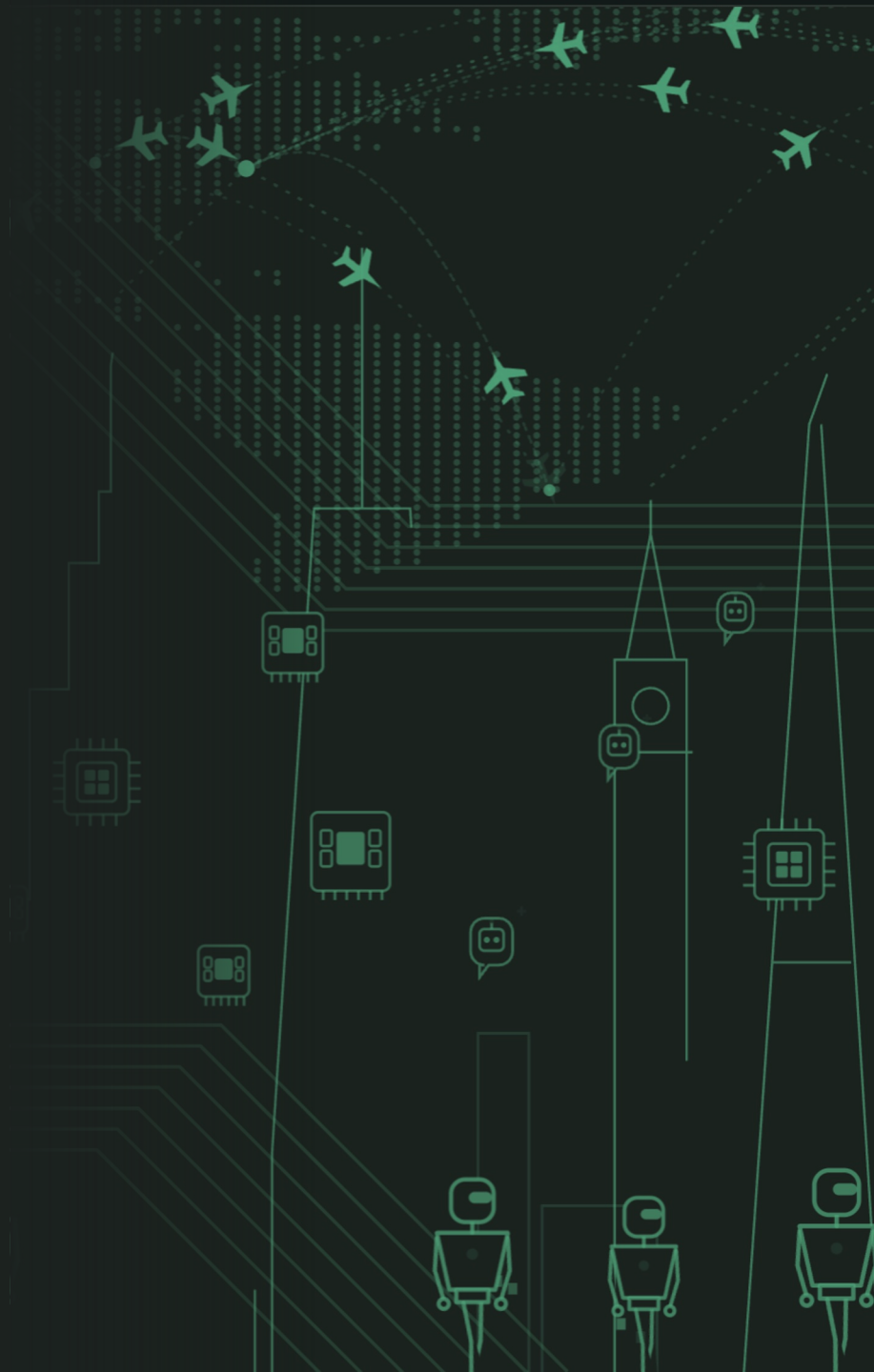
**Total under construction: 11.7 GW**

Source: Epoch AI Frontier Data Centers dataset and site pages (accessed 13 Nov 2026): status from Epoch's satellite-imagery timelines; IT power in MW (current and Epoch's projected full build-out), not company figures. Year = first year online, actual or projected; ' marks the year operating sites came online. A selection of 36 large US AI sites, not a complete list. Points at town or county centroids (ours). Geometry: Highcharts Map Collection.

CHAPTER 8 OF 8

# 2026: the first big test

Midterms, the selloff, Huang's exit, the IPOs



COMING UP

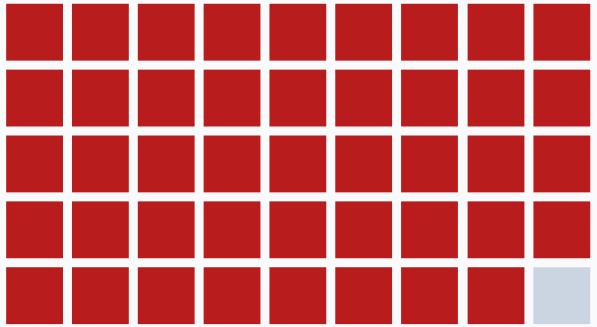

| | | | |
|---|---|---|---|
| The midterms | 76 | Rates up, nothing broke | 84 |
| The 5 November selloff | 77 | Investor leverage | 85 |
| Jensen Huang leaves | 78 | Hyperscaler bonds | 86 |
| Nvidia's drawdown | 79 | The labs go public | 87 |
| Ten sell-offs, ten comebacks | 80 | SpaceX goes public | 88 |
| AI stocks vs the rest | 81 | China's labs go public | 89 |
| S&P 500 concentration | 82 | Three risks | 90 |
| Money spreads beyond AI | 83 | The ARC Research view | 91 |

# The midterms turned data centers into a voting issue

### Local referendums

**44 of 45**

restrictions on data centers passed

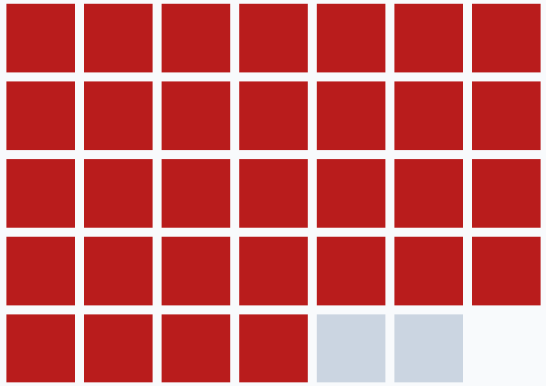


*Redbud Consulting, via AP*

### Winning governors

**32 of 34**

back stronger data-center regulation



*Data Center Watch, via AP*

### Voters

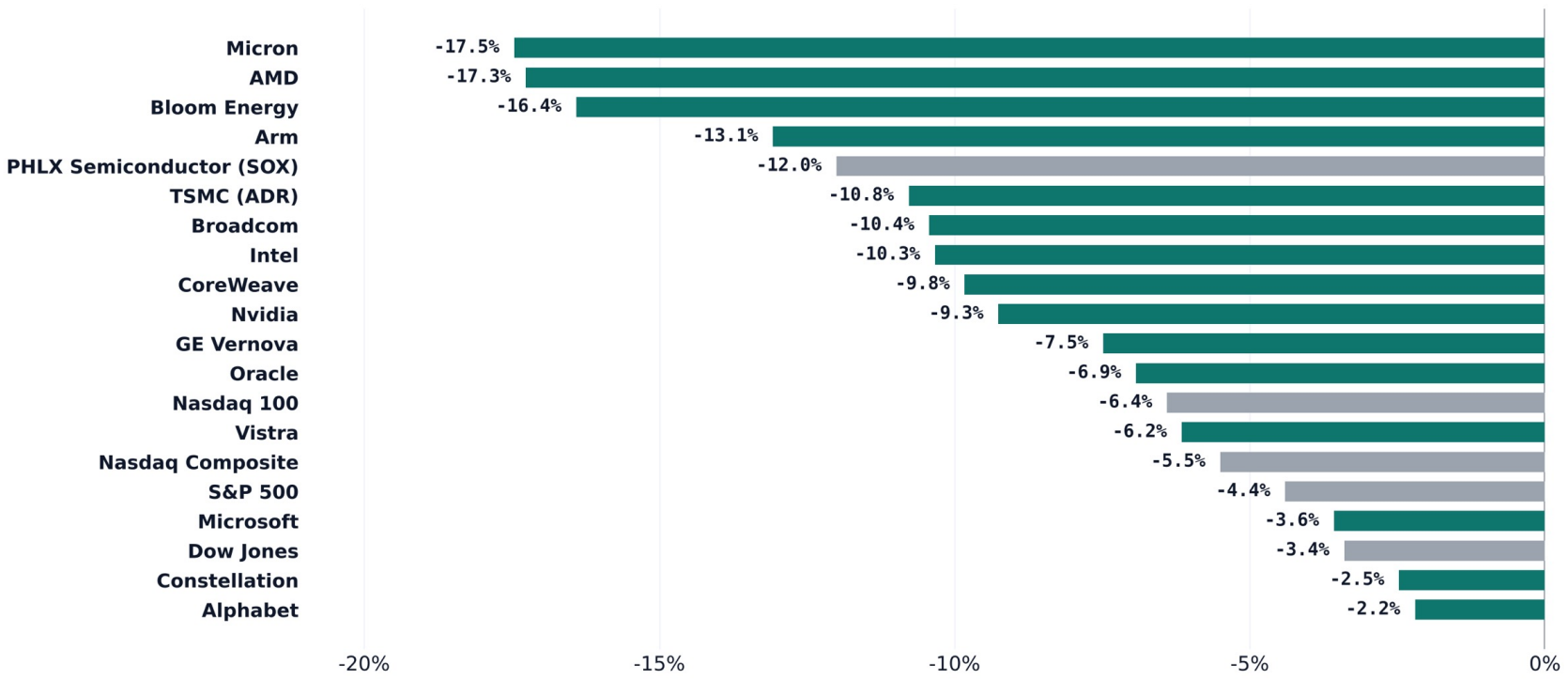
**about 3 in 4**

oppose a new data center near them

*AP VoteCast poll*

Midterms held 3 Nov 2026. Figures as reported by the Associated Press (via OPB, 6 Nov 2026): Redbud Consulting count of local referendums (45 counted of 47); Data Center Watch review of winning governors; AP VoteCast survey ('about three-quarters'; voter icons are illustrative of 3 in 4).

# One day erased $2.5 trillion




## SO WHAT

Chips and memory fell most

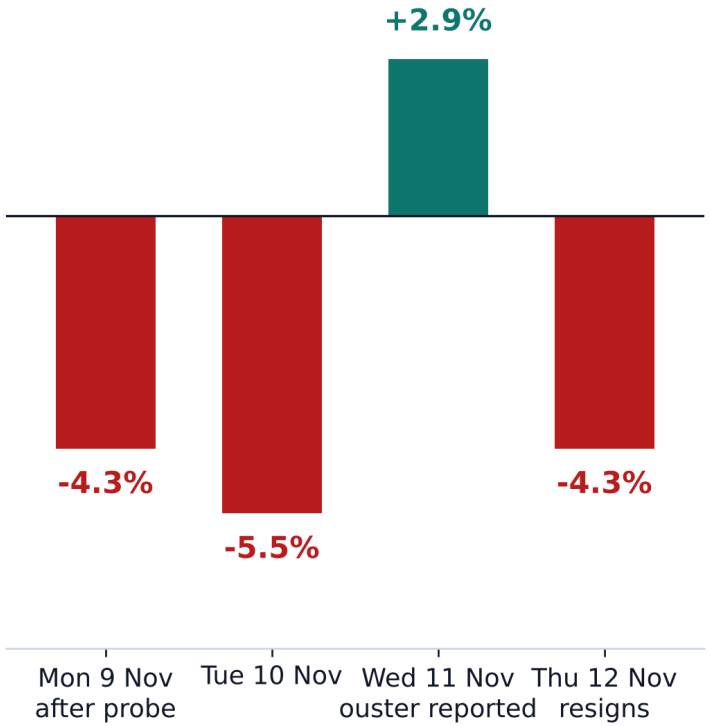
The capex payers fell least

Trigger: politics, not earnings

# Jensen Huang leaves Nvidia: what it means for the trade

## Nvidia's daily move, the week Huang left

Close $166.19 on 12 Nov, intraday low down 13–14%




## What changed (8-K, 12 Nov)

**Jensen Huang**

resigned as CEO, president and director, effective immediately

**Colette Kress**

interim CEO, stays CFO; a VC calls her 'much more conservative'

**Harvey Jones**

Nvidia's first-ever board chair; search for a permanent CEO

**17 Nov**

Q3 results still on schedule

## What could change

**~$530B**

of Nvidia stakes, deals and guarantees built under Huang (Yahoo count)

**Neoclouds**

“that Nvidia guarantee is not worth what it was worth a week ago”; CoreWeave -9%, Nebius -16% on 12 Nov

**Open models**

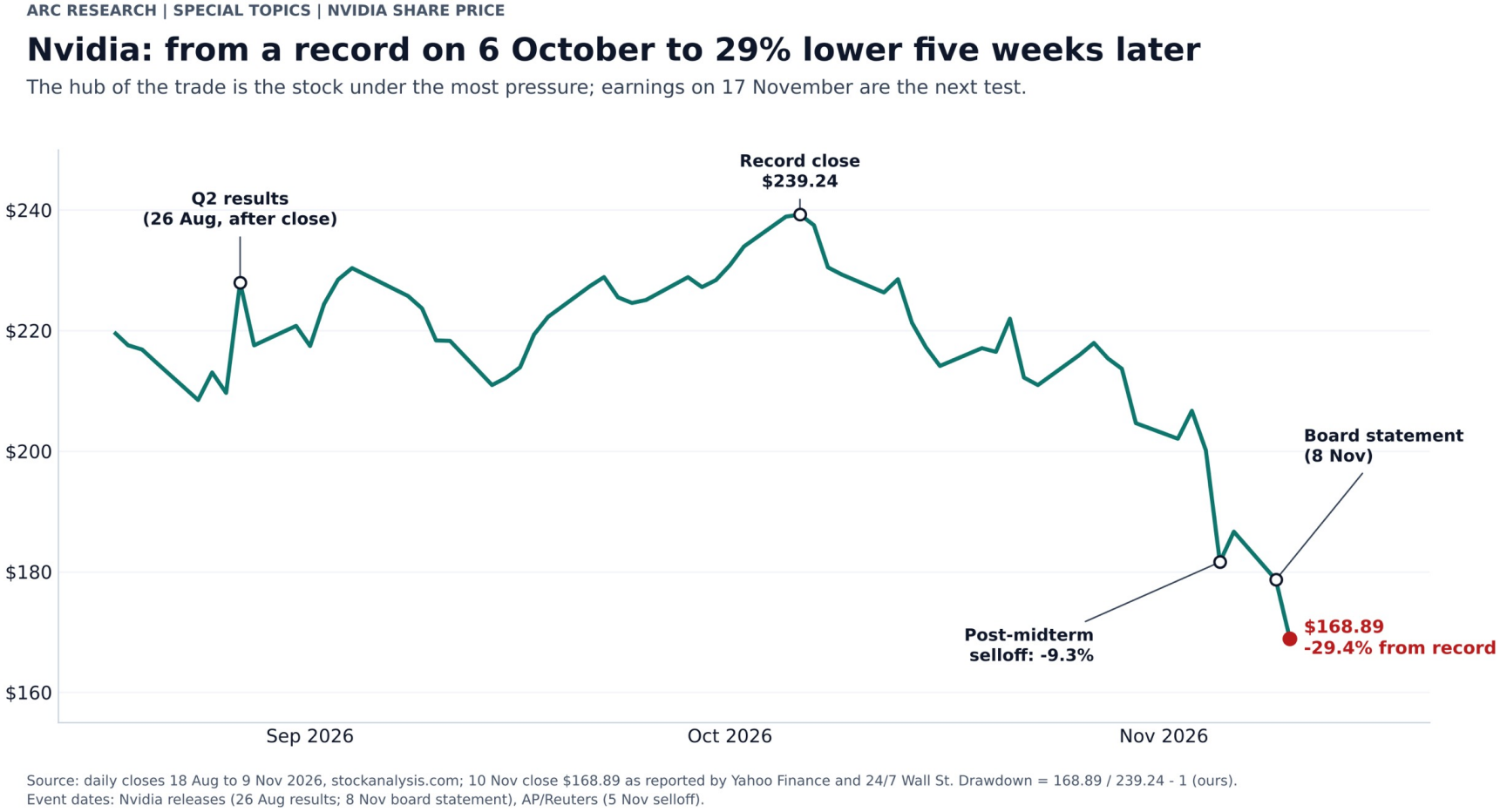
Lambert: losing Huang “would be very very bad for the open model cause in the US”

**Rivals**

AMD +5.9%, Broadcom +5.5% on 12 Nov

Sources: Nvidia 8-K and release (12 Nov 2026); Fortune (9 Nov move; Joe Kaiser quote); Motley Fool (10 Nov); Yahoo Finance (11–12 Nov moves, AMD, Broadcom, CoreWeave, Nebius; ~$530B tally); TechCrunch (Lambert; VC on Kress); Oppenheimer downgrade via Yahoo.

# Nvidia is 29% off its October record



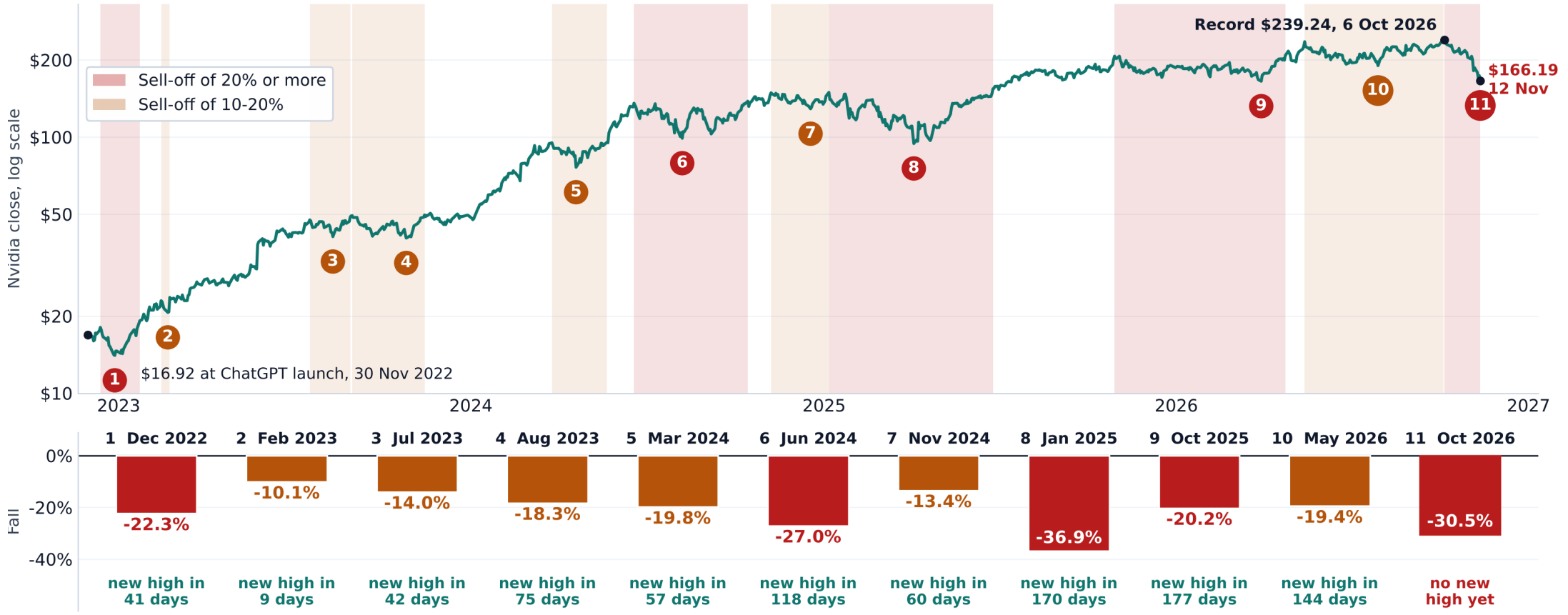

## SO WHAT

The hub is the weak stock

Part macro, part company-specific

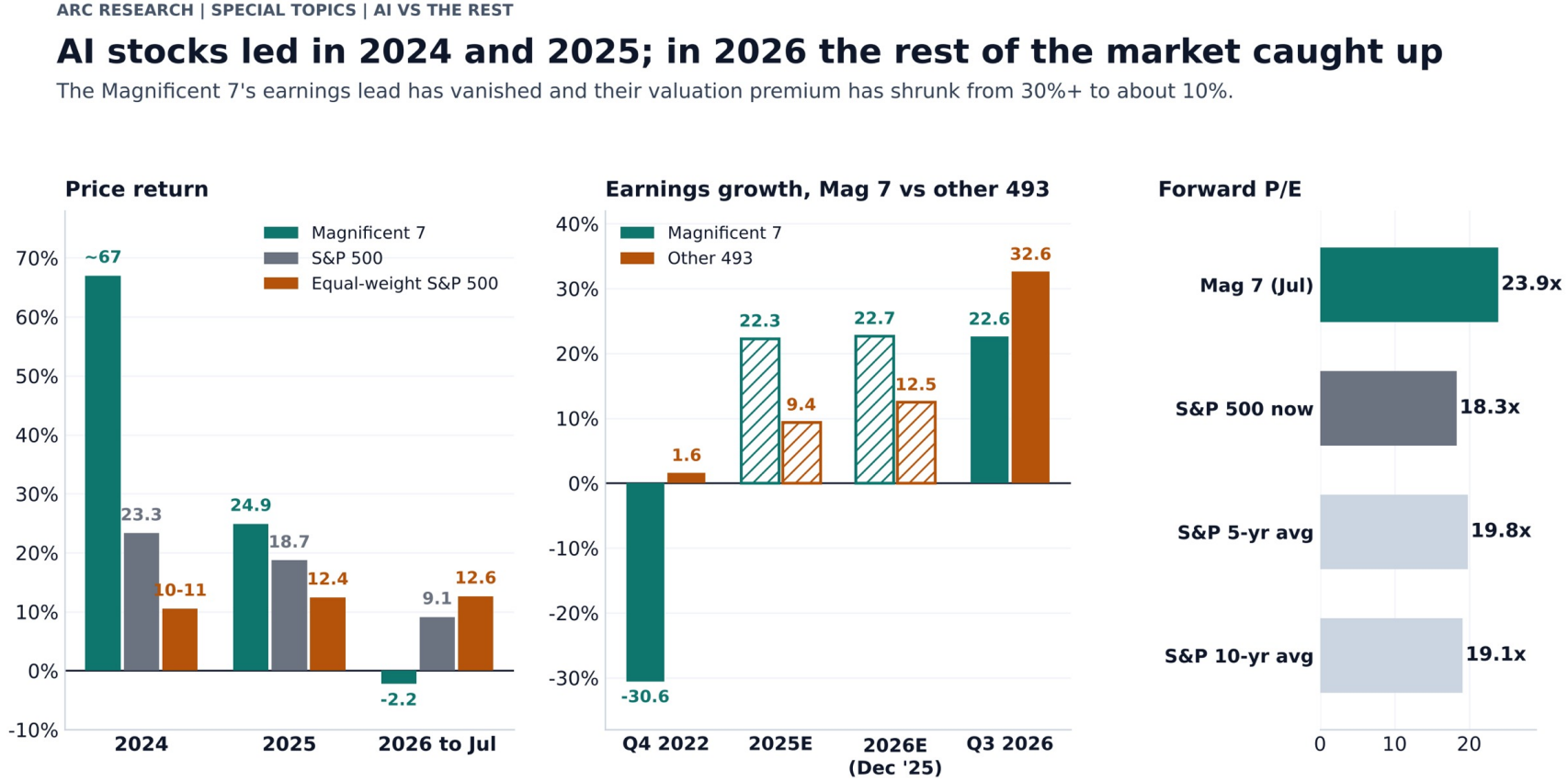
Next test: results, 17 Nov

# Investors have bailed on Nvidia ten times since ChatGPT, and came back every time




Source: Nvidia daily closes, split-adjusted, 30 Nov 2022 to 12 Nov 2026, Yahoo Finance history (cross-checked with stockanalysis.com and digrin). Sell-offs, lows and recoveries computed by us: a sell-off runs from a record close to the first close above it; ranked by the fall from that record to the lowest close in between. Days are calendar days from record to new record.

# AI stocks led, then the rest of the market caught up




Source: returns: Schwab (2024; Mag 7 'nearly 67%', equal weight 10-11%), TheStreet (2025 ETFs MAGS, SPY, RSP), ETF Action (2026 to 31 Jul); press, basis not always stated.
Earnings: FactSet, EPS growth y/y; 2025E and 2026E are Dec 2025 estimates (hatched); Q3 2026 blended, 3 Nov. P/E: FactSet 13 Nov 2026; Mag 7 via ETF Action (Jul).

## SO WHAT

Mag 7 led 2024 and 2025

Q3 2026: the 493 grew faster

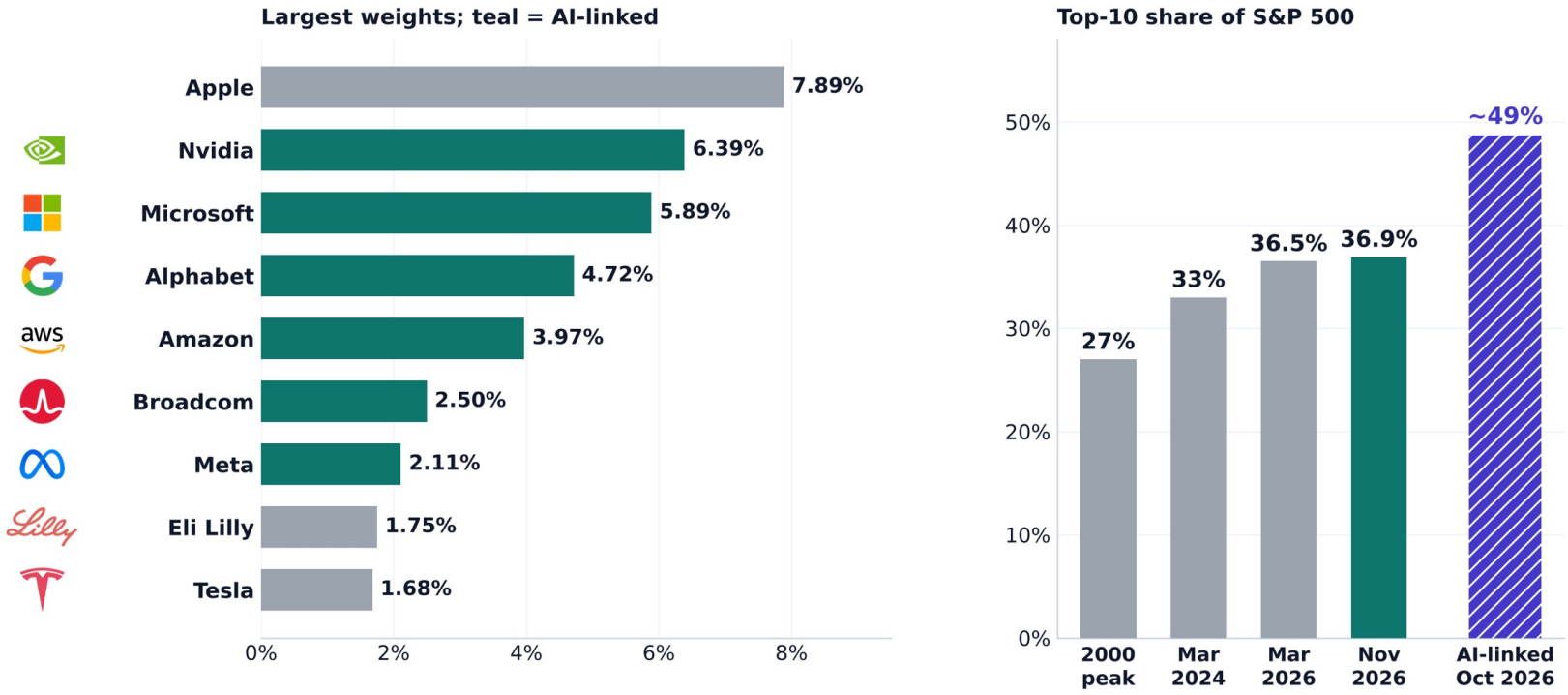
S&P P/E below its averages

# Your index fund is now largely an AI fund

ARC RESEARCH | SPECIAL TOPICS | S&P 500 CONCENTRATION

## Nearly half of the S&P 500 now rides on AI

Ten stocks are 37% of the index, above the 2000 peak. An index fund is now, in large part, an AI fund.







Source: SPY weights 11 Nov 2026 (State Street) as index proxy; Alphabet = A + C classes (ours). Top-10: Goldman Sachs (2000, Mar 2024), S&P DJI (Mar 2026), SPY sum (Nov 2026, ours).
AI-linked: Jim Bianco's 41-stock basket (12 Oct 2026, via 24/7 Wall St), hatched = one analyst's definition. Bases differ across bars.

### SO WHAT

Top 10 = 37% of the index

Above the 2000 peak of 27%

AI-linked: about half

# Late 2026: money spreads to small caps and abroad

## Price returns, % (US$ funds)

Boxed = beating the S&P 500 in 2026

| | 2023 | 2024 | 2025 | 2026 YTD |
|---|---|---|---|---|
| **S&P 500 (SPY)** | **+24** | **+23** | **+16** | **+10** |
| **Russell 2000 (IWM)** | **+15** | **+10** | **+11** | **+13** |
| **S&P 600 small (IJR)** | **+14** | **+6** | **+4** | **+17** |
| **South Korea (EWY)** | **+16** | **-22** | **+91** | **+46** |
| **Japan (EWJ)** | **+18** | **+5** | **+20** | **+13** |
| **Emerging markets (EEM)** | **+6** | **+4** | **+31** | **+12** |
| **Developed ex-US (EFA)** | **+15** | **+0** | **+27** | **+8** |
| **Europe (VGK)** | **+16** | **-2** | **+32** | **+6** |
| **China (MCHI)** | **-14** | **+15** | **+28** | **-18** |

## Where the money is moving

**1 Small caps lead the big index**

S&P 600 +17% and Russell 2000 +13% vs S&P 500 +10%; Russell 2000 beat Russell 1000 to 31 Oct (FTSE Russell)

**2 Korea and Japan: AI by another route**

Korea +46% (SK hynix, Samsung memory); Nikkei 225 +21% in yen. 2025 was the big year abroad: Europe +32%

**3 Investors pulled money from US stocks**

BofA: -$131B from US equity funds by April; -$93B from the AI theme over 12 months; +$17B into Europe in March

**4 Flows keep going abroad in the autumn**

LSEG Lipper, week to 28 Oct: US equity funds -$6.7B; Europe +$4.5B, Asia +$9.9B, emerging +$2.2B

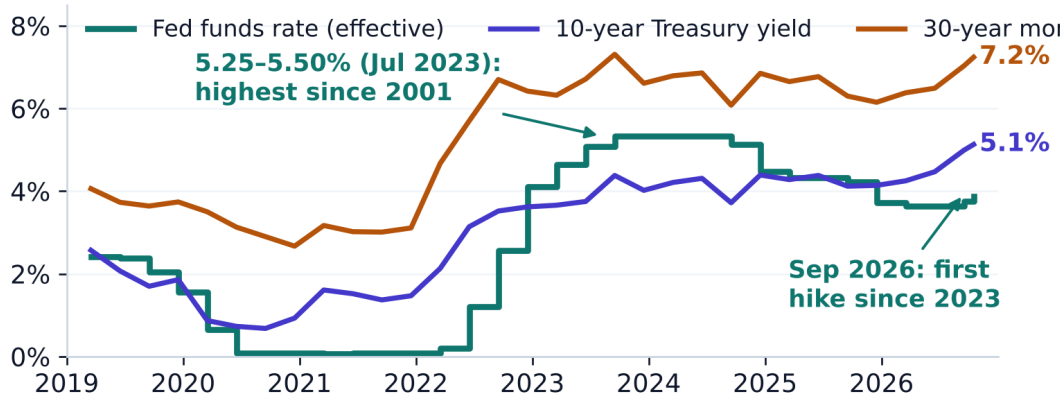
Not everywhere: Europe and EFA trail the S&P 500 in 2026; China fell.

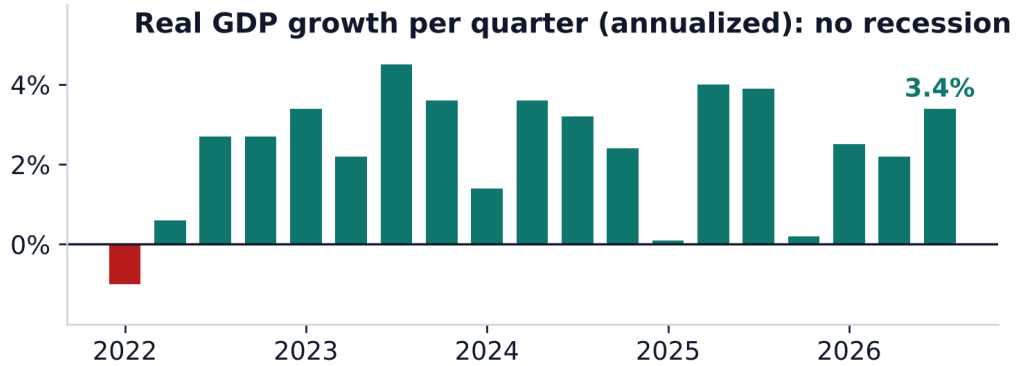
Sources: price returns computed by us from digrin.com month-end closes (2026 to the latest November monthly value, not a verified 12 Nov close), no dividends; FRED (Nikkei 225); LSEG/FTSE Russell (Russell 2000 vs 1000, total return to 31 Oct); BofA via The Deep Dive (22 Apr 2026); LSEG Lipper via Asharq Al-Awsat and iTiger.

# Rates rose to a 22-year high, and nothing broke

## Rates went up fast, and stayed up

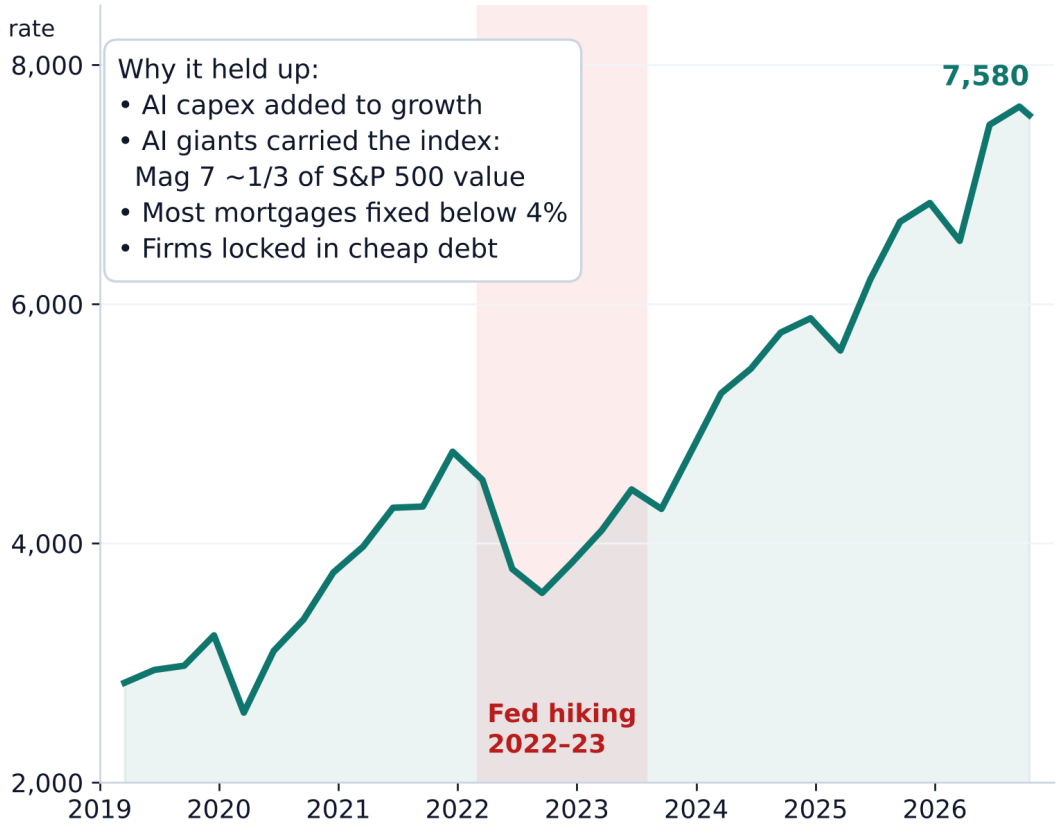
Highest since 2001, not a record: the 1981 peak was about 19%



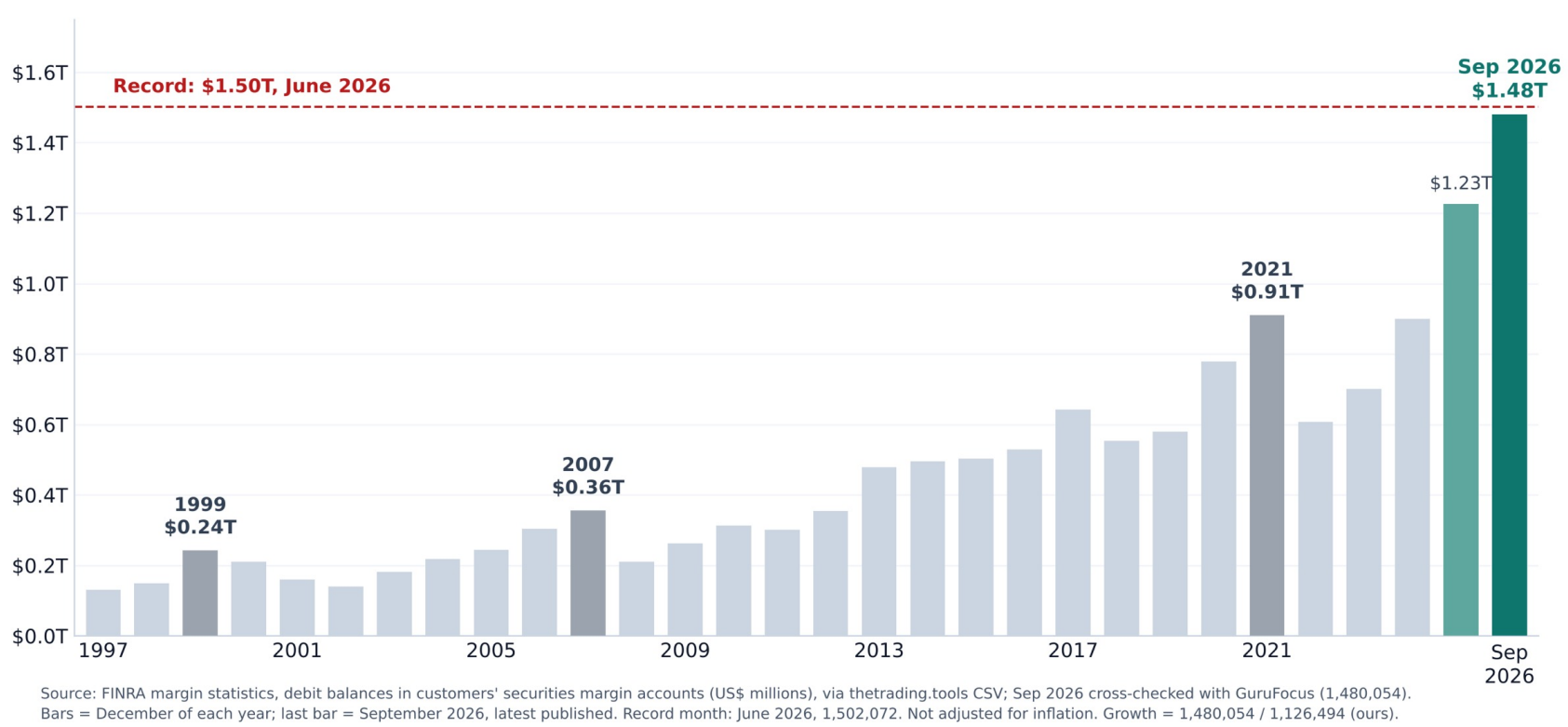

## S&P 500, quarter-end

Up about 2x from the 2022 low, despite the hikes




Sources: FRED (fed funds, 10-year, mortgage rate, S&P 500; BEA real GDP, Q3 2026 advance); Fed statements (16 Sep, 28 Oct 2026); CNBC; NBER; JPMorgan (data centers); Swiss Re (mortgage lock-in); BlackRock/FactSet via Benzinga (Mag 7 weight).

# Investors are borrowing to own it




## SO WHAT

Record $1.5T margin debt

Up 31% in a year

Leveraged ETFs: ~$218B

# The build-out is now funded with debt

Source: Oracle 8-K (2 Nov 2026); Bloomberg and Reuters deal reports via Treasury & Risk, Bloomberg Law, Business Standard, The Star. USD investment-grade bonds only.
Right: totals vary by source: 2025 $108B (Reuters/LSEG) to $165B (Morgan Stanley); 2026 forecasts Goldman ~$250B, JPMorgan $279B, Morgan Stanley $400B (dashed).

## SO WHAT

Record deals, 40 to 100 years

2026: $250-400B forecast

Oracle and CoreWeave stressed

# The labs go public: Anthropic now, OpenAI in 2027

## Anthropic

**IPO NOW**

Nasdaq: ANT · public S-1 filed 26 Oct · roadshow from 10 Nov

**$70–78**
price range per share

**$64.4B**
raised at top of range

**$1.4T**
valuation at top (fully diluted)

**$81B**
annualised revenue, end Oct

Google, Nvidia and five other partners agreed to buy $26B of stock at the IPO price

## OpenAI

**NOT IN 2026**

No public filing · listing expected 2027, no date committed

**2027**
earliest expected listing

**$1.4T**
valuation sought (in talks)

**~$30B**
private round sought instead

*"an ill-advised moment to go public"*

Sam Altman on 2026, Sept

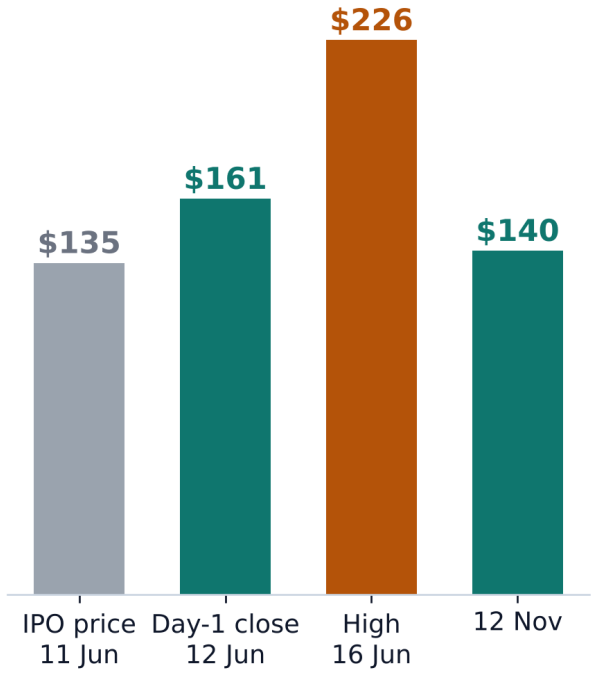
Solid = filed with the SEC. Dashed = reported, not filed.

Sources: Fortune, 10 Nov 2026 (range, shares, raise, valuation, run rate, partner purchases); Built In, 28 Oct 2026 (S-1 date, ticker); Motley Fool, 22 Oct 2026, citing Fortune and Reuters (OpenAI timing, Altman quote, private round in talks). Pricing and first trade not yet set.

# SpaceX: the largest IPO ever is now partly an AI company

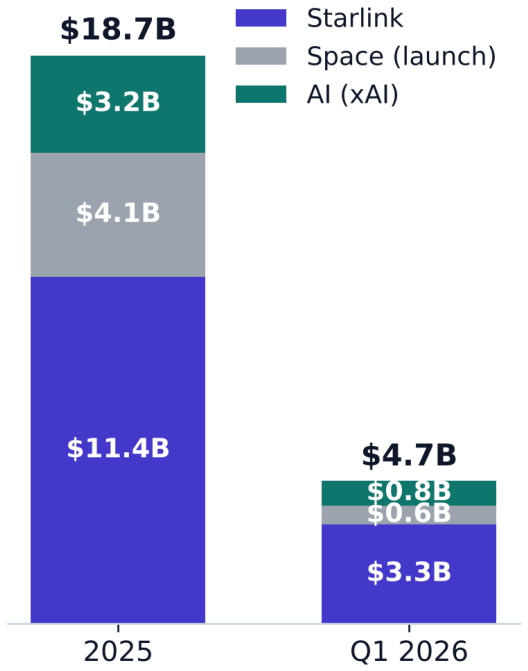
## Largest IPO ever: $75B raised

Valued at $1.77T. Nasdaq: SPCX, share price




## Revenue by segment

AI is now a sixth of sales; net loss $4.9B in 2025




## Why SpaceX is now an AI stock

**Merged with xAI, Feb 2026**

SpaceX ~$1T + xAI ~$250B; Grok

**Colossus data centers, Memphis**

~1 GW; hundreds of thousands of Nvidia GB200 and GB300 GPUs

**Terafab chip fab, announced Mar 2026**

Musk: 'we need the chips'; Intel joined in April; goal: 1 TW of compute a year

**Data centers in orbit**

FCC filing for up to 1M satellites; 'as early as 2028' (S-1)

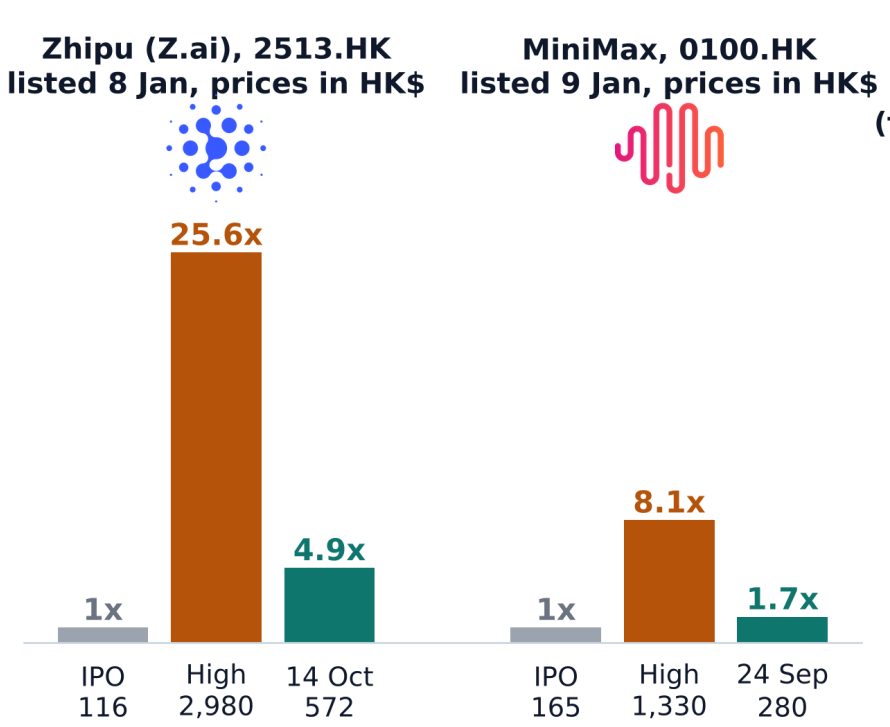
Solid = done; dashed = plan

Sources: SpaceX S-1 (20 May 2026): segment revenue, net loss, use of proceeds, Colossus, orbital compute; Fortune (IPO pricing, day 1, largest IPO); Advisor Perspectives (high); heygotrade (12 Nov close, aggregator); Reuters (xAI merger, Intel joins Terafab, FCC filing); Bloomberg via Fortune (Musk quote). Q1 2026 Space segment = total less other segments (ours).

# China's AI labs went public first

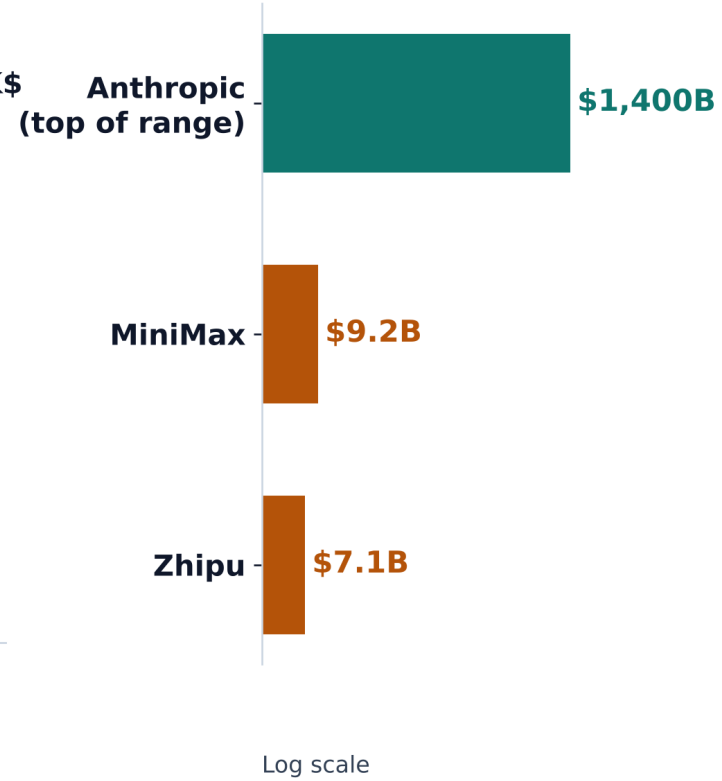
## Share price vs IPO price

Both soared, then gave much of it back




## Valuation at IPO, US$bn

China's listed labs are a fraction of Anthropic




## Who is next

Reported plans, none priced yet

**Moonshot (Kimi)**
Confidential HK filing, Sep; seeking ~$50B valuation

**DeepSeek**
Hired CITIC for a Shanghai listing; private talks ~$75B

**StepFun**
Restructuring to list in Hong Kong

**01.AI**
Targets Hong Kong in 2027

Chipmakers went first: on debut, Moore Threads rose ~400% and MetaX 693% (late 2025)

Sources: Caixin (IPO prices, valuations, 8–9 Jan 2026); stockanalysis.com (52-week highs, latest prices; Nov prices not retrieved); Fortune (Anthropic range, 10 Nov 2026); HK01 via TechNode (Moonshot); Reuters (DeepSeek, StepFun, MetaX, Moore Threads); Deep News (01.AI). Multiples are ours. Press-reported, no HKEX filing read.

# Three risks, each with one number

1 · POLICY

## 44 of 45

local votes to restrict data centers passed

UBS: about 30% of 2028 US additions at risk

2 · FINANCING

## $366B

of commitments at Nvidia alone

CoreWeave: $37–39B capex on $13B revenue

3 · CONCENTRATION

## 44%

of Nvidia’s sales from three customers

TSMC: 72.5% of foundry revenue

ARC RESEARCH VIEW (A VIEW, NOT ADVICE)

# The spending is intact. The winners are spreading down the chain.

## WHAT WE SEE

- Capex at a record: $202B a quarter
- Scarcity has moved to memory and power
- Agents pull in CPUs and DRAM
- The buyers are still the same few

## WHAT TO WATCH

| | |
|---|---|
| **17 Nov** | Nvidia results vs $108B guide |
| **30 Nov–3 Dec** | Nvidia GTC DC |
| **Early Dec** | TSMC November revenue |
| **December** | Fed decision; state data center rules |

SOURCES 1 OF 4

# Company, filed and official

**Capex:** SEC filings (XBRL) for Alphabet, Amazon, Meta, Microsoft, Oracle

**Nvidia:** quarterly releases, CFO commentaries, 10-Q; board statement 8 Nov 2026; Vera Rubin NVL72, GB300, Vera CPU pages

**AMD:** earnings releases Q1 2024–Q3 2026; Instinct MI400 page; deal releases

**Intel, Broadcom:** earnings releases and call transcripts

**TSMC:** monthly revenue reports; quarterly management reports 1Q23–3Q26; earnings calls

**Memory:** SK hynix results; Samsung Q3 2026 release; Micron Q4 FY26 remarks and 10-K; JEDEC HBM4

**Models:** model cards and papers (OpenAI, Google, Meta, DeepSeek, Moonshot, Alibaba); lab pricing and model pages

**Tokens:** Alphabet Q3 2026 CEO remarks (28 Oct); OpenAI (31 Mar 2026)

**Agents:** NVIDIA developer blog (5 May 2026)

**Electricity:** DOE / LBNL data center reports (Dec 2024; 2025 Update, Jun 2026); EIA Short-Term Energy Outlook (Oct 2026)

**Grid:** PJM 2028/29 auction report (14 Jul 2026); ERCOT Senate presentation (29 Jul 2026)

**CoreWeave, GE Vernova, Meta, Google:** company releases and filings

SOURCES 2 OF 4

# Independent research and press-reported

**Epoch AI:** Capabilities Index (10 Nov 2026); Stargate site review; data center directory

**TrendForce:** DRAM contract forecasts 2025–2026; foundry share (9 Sep 2026); CPU:GPU ratios

**Other research:** Counterpoint (HBM share); Mercury Research via press (server CPU share); METR; Bernstein; UBS; Goldman Sachs via Bloomberg; Wood Mackenzie; CBRE; BloombergNEF; Grid Strategies; Global Energy Monitor; Cleanview

**Prices:** stockanalysis.com daily closes; Yahoo Finance; 24/7 Wall St.

**Markets and politics:** AP, Reuters, CNN (5–6 Nov 2026); Yahoo Finance (9–11 Nov)

**Press-reported only:** Google I/O token counts (Shacknews, Digit); Rubin Ultra 1 TB (Tom's Hardware); Google TPU 8 node (The Next Web); Bloom–Oracle (IWR Online); Hyperion details; Nvidia–Groq price; reports about Nvidia's CEO

**Energy deals:** Utility Dive, ESG Dive, POWER Magazine, Power Engineering, The Register, Latitude Media

**Logos and map:** LobeHub Icons, Simple Icons; Highcharts Map Collection (Natural Earth)

SOURCES 3 OF 4

# US vs China, chips, models and the money loop

**Filed:** Nvidia 10-K FY2021–FY2026 and 10-Qs Q1–Q2 FY2027 (China revenue); Alibaba, Tencent, Baidu quarterly results (capex)

**Vendor pages:** Nvidia, AMD Instinct, Google Cloud TPU, AWS Trainium, Microsoft Maia 200 (26 Jan 2026), Meta AI, Huawei

**Deals:** OpenAI (27 Feb 2026), Anthropic (Series H, 28 May 2026), Microsoft (28 Oct 2025), Amazon, AMD, CoreWeave, SoftBank releases; Alphabet, Amazon, Broadcom calls

**Policy:** BIS rules 2022–2026; Nvidia 8-K (9 Apr 2025); MOFCOM announcements 2023–2025; SAMR (15 Sep 2025)

**Research:** Epoch ECI scores; Epoch AI compute by country; Stanford AI Index 2026; ATOM Report (arXiv 2604.07190); Hugging Face; TrendForce (20 Jan 2026); Moody's via TechNode

**Power:** Ember Global Electricity Review 2026; EIA via SUN DAY Campaign; China NEA via Xinhua; Carbon Brief; World Nuclear Association; BloombergNEF

**Press-reported:** Meta MTIA (Tom's Hardware); FT; Reuters; SCMP; WSJ (Oracle); Bloomberg (Google–Anthropic); HotHardware (SpaceX); NeoTeo (CXMT Q3)

**China:** IDC via Reuters; Bernstein; SemiAnalysis; TrendForce DRAM ranking; CXMT prospectus via TrendForce; Commercial Times. Logos: Simple Icons

SOURCES 4 OF 4

# Year-by-year returns, growth, HBM financials, IPOs

**Stock returns:** Year-end and 12 Nov 2026 closes from digrin.com and stockanalysis.com, split-adjusted, price only; cross-checked with macrotrends; SPY and QQQ as index proxies; Yahoo Finance daily closes for Nvidia

**US growth:** BEA national accounts, contributions to real GDP growth (information processing equipment, software), via FRED; Furman; Deutsche Bank; St Louis Fed; Federal Reserve Board; Goldman Sachs; ING

**Data centers:** Epoch AI US sites tracker; OpenAI Stargate sites; company releases

**Cloud and own chips:** Amazon, Alphabet, Microsoft, Oracle earnings releases and calls (cloud revenue, Azure growth, RPO, backlog); Google Cloud, AWS, Microsoft and Meta chip announcements

**China IPOs:** Caixin; Reuters; HK01 via TechNode; stockanalysis.com; Deep News

**Late-2026 additions:** Marvell/Lumentum 8-K; SpaceX S-1; Cerebras 8-K; METR; FRED and Fed statements; Silicon Data, SemiAnalysis; neocloud filings; Nvidia 10-Q; neolab press (TNW, Forbes)

**HBM makers:** SK hynix, Samsung Electronics and Micron quarterly results and earnings calls; Korea Exchange; FactSet

**AI vs the rest, breadth:** FactSet Earnings Insight; S&P Dow Jones Indices; First Trust; Morgan Stanley via Yahoo Finance; SPY, RSP and sector SPDR closes (digrin, stockanalysis, StatMuse)

**IPOs:** Fortune, 10 Nov 2026; Built In, 28 Oct 2026 (Anthropic S-1); Motley Fool, 22 Oct 2026, citing Fortune and Reuters (OpenAI)

**2022–23 story:** openai.com (ChatGPT, OpenAI LP, ChatGPT Plus, GPT-4); Microsoft (2020, 2023); UBS via Reuters; Epoch notable models; Meta LLaMA paper; Inflection; Nvidia results releases FY2023–24 and H100, Mellanox releases; TSMC via Counterpoint; TechInsights via HPCwire